Hierarchical Structural Realism and Universality Classes of Embedded Observers

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Abstract

This paper articulates the structural commitments of the Incompleteness of Observation (OI) framework beyond its selected cubic-lattice branch, and develops comparisons with broader unification programs. An observation axis distinguishes the foundational axiom, observer admission, specified observational equivalence classes and selected model families; a gauge axis distinguishes Hamiltonian rephasing, emergent gauge structure, proposed substratum identifications and algebra/channel equivalence. These are levels of comparison rather than a proved chain of group inclusions. *Part I* finds structural incompatibilities between the standard matrix-model formulations of string theory and OI's microscopic A1–A6 model family, particularly at A2 and A5; this does not by itself decide observational equivalence. *Part II* defines embedded observers using a Schrödinger channel and its invariant Heisenberg adjoint restriction, and defines universality classes by normal -*isomorphism and intertwining of those restricted pairs. Proposition 9.5 proves inclusion for two specified fixed-interface channel-preserving transformations, while the full substratum gauge classification remains open. Minimal prepared-isometry uniqueness is separated into finite-dimensional kernel results and a general separable-space proof; neither determines full hidden dynamics from one channel.* Part III* audits shared probability-representation, dilation, memory and symmetry mechanisms across frameworks. Shared features motivate comparisons but do not establish membership in one observational class. Observer admission and the conditional Standard-Model predictions remain distinct questions, with prediction transport requiring the observables and dynamics relevant to each prediction to be preserved.

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1. Introduction

1.1 Scope and structure

The framework developed in [Main, SM, GR, Substratum] supplies **conditional outputs for a selected cubic local branch**, not a unique local substratum

modulo a completed gauge group. The original semigroup-transfer inference is refuted and the classification of all admissible observer partitions remains open [Substratum §§3–4]. H-Bell separately conditions Bell-inclusive existence, whose uniqueness also remains open.

The first layer (§2) articulates the *hierarchical structure* of the framework's structural realism. Its observation axis runs from the foundational axiom through observer admission and specified observational equivalence to the selected OI model branch; its gauge axis distinguishes Hamiltonian rephasing, emergent gauge structure, proposed substratum identifications and algebra/channel equivalence. Different predictions use different carriers and hypotheses. The diagram organizes those dependencies without asserting a complete nested gauge classification.

The second layer (Parts I, II, III) develops the framework's relationship to broader unification programs through two related but distinct questions:

Question 1: Substratum-level equivalence. Are SM-reproducing string compactifications gauge-equivalent representatives of OI's substratum equivalence class? An affirmative answer would dissolve the landscape multiplicity problem. Part I (§§3–7) develops this through technical analysis of matrix-model formulations, with negative outcome — OI's structural conditions A2 (determinism) and A5 (linearity) fail.

Question 2: Cross-framework structural realism. If the microscopic model families differ, can specified embedded observers nevertheless be observationally equivalent? Part II (§§8–13) defines this through normal *-isomorphism and intertwining of the restricted Heisenberg pairs. Proposition 9.5 supplies a restricted inclusion of established channel-preserving OI transformations; comparison with the full proposed $\mathcal{G}_{\text{sub}}$ remains open.

Part III (§14) synthesizes both questions through a four-feature audit (Born rule, channel-level unitarity, memory/divisibility, commutant gauge-invariance) characterizing the OI-string universality-class relationship.

The two parts have different epistemic statuses. Part I produces a *definitive structural result* (negative — matrix-model formulations fail OI's A2 and A5). Part II is *programmatic* — it articulates the universality-class proposal precisely, formalizes it via definitions and propositions in §9, and identifies open technical questions whose resolution would establish or refute the framing. Definitive identification of specific universality classes — including the OI-string cross-class — requires substantial follow-up work along the lines flagged in §§11.2 and §13.1.

1.2 The string landscape as a multiplicity problem

For four decades, string theory has aspired to derive the Standard Model from compactification of higher-dimensional spacetime. The state of the art is summarized by Marchesano, Shiu, and Weigand [MSW2024]: many compactifications

produce gauge groups that include $SU(3) \times SU(2) \times U(1)$ as a subgroup; some produce three generations of chiral matter; a smaller subset produces the observed hypercharge structure; a still smaller subset produces approximately the observed Yukawa pattern. No compactification has been demonstrated to reproduce the Standard Model uniquely with all parameters matching observation, and no general scheme is known to reproduce the flavor hierarchy [MSW2024, §2.3].

The conventional response to this state of affairs has been the landscape picture: string theory has approximately 10^{500} metastable vacua, each corresponding to a different low-energy effective theory, and our universe corresponds to whichever vacuum produces the SM with the observed parameters. The vacuum-selection problem — why our universe is in this particular vacuum — has been variously addressed by the anthropic principle, by eternal inflation populating all vacua, or by sharper conjectures (the swampland program) attempting to delineate which vacua are consistent with quantum gravity [Vafa2005, Palti2019, OoguriVafa2007].

The swampland program in particular has produced significant progress in constraining the landscape from above [Lehnert2025, MSW2024]. Conjectures such as the Weak Gravity Conjecture, the de Sitter Conjecture, the AdS Distance Conjecture, and the Cobordism Conjecture aim to identify which low-energy effective theories admit consistent UV completion in quantum gravity. These conjectures shrink the apparent multiplicity of vacua but do not, in their current form, single out a unique SM-reproducing vacuum.

1.3 The substratum picture

In a separate program [Main, SM, GR, Substratum], the Incompleteness of Observation (OI) framework approaches fundamental physics from a different direction. The starting point is the observation that quantum mechanics is the marginal-statistics description used by an embedded observer with finite distinguishable states; the trace-out over the hidden sector generates the reduced observable law — universally admitting the quantum representation — from a deterministic classical substratum. The framework develops two technical results:

Conditional local reconstruction (Theorem 23, [Substratum §3]). The selected $d = 3$ nearest-neighbor cubic wave-rule branch, H-link with $m = 1$, the octahedral representation, amplitude-scale gauge and the separate [SM] matter premises give the stated local $K = 6$, $(3, 2, 1)$ and representation outputs under their own conditions. The action-scale relation $\hbar = c^3\epsilon^2/(4G\sigma)$ also requires H-slope and the [GR] horizon/frame hypotheses. E1–E7 and A1–A6 **alone** do not uniquely select this branch, and full gauge uniqueness is not established. M1-B/H-Bell are independent Bell-composite inputs.

Substratum observational equivalences (Theorem 24, [Substratum §4]). The appropriate transformations depend on the observable carrier; four

proposed families are not an established complete gauge-group classification:

(i) *Hidden relabellings.* A uniform hidden permutation preserves the full fixed-interface coherent-channel family; a visible-fibre-dependent hidden permutation preserves rooted populations but need not preserve coherences. An arbitrary full-state permutation is not gauge at a fixed readout.

(ii) *Alphabet-size comparisons.* Structural results proved for all q in [SM §2.7] need no additional hypothesis; equality of transition probabilities across distinct q requires its separate hypothesis.

(iii) *Decoupled hidden enlargement.* Independent normalized ancillary dynamics preserve coherent channels when the partial trace is defined, including trace-class infinite-dimensional cases; coupled extensions do not have this exact invariance.

(iv) *Graph structural comparison.* Theorems 1–6 retain their specified dimension, mode and gravitational/quantum structural content under their own hypotheses; quasi-isometry does not itself imply equality of all channels or preserve the cubic octahedral SM decomposition.

The proposed structural-realism interpretation concerns structures actually invariant under a specified observer/readout equivalence. The complete quotient $[(S, \varphi)]/\mathcal{G}_{\mathrm{sub}}$ is not yet characterized; the 22 classified SM comparisons and horizon normalization are conditional evidence for selected mathematical branches, not the proof of a unique substratum.

1.4 Question 1: the equivalence-class proposal

The two pictures describe the same observational target — SM physics — from very different directions. String theory begins from quantum gravity in $d = 10$ or $d = 11$ and seeks to reach the SM through compactification. OI begins from the embedded-observer ontology and reaches the SM through reconstruction. Question 1 asks whether the two descriptions are independent, complementary, or equivalent.

The core proposal investigated in Part I is:

Proposal (observational comparison). A string compactification reproducing the observed Standard Model could be compared with a selected OI realization only after specifying common observable structures, preparations and readouts. Agreement on gauge factors, generations or measured constants does not establish equality of the full channel family or a transformation between the substrata. Theorem 24 does not supply a complete four-generator classification: Pair C refutes the restricted coherent-channel proposal and all-partitions equivalence remains open. Assigning compactification moduli or flux choices to OI hidden freedoms therefore requires an explicit comparison map.

This proposal has three substantive consequences if defensible:

(A) *Conditional reduction of multiplicity.* If an explicit map proved that a specified set of compactifications produced identical data for the chosen

carrier, those differences would be redundant for that carrier. Reproducing the same low-energy Standard Model structure is not sufficient to establish this equivalence for all physical observations.

(B) *Conditional transfer of predictions.* A prediction transfers between models only when a comparison map preserves the data that define that prediction and its hypotheses. Matching some Standard Model outputs does not transfer every OI prediction to every SM-reproducing compactification, or conversely.

(C) *Computational gateway.* OI's cubic-lattice substratum admits direct lattice-Monte-Carlo and lattice-perturbation-theory computation. String compactifications are mathematically rich but computationally demanding. Gauge-equivalence under the proposal would let OI's lattice serve as a computational tool for string-theoretic questions, with results lifted to string-theory predictions through the equivalence.

Pre-conditions for the proposal. The proposal is suggestive but requires three pre-conditions:

Pre-condition 1: Generator (ii) extension to continuum. OI's generator (ii) operates on the alphabet $\mathbb{Z}/q\mathbb{Z}$, a discrete structure. String theory uses continuous degrees of freedom — string positions on Calabi-Yau, complex-structure and Kähler moduli, RR flux quanta. The technical question is whether OI's alphabet-size hypothesis [SM §2.7] extends to continuum theories, or whether OI's discrete alphabet is essential to the reconstruction. If discrete nature is essential, the bridge between OI and string theory passes through finite-dimensional matrix-model formulations of strings (BFSS, IKKT), not the continuum string description directly.

Pre-condition 2: Existence of SM-reproducing compactifications. Despite forty years of effort, no string compactification has been demonstrated to uniquely reproduce the SM with all parameters matching observation [MSW2024]. The "SM-reproducing subset" of the landscape is conjectured to exist but has not been explicitly exhibited.

Pre-condition 3: A1-A6 compatibility under matrix-model bridge. Even with matrix-model finite-dimensional formulations as intermediaries, OI's structural conditions A1 (finiteness), A2 (deterministic bijection), A3 (bounded coupling), A5 (linearity) may be violated by string-theoretic structure. Whether matrix models satisfy A1-A6 is not obvious and requires technical verification.

§§3-6 of Part I address these pre-conditions; §7 considers the consequences both if they hold and if they fail.

1.5 Question 2: why universality classes are substantively new

The negative answer to Question 1 (Part I) does not rest on a complete substratum gauge classification. It concerns the stated model-class differences between

OI and string-theoretic constructions; a conjectured broader universality equivalence is a different question.

Question 2 asks whether distinct microscopic descriptions produce equivalent embedded observers for a specified observational carrier. Definition 9.3 gives the precise criterion: a normal unital *-isomorphism intertwining the restricted adjoint dynamics. Agreement on a list of broad features is evidence for a comparison, while equality of the restricted pairs is the class-membership question. These classes can cut across microscopic model families, including families specified by A1–A6.

Part I's findings make this question concrete. The four-feature audit in §14 compares Born-form representation, channel dilation, memory/divisibility and symmetry restriction across OI and string-theoretic constructions. The memory mechanism has analogs across frameworks, while P-indivisibility is not established cross-framework (§14.3, §14.5). These features define useful comparison families; they do not determine a single universality class in Definition 9.4.

Part II develops the formal framework for this question. Section 8 compares categorical, operational and algebraic formalizations; §9 adopts the algebraic one, defines observational equivalence, proves the restricted inclusion of Proposition 9.5 and separates finite minimal-isometry uniqueness (Propositions 9.7a–9.7b) from its separable extension (Proposition 9.7c). Observer admission is addressed by Definition 9.9, Proposition 9.10 and Corollary 9.11. Section 10 supplies a finite-cutoff OI construction and a conditional EFT target; §11 formulates the comparison with string constructions; §§12–13 develop interpretation, prediction transport and open comparison tasks.

The two questions are not independent. Question 1's negative answer motivates Question 2's broader framework: if substrata cannot be gauge-equivalent at the substratum level, the next question is what equivalence does hold. Question 2's universality-class framework then provides the language for Part III's cross-framework synthesis.

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2. The framework's hierarchical structure

The framework's structural-realism stance uses two axes of comparison. The *observation axis* records the scope and assumptions of a model or observational carrier. The *gauge axis* records the transformations shown to preserve specified data. Calling them orthogonal distinguishes these questions; it does not supply inclusion maps between all the equivalences or identify a microscopic model family with one observational class.

This section articulates the two axes, their orthogonality, and the multi-level structural realism that follows. The articulation is canonical for the framework — it locates where each technical result lives and stratifies the framework's predictions by hierarchy intersection. Subsequent sections of this paper develop

content at specific levels (Part I at Level D × Level G3; Part II at Level C × Level G4; Part III at the partial-trace observational features sub-class within Level C).

2.1 The observation hierarchy

The observation hierarchy nests by inclusion: each deeper level is a more specific commitment.

Level A — The observation axiom. The framework begins with the fact that *observation occurs* [Main §1]. An observer records distinguishable outcomes of interactions with a system not wholly under the observer's control. Level A commits to: there is something to observe; some part of reality is structured to record outcomes (observers are embedded in reality, not external to it); observation has structural consequences. This level is shared with mainstream observer-essentiality programs since 2022 (CLPW, Maldacena, HUZ, DEHK, Slagle-Preskill); the framework's distinguishing feature is producing the consequences from a finite deterministic substratum without extra dimensions or stochasticity.

Level B — Observer admission. Not every substratum admits observation. The framework identifies four conditions on a substratum's bijection φ that are necessary for an embedded observer to exist [Main §3]:

(C1) **Coupling.** The interaction between the observer's degrees of freedom and the rest must be substantial.

(C2) **Record persistence.** The written record must survive to readback — in our universe realized by a slow bath, the hidden-sector correlation time much longer than the observer's measurement timescale (one sufficient route).

(C3) **Sufficient capacity.** The hidden sector must have information capacity enough to store the interaction history.

(C4) **History readback.** The interaction must read back what was written — the load-bearing memory condition.

A substratum (S, φ) is *observer-admitting* if there exists a partition $V \subset S$ such that (S, φ, V) satisfies C1–C4, and *observer-lacking* otherwise (formal Definition 9.9 in §9.6). Level B commits to: the space of substrata is structurally partitioned into observer-admitting and observer-lacking subsets; most substrata are observer-lacking (random permutations on finite sets typically lack the slow-mixing, sufficient-capacity, coupled-subsystem structure required); the framework's content applies only to observer-admitting substrata.

This has structural-anthropic content. Standard anthropic reasoning treats parameters as contingent and explained by selection: we observe a universe with parameters compatible with us because we couldn't observe a universe without them. The observer-admission reading sharpens this: observer-lacking

substrata are not candidates for "our universe" at all — they are mathematical objects of a different kind, without observer-extraction structure. Within observer-admitting substrata, anthropic reasoning operates normally.

Level C — Universality classes. For each admitted embedded observer, choose an invariant observable algebra and the channel or channel family to be compared. Two observers are observationally equivalent when a normal unital *-isomorphism intertwines their restricted Heisenberg adjoint dynamics (Definition 9.3). A universality class is the isomorphism class of $(\mathcal{A}_V, \Phi^*|_{\mathcal{A}_V})$ in Definition 9.4; a proper observable algebra can retain less information than the full ambient channel.

Broad comparison families and exact observational classes serve different purposes (§9.5). Three useful descriptions are:

- *Restriction-interface family:* systems sharing quantum-probability representation or dilation architecture; memory and symmetry restriction require their additional hypotheses.
- *Gauge-target family:* models reproducing specified gauge and matter data, without an assumed isomorphism of their full observational carriers.
- *Algebra/channel class:* systems satisfying the exact restricted-pair equivalence of Definition 9.3.

Level C supports an observation-relative structural realism: the chosen restricted-pair isomorphism class is the common content. Proposition 9.5 places its two established OI channel invariances inside that class under their stated preparation and algebra-invariance hypotheses. It supplies no inclusion theorem for every proposed substratum transformation or complete ordering of observational quotients.

Level D — OI's selected model family and branch. OI adds microscopic structural conditions A1–A6 [Substratum §3] and then selects a cubic local branch with the further hypotheses stated below. These conditions organize a model family; they do not by themselves select one class of Definition 9.4 or a unique microscopic representative:

(A1) Finiteness; (A2) Determinism (bijection); (A3) Bounded coupling degree; (A4) Center independence; (A5) Linearity; (A6) Background independence.

Together with the additional selected wave-rule, H-link and representation hypotheses made explicit in [Substratum Theorem 23], these give conditional local lattice/gauge outputs rather than a proof that E1–E7/A1–A6 uniquely determine a full microscopic gauge class. H-Bell separately conditions Bell-inclusive existence, whose uniqueness is open.

A1–A6 and observer-admission C1–C4 have different roles. C1–C4 describe the observer-admission and memory-bearing sector used in [Main]; A1–A6 constrain the selected microscopic model family. A1 concerns finiteness rather than the capacity clause C3, and A2 specifies deterministic microscopic dynamics. A3–A6 enter the selected local branch and its conditional gauge derivation. Failure

of these microscopic conditions in another formulation does not exclude equivalence of its chosen observable pair with an OI realization.

Level D studies the selected $d = 3$ cubic branch, its exact signed-link representation and its conditional physical outputs. Under H-link single-copy and H-cust, a generic equivariant condensate has stabilizer $\mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$; the physical local gauge reading and central reduction use the additional hypotheses in [SM §4.6]. The quantitative comparisons, including Cabibbo, Koide and the horizon $A/4$ consistency relation, retain the assumptions and classifications in [SM §7] and [GR §5]. The realist proposal concerns the invariant content of a specified observational quotient, rather than an already completed $[(S, \varphi)]/\mathcal{G}_{\mathrm{sub}}$ classification.

The levels organize increasingly specific commitments, but observational classes and microscopic model families need not be nested sets: one family can contain several observational classes, and one class can contain realizations from different families.

2.2 The gauge hierarchy

The gauge hierarchy is the breadth axis: which transformations are observable-preserving, with broader levels absorbing more degrees of freedom as gauge.

Level G1 — D-gauge on the emergent Hamiltonian. The diagonal-unitary equivalence on the emergent Hamiltonian: $H \to DHD^\dagger$ with D diagonal-unitary in the eigenbasis of measurement outcomes [GR §3.3]. This is an exactly invisible freedom of the transition data. Energy shifts and the antiunitary reflection are also invisible; their completeness under the stated gap and overlap hypotheses is conditional on H-BG [GR §3.3]. Two Hamiltonians H and $DHD^\dagger$ produce identical $T_{ij}(t) = |\langle j|e^{-iHt/\hbar}|i\rangle|^2$ in the measurement basis. G1 is the smallest gauge group; it operates within a single Hamiltonian / single QM realization.

Level G2 — Gauge structure on the selected physical branch. The six signed cubic links have the exact multiplicity-free decomposition $T_1(3) \oplus E(2) \oplus A_1(1)$. Under H-link single-copy and H-cust, a generic equivariant condensate has stabilizer $G_0 = \mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$; the commutant of the cubic action itself is block-scalar. With the additional central choice $K = \{(cI_3, dI_2, c^2/d) : c, d \in \mathrm{U}(1)\}$, [SM §4.9, Proposition 14a] identifies G_0/K with $S(\mathrm{U}(3) \times \mathrm{U}(2))$, hence the standard $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$ global form; Proposition 15a gives a faithful charged-family action. The native conjugation action on $\mathrm{End}(V_3 \oplus V_2 \oplus V_1)$ has only the common scalar U(1) kernel, so the second central circle and charged physical observer action are additional premises. Local gauge covariance and physical matter selection retain the hypotheses stated in [SM]. Relating this action to G1 or G3 requires an explicit map; matching local factors alone does not identify observational classes.

Level G3 — Proposed substratum observational gauge. The observa-

tional equivalence is not a single fully classified group without specifying whether it compares coherent channels, rooted populations, or all admissible partitions. For a fixed coherent-channel interface, hidden-only relabellings and decoupled enlargement preserve channels but **do not exhaust** channel-equivalent dynamics (Pair C); for population data the hidden-fibre relabelling freedom is broader. The all-partitions classification and its mapping onto the SM gauge group are open [Substratum §4].

Level G4 — Universality-class equivalence. Definition 9.3 compares $(\mathcal{A}_V, \Phi^*|_{\mathcal{A}_V})$ through a normal unital *-isomorphism and adjoint intertwining. Proposition 9.5 proves that its two specified fixed-interface channel transformations preserve this datum; it does not show that every proposed element of $\mathcal{G}_{\text{sub}}$ does so. Conversely, the same restricted pair may have distinct microscopic realizations, and equivalence on a proper algebra need not identify their full ambient channels. Cross-framework class membership requires the actual algebra map and intertwining for the chosen carrier.

The sequence G1, G2, G3 and G4 represents proposed levels of comparison rather than a proved chain of subgroup inclusions. G1 concerns Hamiltonian transition data; G2 the conditional physical condensate-stabilizer construction; G3 specified substratum observational identifications; G4 restricted algebra/channel equivalence. Their carriers and comparison maps must be supplied before a nesting theorem can be asserted.

2.3 Orthogonality and multi-level structural realism

The observation axis describes scope and assumptions; the gauge axis describes data-preserving transformations. The selected OI branch supplies examples at G1–G3, while G4 compares chosen observable pairs across model families. This organization keeps microscopic assumptions, observable content and gauge maps distinct.

What is real differs at each level:

- *Level G4 × Level C:* the isomorphism class of the specified invariant observable algebra and restricted adjoint dynamics. Quantum representation, hidden memory and symmetry properties are included only to the extent that the chosen carrier encodes them.
- *Level G3 × Level D:* the substratum modulo a specified observational equivalence; all-partitions completeness remains open.
- *Level G2 × Level D:* the selected physical gauge/matter branch, with its carrier, condensate, central-reduction and matter hypotheses.
- *Level G1 × Level D:* the emergent Hamiltonian's specified transition data modulo the established invariances.

These interpretations concern different carriers and assumptions. Their compatibility and empirical reach are determined by the maps preserving the data at each level.

2.4 Predictions stratified by hierarchy level

The framework's empirical predictions live at specific intersections:

- *Level G4 × Level C predictions* are the predictions encoded in a specified restricted observable pair, with compared preparations and readouts transported by its isomorphism. A single-channel pair does not automatically encode a multi-time intervention process.
- *Level G2 × Level D predictions* include the conditional gauge and matter outputs of the selected physical branch: the local gauge factors, the $1 \oplus 3$ candidate-sector count (physical generations under H-spin') and the anomaly calculation under its matter assignment.
- *Level G3 × Level D calculations* include the Cabibbo, Koide and horizon relations under their stated branch assumptions. A value measured on the chosen carrier is invariant between equivalent representatives; a model variation changing that value belongs to a different class on that carrier.

The selected OI branch supplies the distinctive quantitative calculations. Other models may share some of those outputs. Agreement on a gauge target or on several measured values is useful evidence, but establishing one observational class requires the specified algebra map and dynamics intertwining.

Falsification tests the assumptions and observable claims used by a calculation. A failure of a conditional OI branch prediction challenges that branch or its physical identification, while leaving an exact representation theorem or a different model family unaffected unless it uses the same claim. The JUNO relation $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$ is a quantitative selected-branch comparison; the horizon $A/4$ coefficient is a conditional consistency relation with no direct empirical test of the coefficient itself. Different representatives of one class cannot disagree on either quantity when the carrier and compared protocols encode it.

2.5 Where each level's content lives

The framework's papers carry content at different levels of the two-dimensional structure:

- *[Main]:* Levels A and B (observation axiom; embedded-observer characterization via C1–C4)
- *[SM]:* Level D × Levels G1–G2 (Standard Model structure on OI's specific cubic lattice)
- *[GR]:* Level D × Levels G1–G2 (gravitational thermodynamics on OI's specific cosmological-horizon structure)
- *[Substratum]:* Level D × Level G3 (reconstruction theorem; substratum gauge group $\mathcal{G}_{\text{sub}}$)
- *This paper:* Levels B and C × Level G4 (universality-class formalization, observer-admission distinction, OI-string relationship at universality-class level, landscape reframing)

The evidence at each level is separately scoped: Level A is the structural starting

point; Level B uses C1–C4 and the observer-admission results; Level C uses the stated algebra-channel universality definitions and propositions; Level D uses Theorem 23's **conditional selected-branch outputs** and Theorem 24's **restricted invariances and counterexamples**, not a complete substratum gauge reconstruction. G1 and the cubic G2 algebra retain their separate proofs; a full G3 classification and the inter-level maps remain open.

The framework's empirical content (twenty-two classified quantitative predictions — two parameter-free structural and five conditional structural, the remainder layered, chained, or fitted [SM §7.6] — the specific cubic-lattice substratum, and the specific gauge group derivation) is concentrated at Level D × Level G3. The broader levels are where the framework connects to other unification programs and where universality-class structural realism applies — the content of Parts I, II, and III of this paper.

Part I: Substratum-Level Equivalence (Question 1)

3. Per-generator analysis applied to string compactifications

Four proposed comparison families motivate this section: hidden-sector relabelling, alphabet change under the alphabet-size hypothesis, independently decoupled enlargement/reduction, and graph statistical-class comparisons. They are not an established complete generating set for one observational gauge group. Hidden-factor relabelling and decoupled normalized enlargement preserve the fixed coherent-channel interface in Proposition 9.5; alphabet and graph comparisons need additional hypotheses, and population-only fibre relabellings need not preserve coherent channels. Analogies with string constructions—coordinate changes, matrix-rank changes, compactification choices and discretizations—therefore provide questions rather than gauge-equivalence certificates. In particular, a moduli or flux variation changing measured couplings cannot be a change of representative of one class containing those data. Sections 4–6 examine the stated microscopic model-family comparisons separately from this observational test.

The natural identifications are suggestive but the structural incompatibilities documented in §6 prevent direct equivalence-class identification. Per §14.4, the *pattern* of gauge invariance via commutant restriction transports across frameworks, but the specific output (which gauge group, which generators) is forced through framework-specific machinery in each case. This section is preparatory to §6's substratum-level analysis and §14's observational-level analysis; readers who have internalized those sections may treat §3 as supporting documentation.

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4. Generator (ii) and the discrete-to-continuum bridge

This section examines the technical bridge between OI's discrete substratum (alphabet $\mathbb{Z}/q\mathbb{Z}$) and string theory's continuum structures (real-valued matrix entries in BFSS, complex moduli in compactifications, continuous fields on worldsheets). The bridge question is whether OI's alphabet-size hypothesis [SM §2.7] extends to continuum theories, or whether OI's discrete nature is essential to the reconstruction.

4.1 The alphabet-size hypothesis [SM §2.7]

OI's substratum has local state-space $\mathbb{Z}/q\mathbb{Z}$ for some integer $q \geq 2$. That the alphabet size q is a gauge parameter — that no observable depends on the choice of q — is the alphabet-size hypothesis of [SM §2.7], posed for linear dynamics. [SM §2.7] lists the results proved for every q and those that carry q; no proof covers every observable.

This hypothesis bears on OI's claim that A5 (linearity) is forced rather than assumed, conditionally: the forcing runs through *amplitude-scale* gauge (the unobservability of the absolute field scale), which [Substratum §4, Theorem 24, *Status*] classifies as an adjoined operational input rather than a theorem ([SM §2.7]) — its independent warrant by the observer architecture being an open, referee-grade question. Linearity is therefore forced *given* that operational principle, not unconditionally. Non-linear dynamics over $\mathbb{Z}/q\mathbb{Z}$ generically produce q-dependent observable physics (x^2 mod 3 and x^2 mod 5 have different algebraic structure), violating the gauge equivalence. The hypothesis is stated within the discrete alphabet structure; it does not address whether $q \to \infty$ limit produces continuum dynamics or what role real-valued degrees of freedom play.

4.2 Matrix models as candidate finite-dimensional representatives

String-theoretic matrix models — BFSS [BFSS1996], IKKT [IKKT1996] — are finite-dimensional formulations of M-theory and type IIB string theory respectively. BFSS at finite matrix rank N has $9N^2$ continuous bosonic matrix entries plus fermionic counterparts; IKKT has a similar structure with one fewer dimension. These are finite-dimensional in the sense of bounded matrix rank; they are not finite-cardinality in the sense of OI's $|S| < \infty$ requirement.

Whether matrix models can be candidate representatives of OI's equivalence class through the bridge requires a discrete-to-continuum identification. Two natural candidates: (a) the matrix rank N playing the role of OI's alphabet size q, with $N \to \infty$ corresponding to $q \to \infty$; (b) lattice discretization of matrix-model dynamics, producing a finite-cardinality system that approximates the continuum matrix-model in an appropriate limit.

Candidate (a) is structurally suggestive — both q and N parametrize the size of the basic representation space. The disanalogy is that OI's alphabet-size hypothesis asserts q-independence of observables; the analogous claim for N

would be N-independence of matrix-model observables, which is *not* a feature of matrix models. Different N values give different theories (BFSS at $N = 2$ differs from BFSS at $N = 10^9$), and the conjectured M-theory limit is specifically $N \to \infty$ at fixed coupling. The N-dependence of matrix-model dynamics is structural, not gauge.

Candidate (b) is technically tractable through lattice-discretized BFSS [BFSS-Lattice2015]. At fixed lattice spacing and matrix rank, the configuration space is finite-cardinality and OI's A1 holds. The remaining structural conditions A2 (deterministic bijection) and A5 (linearity) still fail per §6's analysis — lattice Monte Carlo is stochastic, and the Yang-Mills commutator-squared interaction remains nonlinear at the lattice level.

4.3 Possible bridges and their status

Three structural possibilities for the discrete-to-continuum bridge:

(a) $q \to \infty$ *limit gives continuous matrices.* Under this reading, OI's substratum at very large q approximates a continuum theory, with continuous matrix entries arising as the limit. This is the most direct continuum extension. The challenge is that the alphabet-size hypothesis is posed for linear dynamics, which the continuum matrix-model violates. The bridge fails at A5.

(b) *Matrix rank N as analog of q.* Under this reading, the matrix-model parameter N plays the same gauge role as OI's q, with N-independence of observables required. This fails empirically — matrix-model dynamics is genuinely N-dependent — and the conjectured M-theory limit is at fixed $N \to \infty$, not at variable N.

(c) *Discrete-and-continuum genuinely incompatible.* Under this reading, OI's discrete structure is essential to its reconstruction theorem, and string theory's continuum structure is essential to its quantum-gravity content. The two frameworks describe physics through structurally distinct mathematical apparatus, with no natural bridge between them at the substratum level. This is the conclusion supported by §6's analysis.

The provisional conclusion is (c). The substratum-level bridge does not exist in the form the equivalence-class proposal of §1.3 required. The bridge does exist at the partial-trace observation level (per §14), but this is a different kind of bridge — it relates structural features of how observation extracts information, not the underlying substratum descriptions. The discrete-to-continuum question at the substratum level is orthogonal to the partial-trace-features question at the observational level.

This connects directly to the framing of §14.5: structural features of observation transport across frameworks (Born rule, unitarity, non-Markovian marginal); substratum-level structure does not (discrete vs continuum, bijection vs path-integral, linear vs nonlinear). The discrete-to-continuum bridge is part of the

substratum-level layer that does not transport.

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5. SM-reproducing compactifications: a catalogue

Three principal classes of string compactifications produce the SM gauge group through structurally different mechanisms. *Heterotic* ($E_8 \times E_8$ or $SO(32)$ on Calabi-Yau threefolds): E_8 broken by CY holonomy and bundle data; three generations from CY topology (Euler character / 2); hypercharge from bundle structure. Specific realizations include the Schoen Calabi-Yau (Bouchard-Donagi 2006) and heterotic line-bundle models (Anderson-Gray-Lukas-Palti 2014). *Type II / D-brane* (IIA or IIB on Calabi-Yau with D-brane configurations): gauge fields from open strings on D-brane stacks producing $U(N)$ factors; chiral matter from D-brane intersections or magnetization; SM gauge group typically as a subgroup of larger Pati-Salam or left-right-symmetric structure. *F-theory* (on elliptically-fibered Calabi-Yau fourfolds): gauge group from singularity structure of the discriminant locus; $SU(5)$ GUT models from A_4 singularities with hypercharge from flux configuration breaking $SU(5)$ to the SM. None of the three classes uniquely determines all SM parameters [MSW2024].

The catalogued compactification families use different gauge constructions—bundles and holonomy, D-brane configurations or singularity/flux data. OI's selected branch instead uses its exact cubic link representation, conditional condensate stabilizer and physical reduction. These constructions can reach a common low-energy gauge target without being equivalent as complete embedded-observer systems. The catalogue supplies candidate comparisons rather than a common observational class.

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6. Structural compatibility: A1–A6 under matrix-model bridges

The equivalence-class reading proposed in §1.3 requires that string compactifications producing the observed Standard Model be gauge-equivalent representatives of $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$. As discussed in §1.4 (Pre-condition 3), the natural bridge between OI's discrete substratum and string theory's continuum description passes through finite-dimensional matrix-model formulations — BFSS [BFSS1996] for type IIA / M-theory, IKKT [IKKT1996] for type IIB, and their close variants. This section examines whether matrix-model formulations satisfy OI's structural conditions A1–A6.

The six structural assumptions A1–A6 [Substratum §3.1] constrain the class of candidate dynamics but do not by themselves select a unique physical branch. Theorem 23's conditional cubic result also uses nearest-neighbor second-order dynamics, H-link with m=1, octahedral representation, amplitude-scale gauge

and separate physical/matter premises. This distinction is crucial when comparing OI's specific choices with string-theoretic substrata.

We address each condition in turn, taking the BFSS matrix model as the principal example. The conclusions extend to IKKT and to BFSS variants with appropriate modifications.

6.1 A1 (finiteness)

Condition. The configuration space S is finite.

BFSS status. The BFSS matrix model at finite N has nine bosonic matrices X^i (with $i \in \{1, \ldots, 9\}$) and sixteen fermionic matrices ψ_α, each of size $N \times N$ with continuous (real-valued bosonic; Grassmann-valued fermionic) entries [BFSS1996, see also [HigherPointBFSS2025]]. The configuration space is $\mathbb{R}^{9N^2}$ for the bosonic sector at fixed N, augmented by a Grassmann algebra for fermions. This is not finite.

Possible repair: lattice discretization. Lattice-discretized BFSS [BFSS-Lattice2015] places the matrix model on a discrete time lattice, and additionally discretizes the matrix entries through standard lattice gauge theory techniques. At fixed N and fixed lattice spacing, with appropriate UV cutoff on matrix-entry values, the configuration space becomes finite. The lattice discretization is the standard tool for nonperturbative numerical study of matrix models and is well-developed in the literature.

Status. A1 fails for continuum BFSS. A1 holds for lattice-discretized BFSS at fixed N and lattice spacing, with the proviso that the continuum and large-N limits — which are believed to recover string-theoretic content — produce $|S| \to \infty$ in a manner that may or may not preserve the equivalence-class structure.

6.2 A2 (determinism / bijection)

Condition. The substratum dynamics $\varphi : S \to S$ is a deterministic bijection.

BFSS status. BFSS dynamics is quantum, defined through the path integral over matrix configurations. At the operational level, lattice-discretized BFSS is studied through Monte Carlo simulation, which samples from the path-integral measure stochastically. Neither the continuum quantum dynamics nor the lattice Monte Carlo is a deterministic bijection on the configuration space.

The classical limit of BFSS — at $\hbar \to 0$, equivalent to taking large coupling λ at fixed N — gives Hamiltonian equations of motion on continuous phase space. These are deterministic but not bijective on a finite configuration space (since the configuration space is continuous). One could discretize phase space and time-step the equations of motion as a deterministic update, but this discretization is artificial and does not respect the underlying continuum dynamics.

Implications for the trace-out construction. OI's emergent-QM derivation [Main §3.2] uses the Stinespring dilation to lift visible-sector CPTP channels to unitaries on $\ell^2(S)$, with the descent restriction selecting permutation matrices and hence bijections. If the substratum dynamics is stochastic (path-integral) rather than deterministic, the trace-out gives a CPTP channel with two sources of randomness — substratum-intrinsic (from the path-integral) and observer-ignorance (from the trace-out) — rather than the observer-ignorance-only structure that OI's framework requires. This breaks the framework's identification of substratum bijection with observable QM unitarity.

Status. A2 fails for matrix models in their standard quantum formulation. The failure is not a discretization artifact but a structural feature: matrix models are quantum theories, and their dynamics is quantum-mechanical (path-integral) rather than classical-deterministic. Repair would require reformulating the matrix model as a classical cellular-automaton dynamics with bijection structure, which is not the standard BFSS conjecture and would represent a substantial re-foundation rather than a discretization.

6.3 A3 (bounded coupling degree)

Condition. Each site is coupled to a bounded number of neighbors in φ's action.

Status. Reframing-dependent. Treating individual matrix entries as sites, BFSS's Yang-Mills interaction $-\frac{1}{4}\mathrm{Tr}[X^i, X^j]^2$ couples each entry to $\sim N^2$ others (unbounded in the large-N limit). Treating each matrix as a single site (9 bosonic sites, 16 fermionic in BFSS), coupling is bounded. The matrix-as-site reframing is in tension with OI's combinatorial substratum (where "site" is a single configuration value); without independent argument that this reframing is correct rather than a definitional convenience, A3 should be regarded as failing on the natural reading.

6.4 A4 (center independence)

Status. Holds. BFSS has manifest invariance under matrix translations $X^i \to X^i + c^i \mathbb{1}$ (the BFSS analog of spatial translation), and matrix indices a, b have no preferred values; the action is symmetric under index permutations. Both features are manifest in the BFSS action.

6.5 A5 (linearity)

Condition. The wave equation governing φ is linear in the configuration variables.

BFSS status. The BFSS action contains the commutator-squared interaction $-\frac{1}{4}\mathrm{Tr}[X^i, X^j]^2$, which is *quartic* in the matrix entries. The equations of motion derived from this action are nonlinear. This is not an accident or simplification:

the nonlinear matrix interaction is what produces the gauge structure of M-theory, the emergence of D-brane dynamics, and the noncommutative-geometry interpretation of matrix theory.

OI's [SM §4.1] necessity argument for linearity rests on amplitude-scale gauge (§4.1 above; [Substratum §4]), an adjoined operational input that the alphabet-size hypothesis of [SM §2.7] does not entail. The hypothesis adds a consistency statement: nonlinear dynamics over $\mathbb{Z}/q\mathbb{Z}$ generically have q-dependent emergent physics (x^2 mod 3 vs x^2 mod 5 are algebraically distinct), violating it.

In matrix models, the analog of q is the matrix size N. Different N values give algebraically distinct theories — the BFSS coupling structure depends on N, and the $N \to \infty$ limit (which is believed to recover the full M-theoretic content) is structurally distinct from finite N. Whether this N-dependence corresponds to OI's q-dependence — or whether N is more analogous to OI's deep-sector cardinality (generator (iii)) — is not obvious.

If N is analogous to q, then the N-dependence of matrix-model dynamics directly violates the OI requirement. If N is analogous to deep-sector cardinality, then N-independence at fixed observable physics is required for N to be gauge under generator (iii); this is the content of the BFSS conjecture (large-N matrix model = M-theory) but is not rigorously established and remains a conjecture.

Status. A5 fails directly: BFSS has nonlinear matrix dynamics, in contrast to OI's linear wave equation. The deeper question — whether the nonlinearity is fundamental or whether some reformulation of BFSS as a linear theory in a different basis is possible — is open but does not have a known answer in the affirmative direction. The most natural reading is that matrix models genuinely have nonlinear substratum dynamics, in structural disagreement with OI's reconstruction.

6.6 A6 (background independence)

Status. Holds in spirit, conditional on the matrix-spatial bridge. BFSS has manifest $U(N)$ gauge invariance under local matrix transformations $X^i \to UX^iU^{-1}$, which is the precursor of the spacetime gauge theory in the long-distance limit. OI's A6 requires covariance under spatially varying internal-index transformations, with the link coupling transformed alongside, but BFSS has no spatial structure beyond the matrix index; the BFSS analog is matrix-internal $U(N)$ gauge invariance. Whether matrix-internal gauge invariance plays the role A6 requires depends on the matrix-spatial bridge, established only in specific limits (e.g., the diagonal-matrix limit where eigenvalues correspond to D0-brane positions).

6.7 Summary of A1–A6 status

The tally across A1–A6:

Condition	BFSS status	Comment
A1 (finiteness)	Fails (continuum) / holds (lattice)	Repairable through lattice discretization
A2 (determinism / bijection)	Fails	Path-integral is stochastic, not bijection; structural mismatch
A3 (bounded coupling)	Fails (entry-as-site) / holds (matrix-as-site)	Reframing-dependent
A4 (center independence)	Holds	Manifest in BFSS
A5 (linearity)	Fails	Commutator-squared interaction is nonlinear; structural mismatch
A6 (background independence)	Holds (matrix-gauge) / conditional (spatial)	Depends on matrix-spatial bridge

The outcome is mixed. Three of six conditions (A4, A6 in spirit, A1 with discretization) are compatible with matrix-model structure. Three of six (A2, A3 on natural reading, A5) fail in ways that are not simple discretization issues but reflect structural commitments of matrix-model dynamics that diverge from OI's substratum.

The key failure is A5 (linearity). OI's reconstruction theorem derives, conditionally on H-link, the Standard Model gauge group through cubic-group decomposition of the $K = 6$ internal components, with the wave-equation linearity playing an essential role: it is the linearity that the alphabet-size hypothesis [SM §2.7] presupposes, and it is the wave-equation structure that produces the cubic-lattice spectrum from which the gauge group is derived. Matrix models reach the gauge group through a different route — emergent gauge fields from open strings ending on D-branes, or matrix-internal $U(N)$ symmetries enhanced in specific limits — and this route is genuinely nonlinear.

The implication for the comparison proposal. The audit finds that the standard matrix formulations do not satisfy the microscopic conditions of the selected OI model family, especially A2 and A5. It therefore supplies no direct realization of those matrix models within that family. This is not a proof that their chosen observable pairs are inequivalent: a microscopic condition is not an invariant of Definition 9.3 unless the specified observational carrier encodes it. That separate comparison still requires its algebra map and restricted adjoint intertwining.

This conclusion is structural rather than technical. Matrix models are the most direct finite-dimensional formulations of string theory, and their structural divergence from OI's substratum at A2 and A5 is informative about the relationship between the two programs. OI and string theory are not gauge-equivalent in the sense the equivalence-class proposal asks.

6.8 What can be salvaged

The negative conclusion in §7.8 admits five partial recoveries, ordered from most to least plausible.

(i) Different bridge. Matrix models are not the only finite-dimensional formulation of string theory. Alternative bridges include: (a) the AdS/CFT correspondence, where the boundary CFT lives on a finite-dimensional spacetime that may be discretized while preserving the bulk string content; (b) tensor-network reformulations of matrix models [TensorNetworkMatrix2024], which translate continuous matrix dynamics into discrete tensor structures; (c) causal dynamical triangulations [Loll2019], a discretization scheme for quantum gravity with manifestly finite configuration space and combinatorial dynamics. Each of these admits independent investigation of A1–A6 compatibility, and the equivalence-class reading might hold for one of them even if it fails for matrix models. This paper does not pursue these alternatives in detail; they remain open directions.

(ii) Emergent string-theoretic features from OI substratum. A reformulation of the relationship asks not whether string theory's substratum-level structures (matrix models) are equivalent to OI's substratum, but whether OI's substratum produces string-theoretic features at appropriate emergent scales. OI's GR §5 recovers the Bekenstein-Hawking $A/4$ entropy through a thermal-self-consistency calibration that is structurally distinct from string theory's microscopic D-brane count [StromingerVafa1996], the coefficient being the ratio of the KMS and first-law normalizations that the calibration matches; the D-brane count yields the same coefficient. If OI's substratum also produces, at appropriate emergent scales, holographic duality structure analogous to AdS/CFT [Maldacena1997], graviton modes from the long-distance limit of the substratum dynamics, and dualities analogous to T-duality and S-duality, then string theory's distinctive technical achievements would be recovered as features of OI's emergent physics rather than as independent substratum-level commitments. The structural barrier identified in §6 (linearity and determinism at the substratum level vs.~Yang-Mills nonlinearity and path-integral structure) does not preclude this recovery: effective field theories with nonlinear interactions and stochastic descriptions are routinely derived from underlying linear-deterministic dynamics through coarse-graining and emergent-scale phenomena. Whether OI specifically produces the right emergent-scale features to recover string-theoretic content is an open technical question requiring direct investigation; this paper does not pursue the analysis but flags it as the most consequential research direction left open by the negative result.

(iii) Approximate equivalence. The equivalence-class reading might hold *approximately* — in the sense that SM-reproducing matrix-model compactifications agree with OI's predictions to within experimental precision, even if the underlying structural conditions differ. This would be an empirical equivalence rather than a structural one and would be testable by computing OI-style predictions (Cabibbo angle, Koide ratio) within specific compactifications and

comparing to OI's results. This is an empirical research direction not pursued here.

(iv) Reformulation of A2/A5. OI's A2 (determinism) and A5 (linearity) might admit reformulations that accommodate matrix-model structure. For A2: a path-integral over deterministic bijection trajectories produces ensemble-stochastic behavior at the coarse-grained level; if BFSS can be reformulated as such an ensemble, A2 might hold at the trajectory level. For A5: the commutator-squared interaction is linear in the *combination* $[X^i, X^j]$, with the nonlinearity arising from the commutator structure rather than higher-order terms; if "linearity" is read at the combination level, A5 might hold. Both readings are speculative and have not been worked out in detail. The current state of OI's framework treats A2 and A5 in the standard sense (bijection at the configuration level; linear in single-variable expansion), and any reformulation must be argued explicitly.

(v) Shared observational invariants. A further comparison asks which observable structures distinct microscopic constructions share. Section 14 identifies quantum-representation/dilation architecture and candidate memory or symmetry mechanisms. Proposition 9.5 gives a restricted OI invariance theorem, while a complete substratum quotient remains open. Agreement on generic features motivates a cross-framework comparison, but equality of an observable pair must be established at a specified carrier rather than inferred from the act of observation itself.

6.9 Summary

Matrix-model formulations of string theory do not satisfy OI's structural conditions A1–A6 in their standard formulation. The failures at A2 (determinism / bijection) and A5 (linearity) are deep — they reflect structural commitments of string theory that diverge from OI's substratum, not artifacts of the discretization bridge. The equivalence-class proposal of §1.3 therefore does not hold for matrix models as currently understood.

This negative result is the central technical finding of this paper. Section 6 develops its implications: that OI and string theory are independent programs with the same observational target, and that the conventional approach to combining them — string theory as a UV completion of the SM, with OI as a foundational reformulation — is structurally distinct from the equivalence-class reading that this paper investigated.

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7. Implications and discussion

The structural analysis of §6 finds that the standard matrix-model formulations examined there fail OI's stated microscopic A1–A6 conditions, with A2 and A5 reflecting substantive differences in description. Thus it does not provide a

direct realization of those matrix models as members of the selected OI model family. Observational equivalence is a separate question: different microscopic structures can still induce equivalent restricted observable pairs. No complete $\mathcal{G}_{\text{sub}}$ classification or cross-framework comparison map has been supplied to decide that question.

7.1 OI and string theory as independent programs

OI and string theory remain independent research programs using structurally distinct microscopic formulations to address overlapping observable targets. Section 6 does not identify them as gauge-related formulations or as stages of one derivation. It also does not rule out equivalence of specified observable pairs; that would require the comparison proposed in Part II.

This independence has methodological consequences. A successful prediction of OI (e.g., the Cabibbo angle $1/(\pi\sqrt{2})$, the Koide ratio $2/3$) is not automatically a prediction of string theory; conversely, the swampland program's identification of consistent UV completions of the SM does not automatically constrain OI's substratum. Each program is responsible for its own empirical match, with no cross-attribution beyond what shared empirical inputs (E1–E4 in OI's notation) provide.

The independence is also informative about the foundational landscape of fundamental physics. OI and string theory are two of several active programs (alongside loop quantum gravity, causal sets, asymptotic safety, and others) that propose substratum descriptions for the SM. The structural divergence between OI and string theory at A2 and A5 indicates that these are deep ontological choices, not technical details: a substratum framework either commits to deterministic-bijection / linear-wave-equation structure (OI's choice, with the foundational economy this implies) or to path-integral / gauge-Yang-Mills structure (string theory's choice, with the gravitational-unification this provides). The choice cannot be made on technical grounds alone; it requires foundational arguments.

The structural divergence leaves several possibilities: distinct descriptions may be empirically distinguishable on a sufficiently rich carrier; they may agree on a restricted carrier despite different microscopic assumptions; or a further construction may relate them as limits or effective descriptions. The present audit does not select among these possibilities. Conditional prediction tests and explicit algebra/channel comparisons can distinguish them without assuming that microscopic difference alone settles observational equivalence.

7.2 The string landscape multiplicity problem is not dissolved by OI's gauge group

The proposed landscape reduction would require a map identifying the selected compactifications' observational data. The matrix-model audit supplies no such map, and the restricted OI invariances do not identify the landscape by themselves. A reduction of multiplicity therefore does not follow from the results

here.

The landscape multiplicity problem remains open. The established OI channel invariances do not identify all compactifications with matching low-energy targets, and Theorem 24 does not provide a complete classification even within the selected OI model family. The A1–A6 mismatch found for the standard matrix models therefore supplies neither a landscape reduction nor a classification of their observable equivalence classes.

This limits OI's proposed gauge classification twice: the full all-partitions observational equivalence within the OI class remains open, and the broader equivalence between OI-class substrata and non-OI candidate models is a separate question. Pair C already refutes the proposed restricted hidden-only/decoupled-generator completeness for one fixed coherent-channel interface. Neither result identifies all observationally equivalent substrata outside the selected cubic branch.

7.3 Cross-attribution of predictions does not follow

The second consequence flagged in §1.3 — that OI's predictions become string-theory predictions through gauge equivalence — likewise does not follow. OI's prediction set (twenty-two classified quantitative SM relations, the sharpest matching observation at sub-percent precision) is a feature of OI's specific substratum derivation chain, which uses linearity of the wave equation, the alphabet-size hypothesis, and cubic-group decomposition. None of these is present in matrix models in the form OI requires, so the prediction chain does not transport.

This means the empirical match of OI's predictions is OI-specific evidence, neither confirmed nor refuted by string-theoretic considerations. If OI's predictions hold up under further experimental scrutiny (DESI Year 5 for the cosmological prediction; JUNO for $\sin^2\theta_{12}$; lattice tests for additional Layer-2(a) predictions), this is evidence for OI's substratum description specifically. If they fail, it's evidence against OI specifically. String theory is not implicated either way through the gauge-equivalence route examined here.

7.4 Computational gateway via OI's lattice does not extend to string theory

The third consequence — OI's computationally-tractable lattice as a tool for string-theoretic computations — also does not follow. Without gauge equivalence, OI's lattice computations produce predictions of OI's substratum, not of string compactifications. They cannot substitute for direct string-theoretic computation (Calabi-Yau geometry, brane dynamics, flux integration) within a specific compactification.

This limitation does not, however, diminish OI's own computational tractability. OI's predictions remain computable from the cubic-lattice substratum without

needing to engage string-theoretic structures. The framework's empirical successes are independent of string-theoretic computation, and that independence is preserved by the negative result.

7.5 Connection to swampland conjectures

The swampland program [Vafa2005, OoguriVafa2007, Palti2019, Lehnert2025] aims to identify which low-energy effective field theories admit consistent UV completion in quantum gravity, by formulating conjectures (Weak Gravity, de Sitter, AdS Distance, Cobordism, etc.) that consistent EFTs must satisfy. The program's success would be a constructive characterization of the SM-reproducing subset of the landscape.

OI's reconstruction theorem provides an alternative characterization of the same SM-reproducing target, but through different methodology: OI derives the SM from the embedded-observer ontology, whereas the swampland program identifies which UV completions are compatible with consistent quantum gravity. The two characterizations are independent in the sense established by §6, but they may agree on which low-energy effective theories are admissible. This section examines OI's compatibility with the major swampland conjectures, identifying both the consistency results and the cross-validation pathways the conjectures provide for OI's specific predictions.

7.5.1 Weak Gravity Conjecture The Weak Gravity Conjecture [ArkaniHamedMotlNicolisVafa2007] states that any consistent quantum-gravity EFT containing a $U(1)$ gauge field must contain a particle whose charge-to-mass ratio in Planck units is ≥ 1 — equivalently, gravity is the weakest force.

OI's emergent QFT contains the SM gauge structure $SU(3) \times SU(2) \times U(1)$ with the standard charged-particle spectrum. The lightest electrically charged particle is the electron, with charge-to-mass ratio $q/m \sim 10^{21}$ in Planck units — vastly exceeding the WGC threshold. The conjecture is satisfied robustly by OI's emergent QFT for the standard $U(1)$ gauge sector.

The neutrino sector provides a sharper test. [IbañezMartinLozanoValenzuela2017] uses the WGC applied to a $U(1)_{B-L}$ gauge field to constrain neutrino-sector physics: if neutrinos are Dirac, the WGC requires a tower of charged states with masses below the electroweak scale, which has not been observed; if neutrinos are Majorana, the WGC constraint is satisfied via the Majorana mass term itself. OI's [SM §8.4] predicts Majorana neutrinos, putting OI's prediction in the WGC-allowed region. The cross-validation pathway is direct: WGC + observed neutrino phenomenology → Majorana neutrinos; OI → Majorana neutrinos. Both routes converge on the same prediction through structurally distinct reasoning.

7.5.2 de Sitter Conjecture The de Sitter Conjecture [ObiedOoguriSpodyneikoVafa2018] in its refined form [OoguriPaltiShiuVafa2019] states that the

scalar potential V of a consistent quantum-gravity EFT satisfies $|\nabla V|/V \geq c \sim O(1)$ at any de Sitter critical point, OR $\min(\mathrm{Hess}\, V) \leq -c'/M_P^2 V$, where c, c' are positive $O(1)$ constants. The conjecture forbids stable de Sitter vacua and is in tension with the standard ΛCDM cosmology with a positive cosmological constant.

OI's [GR §7] carries dark energy in Type II running-vacuum (RVM) form rather than as a fixed cosmological constant — the functional form the local expansion shared by every smooth model, the content the Type II channel, a structural reading rather than a derived law — with the parameter ν encoding the running and the framework's expected magnitude $|\nu| \sim 10^{-32}$. Type II RVM is parametrically away from a static de Sitter vacuum: the cosmological "constant" runs with the Hubble rate, and the late-universe phase is asymptotically de Sitter in the limit $H \to 0$ but not strictly so at any finite time. The dynamical character is consistent with the de Sitter Conjecture's structural content; the small predicted $|\nu|$ value is consistent with the empirical near-ΛCDM behavior. Strict static de Sitter would violate the conjecture; dynamical near-de Sitter satisfies it.

The recent DESI DR2 results, rejecting the cosmological constant at 2.8σ–4.2σ in extended cosmological models with potential phantom crossing [DESI_DR2_2025], provide independent empirical pressure on static dark-energy models — the same direction the de Sitter Conjecture pushes. OI's position relative to this data must be stated carefully, consistent with [GR §6.2, §7.1]: the framework's predicted running $|\nu| \sim 10^{-32}$ is observationally indistinguishable from a cosmological constant, so the DESI phantom-crossing *magnitude* is in 3–4σ tension with OI's prediction even though the running-vacuum *form* is favoured qualitatively. The de Sitter Conjecture consistency claim rests on the strict-static distinction — OI is nominally dynamical but observationally Λ-like — and should be read at that level of strength.

7.5.3 Distance Conjecture The Distance Conjecture [OoguriVafa2007] states that as one moves to large distance in the moduli space of a consistent quantum-gravity EFT, an infinite tower of states becomes light, with masses scaling exponentially with the distance traveled.

OI's framework has no fundamental moduli: the substratum bijection is fixed, and the emergent QFT's parameters are derived from the cubic-lattice structure rather than being moduli to be selected. The Distance Conjecture is therefore satisfied trivially in a degenerate sense — there is no moduli space to traverse.

This is a structural difference between OI and string-theoretic frameworks. In string theory, the landscape's structure is the moduli space of compactifications; the Distance Conjecture provides a non-trivial constraint on EFTs derived from that moduli space. In OI's framework, the analog of the moduli space is the universality-class structure of Part II — different substrata in different universality classes are not connected by continuous moduli but by structural-class

membership. The conjecture's content does not directly transport, but a weakened analog might apply: substrata "far" from OI's class in the universality-class space might be connected to OI's class only through transitions involving infinite towers of states. This is speculative and open.

7.5.4 AdS Distance Conjecture The AdS Distance Conjecture [LustPaltiVafa2019] states that AdS minima of consistent quantum-gravity EFTs are accompanied by an infinite tower of states becoming light, with masses scaling with the AdS radius. The conjecture excludes scale-separated AdS vacua in the strict sense.

OI's emergent QFT does not predict an AdS minimum: the cosmological structure is positive-cosmological-constant (de Sitter-like), with the small $|\nu|$ deviation per §7.5.2. The AdS Distance Conjecture is therefore not directly applicable to OI's framework. If a sub-cluster of observer-admitting compactifications reproduced OI's predictions but admitted an AdS minimum, the conjecture would constrain that sub-cluster — but this would be a constraint within the universality-class space, not on OI itself.

7.5.5 Cross-validation pathways The swampland conjectures collectively define a region of EFT parameter space compatible with consistent quantum gravity. OI's emergent QFT lives within this region across the conjectures examined (WGC, dS, Distance, AdS Distance). This is non-trivial cross-validation: OI's structural predictions (Majorana neutrinos, dynamical dark energy via Type II RVM, no fundamental moduli, no AdS minimum) align with the swampland-allowed region without being designed for it.

The cross-validation operates at the level of structural prediction, not numerical agreement. OI predicts Majorana neutrinos because the substratum's structural conditions force lepton-number violation; the WGC predicts Majorana neutrinos because Dirac neutrinos require an unobserved tower of states. Both routes converge on the same prediction through different reasoning. Similar structural convergence operates for dark energy: OI predicts running cosmological constant from the GR cluster; the de Sitter Conjecture forbids static de Sitter vacua. Both favor dynamical dark energy.

The pathway provides empirical leverage. If LEGEND-1000 [LEGEND2024] confirms Majorana neutrinos, both OI and the WGC-derived prediction are confirmed. If DESI Year 5 strengthens the rejection of the cosmological constant, both OI's Type II running-vacuum channel and the de Sitter Conjecture are favored. The empirical tests are joint tests of OI and the swampland-favored EFT structure — confirmation supports both; refutation challenges both.

This section does not resolve whether OI's specific predictions can be derived from swampland reasoning, or whether swampland-consistent EFTs uniquely determine OI's universality class. Those are open questions. What this section establishes is that OI's framework is consistent with the major swampland con-

jectures examined, and that cross-validation pathways exist for OI's structural predictions through swampland-derived constraints.

7.6 Partial recoveries and what would be required to revisit the equivalence-class reading

§6.8 identified five partial recoveries, ordered roughly from most to least consequential. *Emergent string-theoretic features from OI's substratum* (recovery (ii)) is the most consequential if it holds: OI's GR §5 already produces the Bekenstein-Hawking $A/4$ entropy through thermal self-consistency [StromingerVafa1996 has the microscopic derivation]; whether OI similarly produces holographic-duality structure analogous to AdS/CFT, graviton modes from long-distance dynamics, and T/S-dualities at appropriate emergent scales is an open technical question. The structural barrier identified in §6 does not preclude this recovery: emergent-scale physics can have nonlinear and stochastic features arising from underlying linear-deterministic dynamics through coarse-graining. *Shared observational invariants* (recovery (v)) is structurally distinct: it asks whether OI and string theory agree at the observable level because both respect invariants of the observation process, identifying invariants across structural classes rather than within OI's class. *Alternative bridges* (AdS/CFT with discrete boundary, tensor networks, causal dynamical triangulations) is tractable but would constitute substantial follow-up work. *Approximate-equivalence via empirical agreement* is testable: computing OI's predictions (Cabibbo, Koide) within a specific SM-reproducing compactification and comparing. *A2/A5 reformulation* is the most speculative — it would require substantial reworking of either OI's conditions or string-theoretic structure.

The two most consequential — emergent features and shared invariants — are not in competition. They address structurally distinct questions: the first treats OI as a candidate foundational description from which string-theoretic content emerges; the second treats observation itself as the locus where different substratum descriptions are forced into agreement. Both extend OI's structural-realism stance — that what is real is invariant structure, not specific representatives — to the comparison with other unification programs.

7.7 Recovery (ii) developed: emergent string-theoretic features from OI's substratum

Recovery (ii) of §7.6 — that OI's substratum produces emergent string-theoretic features at appropriate emergent scales — is the most consequential of the five partial recoveries if it holds. This section develops what is currently tractable in this direction, distinguishing structural features of string theory that plausibly emerge from OI's substratum from features that do not, with honest scoping of which features are open research questions.

The structural barrier identified in §6 (A2 and A5 violations in matrix-model formulations) does not preclude this recovery. The barrier operates at the level

of substratum descriptions: OI's substratum is deterministic-bijection-on-cubic-lattice; matrix models' substrata are Hermitian-matrix-path-integral. These differ as substratum descriptions. However, emergent-scale physics derived from each substratum can have shared features, in the same way that two different microscopic theories can share long-distance behavior through RG universality. The question is which string-theoretic features are robust at the emergent scale and which are tied specifically to string-theoretic substratum content.

7.7.1 Horizon holography vs full AdS/CFT The strongest emergence claim is for *horizon holography* — the Bekenstein-Hawking $A/4$ area law and the holographic information bound. OI's [GR §5] recovers the $A/4$ coefficient from thermal self-consistency at the cosmological-horizon scale, as the ratio of the KMS and first-law normalizations that the calibration matches, with the entropy density emerging as a partial-trace invariant of the embedded-observer marginal. The derivation does not appeal to specific string-theoretic content; it invokes only the framework's structural features (the trace-out, the cosmological-horizon information capacity, the thermal equilibrium of the hidden sector). The $A/4$ coefficient transports to OI from the partial-trace structure, not from string-theoretic content.

This produces a structural feature shared with string-theoretic horizon physics: any horizon entropy in OI's framework is bounded by $A/4$, and the holographic principle (information content of a region bounded by area, not volume) holds. The shared feature is at the level of the partial-trace observational structure, consistent with §14.2's transport assessment for channel-level unitarity. Recent confirmation comes from GW250114 [GW250114], where Hawking's area theorem was confirmed at 99.999% confidence — a classical-horizon-dynamics result that both OI and string theory preserve; the 1/4 coefficient itself (in OI as the ratio matched by the horizon calibration, under the horizon conditions of [GR §2, §8.5], in string theory by microscopic state counting [StromingerVafa1996]) has no direct empirical test.

The harder question is whether OI produces *full AdS/CFT-style holography* — bulk-boundary correspondence with specific dictionary between bulk fields and boundary operators. Here the answer is more nuanced. OI's cosmological structure is positive-Λ (de Sitter-like), not anti-de Sitter; the cosmological horizon is a de Sitter horizon, not an AdS boundary. The bulk-boundary correspondence in standard AdS/CFT relies on the asymptotic AdS structure to define boundary operators uniquely, and this asymptotic structure is absent in OI's framework.

Two possible directions for emergence remain open:

- *de Sitter holography.* Work on de Sitter holography [StromingerDS2001] develops bulk-boundary correspondence for de Sitter spacetime with a CFT defined on the asymptotic past or future de Sitter boundary. OI's cosmological structure is asymptotically de Sitter (under the framework's

$|\nu| \sim 10^{-32}$ Type II RVM), so de Sitter holography would be the natural framework. Whether OI's substratum produces a specific de Sitter holographic dictionary — bulk graviton fields $\leftrightarrow$ boundary CFT operators with specific scaling dimensions — is open. The structural features (asymptotic de Sitter, finite horizon entropy, holographic bound) are present; the specific dictionary requires further analysis.

- *Approximate AdS in deep IR.* In some limits, de Sitter spacetime admits approximate AdS structure (e.g., near the horizon, in certain coordinate patches). Whether OI's framework produces an effective AdS region with corresponding effective AdS/CFT structure is similarly open.

The honest assessment: OI naturally produces horizon holography (the $A/4$ entropy law and the holographic bound) but does not naturally produce full AdS/CFT-style correspondence as an emergent feature. A is a structural consequence of the partial-trace observational structure; the latter requires specific compactification-dependent content that OI's substratum does not directly provide.

7.7.2 Graviton modes from long-distance dynamics OI's [GR §3–§5] derives linearized Einstein gravity from the partial-trace structure of the substratum bijection at the cosmological-horizon scale. The mechanism: the substratum's coherent excitations include long-wavelength fluctuations of the visible-sector metric, which are the emergent graviton modes. The framework reproduces the standard GR structure at the macroscopic level with Type II RVM corrections, including the linearized-graviton sector with the right propagator structure (massless spin-2 modes, transverse-traceless polarization, lightlike propagation).

This emergence is structurally analogous to the string-theoretic graviton emergence: in both frameworks, gravitons arise as collective modes of the substratum dynamics. In string theory, the graviton is a closed-string vibrational mode with worldsheet σ-model structure. In OI, the graviton is an emergent collective mode of the substratum's coherent excitations on the cubic lattice, derived through the partial-trace-and-coarse-grain procedure.

The *structural similarity* is at the level of the IR effective theory: both frameworks produce massless spin-2 modes coupled to the stress-energy tensor in the linearized limit, satisfying the equivalence principle and gauge-invariance under linearized diffeomorphisms. This is what physicists mean by "the emergent low-energy theory contains gravitons."

The *structural difference* is at the UV completion. In string theory, the graviton is a fundamental string mode with worldsheet origin; in OI, the graviton is an emergent collective mode with no string-mode interpretation. The differences become important at high energies — string-scale corrections to graviton propagators in string theory vs Planck-scale corrections from substratum-level lattice structure in OI; the existence of an infinite Regge tower of massive string modes

vs OI's finite-but-large substratum modes. These UV differences are testable in principle but typically inaccessible at current and near-future energy scales.

The recovery (ii) claim is supported for graviton-mode emergence: OI produces emergent graviton modes with the structural features that string-theoretic graviton modes share — namely the IR effective-theory content. The framework does not produce the specifically string-theoretic UV completion (worldsheet structure, infinite massive tower) but produces the universally-required IR structure that any successful emergent-graviton framework must reproduce.

7.7.3 T-duality and S-duality analogs T-duality in string theory is the equivalence of closed-string theories on a circle of radius R and radius α'/R, with momentum modes $\leftrightarrow$ winding modes. The duality requires a string-mode structure with both momentum and winding quantum numbers, neither of which has a direct analog in OI's substratum.

OI's substratum has lattice-momentum modes (excitations with definite lattice wavevector), but no winding modes (no extended objects winding around compact dimensions). The cubic lattice is the physical spatial structure, not a compactification. T-duality therefore has no clean analog in OI's framework: the duality-pair structure (momentum $\leftrightarrow$ winding) is absent because winding modes are absent.

A weakened analog might exist at the universality-class level. If two SM-reproducing universality classes have algebra-channel pairs related by a specific symmetry — say, an automorphism that exchanges high-momentum and low-momentum content — this would be a T-duality-like feature at the universality-class level rather than the substratum level. Whether such a structure exists in OI's universality-class framework is open. The honest assessment: T-duality at the substratum level has no analog; T-duality-like features at the universality-class level are speculative.

S-duality in string theory is the equivalence of strong-coupling and weak-coupling descriptions in certain string theories (most cleanly in type IIB). The duality maps the string coupling $g_s \to 1/g_s$, exchanging fundamental strings with D1-branes (and other extended objects) under the duality.

OI's framework has emergent gauge couplings derived from the substratum structure. Whether there is a structural S-duality-like feature — e.g., a duality between a strong-coupling regime where the visible-sector EFT description breaks down and a weak-coupling regime in a different visible-sector EFT description — is an open question. The substratum-level dynamics is fixed (the bijection on the cubic lattice), so there is no direct analog of varying the string coupling. The visible-sector EFT couplings can be varied continuously, but whether the strong-coupling limit maps onto a different weak-coupling EFT description is not established.

The honest assessment: S-duality analogs in OI are speculative. Some structural

features of S-duality (the existence of dual descriptions of the same observable physics through different regime variables) might transport to OI's universality-class framework, but no specific S-duality-like map has been identified.

7.7.4 Synthesis: what emerges and what does not Recovery (ii) develops with the following pattern:

Feature	Emerges from OI?	Caveat
Horizon holography ($A/4$ law, holographic bound)	Yes	Structural consequence of partial-trace observation
Bulk-boundary correspondence (full AdS/CFT)	No	OI is dS, not AdS; de Sitter holography open
Linearized graviton modes	Yes	IR effective theory shared with string theory
String UV completion (worldsheet, Regge tower)	No	OI's UV is lattice substratum, structurally different
T-duality	No	No winding modes in OI substratum
S-duality	Speculative	Universality-class-level analogs open

The pattern: OI naturally produces the *IR effective theory* and the *structural observational features* of string-theoretic frameworks (horizon holography, graviton modes, holographic bound), but does not produce the *specifically string-theoretic UV content* (string worldsheet, Regge tower, T-duality, AdS/CFT bulk-boundary dictionary). The IR content transports because it is a feature of the partial-trace observational structure; the UV content does not transport because it is specific to string-theoretic substratum content that OI's substratum does not produce.

This refines the §6 negative result. OI and string theory share the IR effective theory at the partial-trace observational level (recovery (v)), and emergent structural features that follow from the partial-trace structure (the horizon-holography component of recovery (ii)), but they differ on string-theoretic UV content. The partial recovery via (ii) succeeds for the IR features and the horizon-holography component; it does not succeed for the full string-theoretic UV content.

The implication for the §1.4 equivalence-class proposal: the proposal does not hold at the substratum level (per §6), and it does not hold at the level of string-theoretic UV content (per the negative parts of this section), but it does hold

at the level of IR effective theory and partial-trace observational features (per the positive parts of this section and §14). Recovery (ii) is therefore a partial recovery in the precise sense: it identifies which features transport — the structural features of partial-trace observation (Born rule, channel-level unitarity, non-Markovian marginal, horizon holography, linearized graviton modes) and the IR effective theory — but not the substratum-specific UV content.

The work that remains: a rigorous derivation of de Sitter holographic dictionary from OI's substratum (§7.7.1); a quantitative comparison of OI's and string-theoretic graviton-propagator UV corrections (§7.7.2); identification or exclusion of universality-class-level T- and S-duality analogs (§7.7.3). These are the natural follow-up directions to this recovery's development.

7.8 What this paper has established

This paper distinguishes a direct microscopic model-family comparison from the observational comparison. The matrix-model audit supplies no direct A1–A6 realization and no derivation relating the two microscopic descriptions; the proposed recovery directions remain research questions. The §14 audit develops shared representation, dilation and memory mechanisms. Those positive structural parallels motivate the restricted-pair comparison, while neither proving nor refuting equality of the full observational carriers.

7.9 Observer admission and the structure of the landscape

Part II §9.6 formalizes a structural distinction between *observer-admitting* substrata — those that admit a partition satisfying C1–C4 — and *observer-lacking* substrata, where no such partition exists. The distinction operates at the level of the substratum's bijection structure, prior to any observer being instantiated. This section examines how the distinction reframes the relationship between OI and the string landscape.

The landscape, viewed through observer admission. The string landscape comprises approximately 10^{500} distinct compactifications. The standard reading [Vafa2005, Susskind2003] treats the landscape as a multiplicity of candidates for "the universe," with anthropic selection narrowing the candidate set to those compatible with our observations. Under the observer-admission reading, the landscape's multiplicity decomposes structurally:

- *Observer-lacking compactifications.* Compactifications producing four-dimensional vacua without chiral matter, with rapid hidden-sector mixing, with insufficient horizon capacity, or with broken cosmological structure cannot satisfy C1–C4. They correspond to observer-lacking substrata. These compactifications are mathematical objects in string theory's space of vacua, but they are not candidate physical universes in OI's framework — observers cannot be embedded in them.
- *Observer-admitting compactifications.* Compactifications producing four-

dimensional vacua with chiral matter, stable cosmological structure (de Sitter or near-de Sitter with horizon timescales $\tau_B \sim H^{-1}$), and sufficient information capacity (the cosmological horizon's entropy $\sim 10^{122}$) satisfy C1–C4 and are observer-admitting. The SM-reproducing compactifications surveyed in §5 — heterotic on Calabi-Yau with appropriate bundles, Type II / D-brane Pati-Salam constructions, F-theory $SU(5)$ with hypercharge flux — fall in this class.

The observer-admission criterion distinguishes the two classes structurally. Most compactifications in the landscape are likely observer-lacking; the SM-reproducing subset is a small fraction. The numerical multiplicity of the landscape (10^{500}) is largely a measure of observer-lacking substrata, not of competing candidates for our universe.

The translation of C1–C4 to string-theoretic structure. The structural conditions for observer admission [Main §3] are formulated for finite deterministic substrata. Translating them to string-theoretic compactifications requires identifying analogs in string-theoretic structure:

- *C1 (coupling)* requires substantial coupling between visible-sector and hidden-sector dynamics. In string-theoretic terms: the four-dimensional EFT must have non-trivial interaction between observable matter (chiral fermions, gauge fields, Higgs sector) and the gravitational / Kaluza-Klein / hidden gauge sectors. Compactifications with completely decoupled hidden sectors fail C1.
- *C2 (memory persistence)* requires the written record to persist to readback, the slow bath one realization: the hidden-sector correlation time much longer than the observer's measurement timescale. In string-theoretic terms: the cosmological structure must have a stable phase with horizon timescales $\tau_B \sim H^{-1}$ much longer than laboratory timescales τ_S. AdS compactifications without de Sitter structure, or compactifications with rapidly mixing cosmological structure, fail C2.
- *C3 (sufficient capacity)* requires the hidden sector to have sufficient information capacity to store the interaction history. In string-theoretic terms: the cosmological horizon's information capacity (set by the holographic bound at $\sim 10^{122}$ for our observed universe) must be sufficient. Compactifications with horizons of insufficient capacity fail C3.

These translations are intuitively clear: a compactification reproducing our universe's cosmological structure satisfies all three. A compactification producing AdS, anti-de Sitter, or rapidly-mixing dS-with-very-short-horizon-time fails one or more. The translation is plausible structurally, though a rigorous version would require detailed analysis specific to compactification classes.

The OI-string relationship reframed. Under the observer-admission reading, the relationship between OI and string theory is structurally distinct from the standard "two competing programs" framing. OI operates entirely within

observer-admitting substrata — the framework's foundational axiom "observation occurs" is taken to be structurally specific to observer-admitting cases (per §9.6). String theory's landscape encompasses both observer-admitting and observer-lacking substrata, with the observer-admitting subset being where SM-reproducing physics resides.

The relationship therefore takes this form:

- OI's universality class lives in the observer-admitting subset.
- Some SM-reproducing string compactifications also live in the observer-admitting subset.
- The intersection between OI's universality class and the observer-admitting compactification subset is what the partial-trace-features analysis of §14 characterizes.
- Compactifications outside the observer-admitting subset are not candidates for "our universe" in OI's framework — they are observer-lacking substrata, mathematical objects but not physical universes.

This reframing has substantive consequences for the landscape problem. The standard landscape problem treats 10^{500} vacua as competitors; selecting "the right one" appears to require either anthropic reasoning or unprincipled choice. The observer-admission reading dissolves this: most of the landscape is observer-lacking and not in competition for "our universe." The candidate space is the observer-admitting subset, which is much smaller. Within the candidate space, our universe is in some specific universality class, distinguished from other observer-admitting universality classes by specific predictions.

Connection to anthropic reasoning. The standard anthropic argument explains the landscape's empirical match to our universe through selection: we observe a universe compatible with observers because we couldn't observe a universe incompatible with observers. The observer-admission reading sharpens this. Anthropic selection separates into two distinct kinds of work:

- *Structural exclusion of observer-lacking substrata.* Substrata that lack the structural features for observer-admission are not in the candidate space at all. There is no anthropic selection within them — they are not competitors. This is structural exclusion, not selection.
- *Within-candidate-space selection of universality class.* Among observer-admitting substrata, our universe is in some specific universality class. Standard anthropic reasoning applies here: we are in the universality class that produces our observations, and the framework's predictions distinguish OI's class from other observer-admitting classes within the candidate space.

Observer admission asks which candidate models can support the required records. Selecting among those candidates by observed couplings, matter data or dynamics is a further empirical question. The four-feature audit identifies broad comparison mechanisms, while a particular observational class still requires

the data and map of Definition 9.3. This distinction preserves the observer-admission proposal without treating a feature list as a completed classification of the landscape.

Implications for the relationship. The "two competing programs" framing of OI vs. string theory is not the right characterization under the observer-admission reading. The structurally accurate framing is:

- *String theory* supplies a broad collection of candidate microscopic and effective constructions whose observer-level content must be specified for comparison.
- *OI* studies observer-admitting models subject to A1–A6 and the additional selected-branch assumptions.
- *A shared observational class* requires particular constructions and the map of Definition 9.3; the four-feature audit alone does not establish the intersection.
- *Other comparisons* may reveal different observable data or shared coarser targets, depending on the chosen carrier.

This reframing separates observer admission, microscopic model choice and observational equivalence. The selected OI branch retains its conditional predictions, which can distinguish it from other observer-admitting models; deciding that the models share one restricted-pair class requires the explicit comparison.

Open structural questions. The observer-admission framing of the landscape is a structural reading, not a rigorous result. Several questions remain open:

- *Rigorous translation of C1–C4 to string-theoretic structure.* The translations above are plausible but not formal. Working out a rigorous version for specific compactification classes (heterotic, Type II, F-theory) would require detailed analysis specific to each class.
- *Numerical estimate of the observer-admitting subset.* The standard 10^{500} figure is for the full landscape including observer-lacking compactifications. Estimating the observer-admitting subset's size — even to within several orders of magnitude — would substantially clarify the landscape problem's empirical content.
- *Observer-admitting compactifications outside OI's universality class.* If observer-admitting compactifications exist that are not in OI's universality class, they would be alternative candidates for "the universe" — distinct from OI in their predictions, with their own internal structure. Identifying such compactifications would be a substantial result.
- *Connection to swampland criteria.* The swampland program identifies criteria for compactifications consistent with quantum gravity (Weak Gravity Conjecture, de Sitter Conjecture, Distance Conjecture). The observer-admission criterion is structurally distinct but may overlap with swampland criteria — both restrict the candidate compactification space, pos-

sibly with different overlap regions. Working out the relationship would clarify both frameworks.

These questions are not addressed in this paper; they are flagged as research directions opened by the observer-admission reframing. The reframing itself — reading the landscape's multiplicity as containing predominantly observer-lacking substrata, with the observer-admitting subset being structurally smaller — is a substantive contribution that the framework's structural-realism stance makes available to the landscape discussion.

Part II: Universality Classes of Embedded Observers (Question 2)

8. Formalizing embedded observation

The notion of "embedded observer" in OI's framework is informal: an observer with finite distinguishable states, embedded in a substratum, with non-trivial coupling to a hidden sector that has slow dynamics relative to observation, and high information capacity in the hidden sector. For the universality-class question to admit precise formulation, we need a formalization sufficient to define equivalence relations on substratum-with-observer systems. Three candidate formalizations are available.

8.1 Categorical formalization

The categorical approach treats embedded observers as functors $F : \mathbf{Sub} \to \mathbf{Obs}$ from a category of substratum descriptions to a category of observable descriptions, with the universality-class equivalence given by natural equivalence of functors (substrata producing isomorphic observable descriptions, possibly through different observation functors). The formalization is well-defined but requires substantial machinery (explicit construction of **Sub** and **Obs** with their respective morphism structures) that has not been worked out in the OI literature. We do not pursue it further here.

8.2 Operational formalization

The operational approach treats embedded observers as operational protocols $(\mathcal{P}, \mathcal{M}, p)$ — preparations, measurements, and outcome probabilities — without committing to a specific underlying substratum, in the spirit of Mackey's quantum logic [Mackey1963] and the generalized probabilistic theories program [Hardy2001, Barrett2007]. The universality class is the equivalence class under operational indistinguishability. This formalization is closest to empiricist intuition and methodologically clear, but it is empiricist — it does not commit to

substratum-level realism — and may sit uneasily with OI's structural-realism stance, which treats the equivalence class itself as ontologically real.

8.3 Algebraic formalization

The algebraic approach assigns an embedded observer an invariant algebra of observables and a channel on states. In the Schrödinger picture the channel is $\Phi : \mathcal{T}_1(\mathcal{H}_V) \to \mathcal{T}_1(\mathcal{H}_V)$, where $\mathcal{T}_1$ denotes trace-class operators. Its adjoint $\Phi^* : \mathcal{B}(\mathcal{H}_V) \to \mathcal{B}(\mathcal{H}_V)$ is normal, unital and completely positive. The data comprise: - An observable von Neumann algebra $\mathcal{A}_V \subseteq \mathcal{B}(\mathcal{H}_V)$ satisfying $\Phi^*(\mathcal{A}_V) \subseteq \mathcal{A}_V$; - The restricted Heisenberg dynamics $\Phi^*|_{\mathcal{A}_V}$; - When supplied, a unitary realization: a hidden space $\mathcal{H}_H$, a reference state ρ_H, and a unitary U with $\Phi(\rho) = \mathrm{Tr}_H\, U(\rho \otimes \rho_H)U^\dagger$ (Definition 9.2). For pure ρ_H, the associated isometry is the Stinespring dilation.

The realization contains microscopic data beyond this observable carrier. The carrier is the pair $(\mathcal{A}_V, \Phi^*|_{\mathcal{A}_V})$. Taking $\mathcal{A}_V = \mathcal{B}(\mathcal{H}_V)$ always satisfies invariance; a proper subalgebra requires a separate invariance check. Equality of the full ambient channel is stronger than equivalence of these restricted pairs.

The universality-class proposal compares these pairs by a normal unital *-isomorphism intertwining the restricted adjoint channels. For a specified channel family (Φ_t), the chosen algebra must be invariant under every Φ_t^* and the same isomorphism must intertwine every t. Neither a single channel nor its separate-time family alone specifies all joint intervention-word probabilities.

Status. The algebra/channel formalization specifies an observable carrier using standard operator-algebra and channel machinery. A chosen OI realization supplies such data through its observation map. Theorem 23 does not recover a unique realization from the pair, and Theorem 24 does not classify all transformations preserving it. Extending the comparison to a string construction requires its corresponding algebra/channel pair and a map between the chosen observable structures.

8.4 Choosing a formalization

The three formalizations are related but not equivalent. The operational formalization is weakest — it characterizes observers by their data without committing to underlying structure. The algebraic formalization adds structure (algebra of observables, channel dynamics) that operational data alone does not specify. The categorical formalization is heaviest — it requires explicit category-theoretic machinery that the others do not.

For the purposes of this paper, we adopt the **algebraic formalization** as the principal framework. It is well-developed in the operator-algebra and quantum-information literatures; it captures OI's framework and string-theoretic emergence directly; and it provides sufficient structure to define universality classes precisely without requiring the heavy categorical machinery.

We will return to operational and categorical perspectives where they illuminate features the algebraic formalization does not capture, but the principal definitions and arguments use the algebra-channel structure of §8.3.

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9. Universality classes of embedded observers

9.1 Formal definitions

We adopt the algebraic formalization of §8.4. The following definitions make precise the universality-class structure used throughout the rest of the paper.

Definition 9.1 (Embedded observer). *An embedded observer is a triple* $(\mathcal{H}_V, \mathcal{A}_V, \Phi)$ *where:* - $\mathcal{H}_V \neq \{0\}$ *is a separable Hilbert space;* - $\mathcal{A}_V \subseteq \mathcal{B}(\mathcal{H}_V)$ *is a unital von Neumann algebra;* - $\Phi : \mathcal{T}_1(\mathcal{H}_V) \to \mathcal{T}_1(\mathcal{H}_V)$ *is a Schrödinger CPTP channel whose normal unital CP adjoint satisfies* $\Phi^*(\mathcal{A}_V) \subseteq \mathcal{A}_V$.

For a specified family of channels, replace the last condition by invariance under every adjoint in the family. This does not assume that the family is a semigroup.

Definition 9.2 (Substratum-with-observer system). *A substratum-with-observer system is a quintuple* $(\mathcal{H}_V, \mathcal{A}_V, \Phi, \mathcal{H}_H, U)$ *where* $(\mathcal{H}_V, \mathcal{A}_V, \Phi)$ *is an embedded observer,* $\mathcal{H}_H$ *is a separable Hilbert space (the hidden-sector state space), and* U *is a unitary on* $\mathcal{H}_V \otimes \mathcal{H}_H$ *such that*

$$\Phi(\rho) = \mathrm{Tr}_H[U(\rho \otimes \rho_H)U^\dagger]$$

for some hidden-sector reference state ρ_H. *The triple* $(\mathcal{H}_H, \rho_H, U)$ *is a* unitary realization *of* Φ. *When* $\rho_H = |0\rangle\langle 0|$ *is pure, the map* $V_{U,0} : \psi \mapsto U(\psi \otimes |0\rangle)$ *is an isometry* $\mathcal{H}_V \to \mathcal{H}_V \otimes \mathcal{H}_H$ *satisfying* $\Phi(\rho) = \mathrm{Tr}_H[V_{U,0}\, \rho\, V_{U,0}^\dagger]$, *and it is this isometry — not* U *— that is a Stinespring dilation of* Φ.

Remark (why the distinction is load-bearing). A unitary realization carries strictly more data than the channel constrains, in two separate ways, and both matter below. First, when ρ_H is pure the channel sees U only through its action on $\mathcal{H}_V \otimes \mathbb{C}|0\rangle$; the action of U on the rest of $\mathcal{H}_V \otimes \mathcal{H}_H$ is entirely free. Second, when ρ_H is mixed the realization remains a realization rather than a Stinespring dilation, and its hidden-sector dimension is not the Kraus rank — it can be strictly *smaller*. The completely depolarizing qubit channel has Kraus rank 4, yet $\rho_H = \mathbb{1}/2$ with $U = \mathrm{SWAP}$ realizes it on a two-dimensional hidden sector. This is not an edge case for OI: §10.4's construction uses the uniform mixed reference state, so OI's own $(\mathcal{H}_H, \rho_H, U_\varphi)$ is a unitary realization and not a minimal Stinespring dilation, and the uniqueness statements below must not be read as applying to it directly.

Definition 9.3 (Observational equivalence). *Two embedded observers* $(\mathcal{H}_V^{(1)}, \mathcal{A}_V^{(1)}, \Phi^{(1)})$ *and* $(\mathcal{H}_V^{(2)}, \mathcal{A}_V^{(2)}, \Phi^{(2)})$ *are observationally equivalent, written* $(\mathcal{H}_V^{(1)}, \mathcal{A}_V^{(1)}, \Phi^{(1)}) \sim_{\mathrm{obs}} (\mathcal{H}_V^{(2)}, \mathcal{A}_V^{(2)}, \Phi^{(2)})$, *if there exists a normal unital*

-isomorphism $\theta : \mathcal{A}_V^{(1)} \to \mathcal{A}_V^{(2)}$ such that*

$$\theta \circ \Phi^{(1)*}|_{\mathcal{A}_V^{(1)}} = \Phi^{(2)*}|_{\mathcal{A}_V^{(2)}} \circ \theta.$$

Definition 9.4 (Universality class). *A universality class is an equivalence class under Definition 9.3, represented by the isomorphism class of* $(\mathcal{A}_V, \Phi^*|_{\mathcal{A}_V})$*. Two substratum-with-observer systems are in the same class when these restricted pairs are equivalent, irrespective of their unitary realizations. For proper* $\mathcal{A}_V$*, this class need not determine the full ambient channel or its Hilbert-space realization.*

The definitions are deliberately framework-neutral. They invoke standard machinery from operator-algebra theory (von Neumann algebras, CPTP channels, *-isomorphisms) without requiring framework-specific structural commitments. This is essential for the universality-class framework's purpose: characterizing equivalence at the level of observation, independent of the specific substratum machinery producing observable physics.

9.2 Relation to OI's gauge group: a proposition

The established fixed-interface channel invariances give the following restricted observational equivalence. Population-only preservation and structural alphabet/graph comparisons do not alone imply an algebra/channel isomorphism.

Proposition 9.5 (Established channel invariances induce observational equivalence). *Fix a finite nonzero visible Hilbert space* $\mathcal{H}_V$*, a chosen observable algebra* $\mathcal{A}_V$ *invariant under every adjoint in the channel family, and the family induced by a finite permutation realization with uniform hidden preparation. Let a second realization be obtained by a hidden-factor permutation applied identically in every visible fibre, or by an independently decoupled normalized finite ancillary enlargement as in [Substratum Theorem 24(i),(iii)]. The resulting embedded observers are observationally equivalent in Definition 9.3, with the identity on* $\mathcal{A}_V$ *intertwining every restricted adjoint channel.*

Proof. Theorem 24(i),(iii) gives equality of the full visible channels at every time for the stated transformations and preparations. Their adjoints are therefore equal on $\mathcal{B}(\mathcal{H}_V)$. The assumed invariant algebra is fixed, so its identity isomorphism intertwines their restrictions. Population equalities alone are not used to infer this isomorphism. □

The converse claim that every observationally equivalent OI-class substratum is related by the four proposed transformations is **not established**. On the fixed-interface coherent-channel reading it is refuted for the restricted moves by the six-state Pair C certificate [Substratum §4; ST3/ST4]; on the population-only reading the larger fibre-dependent hidden relabellings preserve rooted counts, while the all-partitions reading remains open. Stinespring uniqueness for one channel does not provide semigroup-wide uniqueness of the generating dynamics.

Corollary 9.6 (Universality class contains the established channel-preserving realizations). *The universality class of a chosen embedded-observer triple contains every realization obtained by a finite sequence of the moves in Proposition 9.5, with the visible algebra and preparation/readout carrier held fixed and every stage satisfying the relevant preparation and decoupling hypotheses. It may also contain other realizations. No completeness of these moves or classification across all observer partitions is asserted.*

Each move preserves the chosen restricted pair by Proposition 9.5, so a finite sequence does as well. This inclusion does not classify the full substratum observational gauge structure. Alphabet/graph comparisons require their own hypotheses, and fibre-dependent population preservation need not preserve coherent channels.

9.3 Many-to-one structure of dilations

Many unitary realizations can produce the same full ambient channel. The following uniqueness results assume equality of that full channel on $\mathcal{T}_1(\mathcal{H}_V)$, equivalently equality of its adjoint on all $\mathcal{B}(\mathcal{H}_V)$. Equivalence on a proper invariant observable algebra is weaker and does not supply this hypothesis. We separate the finite-dimensional kernel result from the general separable-space theorem and distinguish Stinespring isometries from full unitaries.

Proposition 9.7a (Minimal Kraus uniqueness). *Let $\mathcal{H}_V$ be finite-dimensional and let*

$$\Phi(\rho) = \sum_{i=1}^{r} A_i \rho A_i^\dagger = \sum_{j=1}^{s} B_j \rho B_j^\dagger$$

with $\{A_i\}_{i\le r}$ and $\{B_j\}_{j\le s}$ each linearly independent. Then $r = s$ and there is a unique unitary $W \in \mathrm{U}(r)$ with

$$B_j = \sum_i W_{ji} A_i.$$

Proof. Vectorize: the map $\rho \mapsto \sum_i A_i \rho A_i^\dagger$ determines, and is determined by, the positive semidefinite operator $M = \sum_i |a_i\rangle\langle a_i|$ where $|a_i\rangle = \mathrm{vec}(A_i)$. Linear independence of the family says the $|a_i\rangle$ are linearly independent, so $r = \operatorname{rank} M = s$ and both families are bases of $\operatorname{ran} M$. Writing X and Y for the matrices with those columns, $XX^\dagger = M = YY^\dagger$ with X, Y of full column rank forces $Y = XW^{\mathsf{T}}$ for a unique unitary W. □

Linear independence is exactly minimality, and it is not removable: without it the family size is not even determined. For any 3×2 isometry M the family $C_k = \sum_i M_{ki} A_i$ realizes the same channel with three members rather than two, and no unitary relates families of different sizes.

Proposition 9.7b (Finite minimal pure-reference dilation uniqueness). *Let the visible and both hidden Hilbert spaces be finite-dimensional, with $\mathcal{H}_V \neq \{0\}$.*

Let $(\mathcal{H}_H^{(1)}, |0_1\rangle\langle 0_1|, U^{(1)})$ *and* $(\mathcal{H}_H^{(2)}, |0_2\rangle\langle 0_2|, U^{(2)})$ *be pure-reference unitary realizations of the same channel* Φ*, and let* $V_a\psi = U^{(a)}(\psi \otimes |0_a\rangle)$ *be the associated Stinespring isometries. If both are minimal —* $\overline{\mathrm{span}}\{(A \otimes \mathbb{1})V_a\psi : A \in \mathcal{B}(\mathcal{H}_V),\ \psi \in \mathcal{H}_V\} = \mathcal{H}_V \otimes \mathcal{H}_H^{(a)}$ *— then* $\dim \mathcal{H}_H^{(1)} = \dim \mathcal{H}_H^{(2)}$ *and there is a unique unitary* $W : \mathcal{H}_H^{(1)} \to \mathcal{H}_H^{(2)}$ *with*

$$V_2 = (\mathbb{1}_V \otimes W)\, V_1, \qquad \text{equivalently} \qquad U^{(2)}(\psi \otimes |0_2\rangle) = (\mathbb{1}_V \otimes W)\, U^{(1)}(\psi \otimes |0_1\rangle)\ \ \forall \psi.$$

No equality of the full unitaries away from the reference subspaces follows.

Proof. In finite dimension with nonzero visible space, cyclic minimality is equivalent to linear independence of the Kraus family $A_i = (\mathbb{1}_V \otimes \langle i|)V_a$. Proposition 9.7a therefore gives equal hidden dimensions and the unique unitary with the stated orientation. This is the scope of the finite-dimensional Lean theorem `StinespringUniqueness.proposition_9_7b`. □

Proposition 9.7c (Separable minimal-isometry uniqueness). *Let* $\mathcal{H}_V \neq \{0\}$ *and* $\mathcal{K}_1, \mathcal{K}_2$ *be separable Hilbert spaces. Suppose* $V_a : \mathcal{H}_V \to \mathcal{H}_V \otimes \mathcal{K}_a$ *are cyclically minimal Stinespring isometries for the same full ambient channel. Then there is a unique unitary* $W : \mathcal{K}_1 \to \mathcal{K}_2$ *such that* $V_2 = (\mathbb{1}_V \otimes W)V_1$*. Consequently the conclusions of Proposition 9.7b about the prepared isometries hold without finite-dimensionality.*

Proof. On the dense cyclic span define

$$T\left(\sum_j (A_j \otimes \mathbb{1})V_1\psi_j\right) = \sum_j (A_j \otimes \mathbb{1})V_2\psi_j.$$

The squared norm on either side is $\sum_{i,j}\langle \psi_i, \Phi^*(A_i^\dagger A_j)\psi_j\rangle$. Equality of the complete adjoint channels makes T well-defined and isometric. Minimality on both sides extends it to a surjective isometry, hence a unitary. Its definition gives $T(A \otimes \mathbb{1}) = (A \otimes \mathbb{1})T$ for every $A \in \mathcal{B}(\mathcal{H}_V)$. Fix a unit vector $\xi \in \mathcal{H}_V$ and set $W\eta = (\langle \xi| \otimes \mathbb{1})T(\xi \otimes \eta)$. Intertwining the rank-one operators $|\psi\rangle\langle\xi|$ gives $T(\psi \otimes \eta) = \psi \otimes W\eta$. Unitarity of T implies unitarity of W. Any other such W agrees on the dense cyclic span, so it is equal to this one. This is the standard minimal Stinespring uniqueness argument [Stinespring1955]; the general separable extension is a written proof here, beyond the finite-dimensional kernel result. □

The full-unitary conclusion is limited to one prepared channel. With $\mathcal{H}_V = \mathcal{H}_H = \mathbb{C}^2$, $U^{(1)} = \mathrm{CNOT}$ and $U^{(2)} = \mathrm{CNOT}(\mathbb{1} \otimes Z)$, both with hidden reference $|0\rangle$, produce the same minimal dephasing channel. The unique isometry intertwiner is $W = \mathbb{1}$, but the unitaries differ off the prepared subspace. This freedom does not automatically preserve the channel family obtained by repeatedly applying U: $(U^{(1)})^2 = \mathbb{1}$, whereas $(U^{(2)})^2 = Z \otimes \mathbb{1}$, so their time-two coherent channels differ.

Remark (the conjugation form). If the unique W additionally satisfies $W|0_1\rangle = |0_2\rangle$, the isometry relation can be written as conjugation on the reference subspace. This is an extra hypothesis, not a freely selectable normalization. For example, $U^{(1)} = \mathrm{CNOT}$ and $U^{(2)} = (\mathbb{1} \otimes X)\mathrm{CNOT}$ with both reference vectors $|0\rangle$ give the same minimal channel but have unique intertwiner $W = X$, which does not preserve $|0\rangle$. Even when the extra hypothesis holds, no full-operator conjugation follows.

Remark (limits of semigroup-transfer uniqueness). Equal channel families need not imply hidden-unitary conjugation of the generating dynamics, as the exact Pair C construction in [Substratum §4] shows (ST3/ST4). Proposition 24.2 instead gives block-word trace rigidity under the full word-trace and block-spanning premises. Proposition 9.7b is a single-channel minimal-isometry uniqueness result, not full generated-unitary uniqueness for arbitrary dilations.

These propositions identify the many-to-one structure for a fixed complete channel: minimal pure-reference isometries are related by a hidden unitary, non-minimal realizations permit enlargement, and a prepared isometry need not fix its full-unitary extension. General hidden unitaries need not be finite hidden permutations, and general channel dilations need not lie in OI's finite-bijection class. These results therefore do not classify $\mathcal{G}_{\mathrm{sub}}$ or its whole-time-family action.

Corollary 9.8 (Realization freedom for a fixed prepared channel). *A fixed full ambient channel on a finite nonzero visible space need not determine a unique unitary realization. Among minimal finite pure-reference realizations it fixes the Stinespring isometry up to the unique hidden unitary of Proposition 9.7b. It does not fix a full-unitary extension off the prepared subspace, and non-minimal realizations allow further enlargement. The universality-class datum of Definition 9.4 is $(\mathcal{A}_V, \Phi^*|_{\mathcal{A}_V})$, which can contain still less information when $\mathcal{A}_V$ is proper.*

For general separable spaces the positive isometry-uniqueness statement follows from Proposition 9.7c. The finite-dimensional kernel result and the explicit finite countercontrols have separate evidential roles; the general proof does not extend the kernel theorem's formal scope.

Scope (relative to the fixed-reference observational interface). Off-reference unitary freedom is invisible to one channel prepared with the fixed reference state. Independently preparing hidden inputs, or reusing the hidden system across several times, can reveal it. Calling this freedom gauge is therefore relative to that single prepared-channel carrier, not a completeness claim about OI dynamics or multi-time records.

A structural-realist interpretation can treat the isomorphism class of the chosen restricted pair as observational content. Whether this exhausts physically meaningful structure depends on the carrier and the available preparations. It does not follow that every microscopic difference is a $\mathcal{G}_{\mathrm{sub}}$ gauge move.

9.4 Relation to OI's gauge group, refined

With the formal definitions in place, the relationship between universality classes and OI's gauge group can be stated more precisely than in §9.2.

OI's proposed $\mathcal{G}_{\text{sub}}$ is intended to organize observationally redundant substratum descriptions. Proposition 9.5 proves inclusion only for its two specified coherent-channel transformations under their preparation and invariance hypotheses. Theorem 24 does not classify every OI realization of the same restricted algebra/channel pair. Definition 9.4, in contrast, defines an equivalence directly on that pair across frameworks; whether non-OI realizations occupy the same class requires the corresponding algebra map and intertwining.

The proposed substratum quotient and the algebra/channel quotient can be compared only after their carriers are fixed. The established transformations act within a class of Definition 9.4, but equality of the two full equivalence relations within A1–A6 is not established. A claim about physical ontology requires more than this restricted inclusion.

9.5 Structure of universality classes

Universality classes have internal structure beyond bare equivalence. Three structural features:

Sub-classes by structural features. Equivalent restricted pairs share the properties actually encoded in the chosen invariant algebra and its adjoint dynamics. They need not share ambient-channel unitarity or every multi-time marginal, which require further data. Realizations in one class may satisfy different microscopic structural conditions; these conditions can define sub-classes without changing the chosen observable carrier.

Sub-classes by gauge content. A gauge comparison requires the observable algebra, channel and the symmetry action being compared. The Standard Model factors are calculated in the selected OI branch under its carrier and physical hypotheses. A string construction can reproduce the same factors through different premises; matching them does not by itself give an isomorphism of the full embedded-observer triples or a complete map of substratum gauge groups.

Refinement by substratum-symmetry. A coarse comparison that omits gauge charges or couplings may group different gauge targets together. A carrier containing their measured values cannot: equivalence must preserve them. SM-reproducing model families can therefore contain multiple classes, and claims at a coarser level must state which data have been omitted.

This permits a hierarchy of observational carriers: partial-trace features alone, specified gauge and matter data, and a fuller dynamics including quantitative predictions. Refining a carrier can split an equivalence class. Varying a measured Cabibbo angle or mass within a model family is not a change of representative of one class whose carrier already includes that quantity.

9.6 Observer-admitting and observer-lacking substrata

The framework's universality classes are populated by *observer-admitting* substrata — pairs (S, φ) that admit a partition $V \subset S$ satisfying the framework's structural conditions C1–C4 (coupling, record persistence, sufficient capacity, history readback) [Main §3]. This restriction defines the scope of the framework's discourse and is structurally distinct from the absence of an observer in any specific substratum. We make the distinction explicit here because it bears on the relationship between OI and other unification programs in §12.

Definition 9.9 (Observer-admitting substratum). *A substratum (S, φ) is observer-admitting if there exists a partition V such that the triple (S, φ, V) satisfies C1–C4. A substratum is observer-lacking if no such partition exists.*

Three structural conditions on φ characterize observer-admission:

- *Coupled dynamics:* φ does not decompose as $\varphi_V \times \varphi_H$ for any candidate partition. Substrata where visible and hidden sector dynamics factorize cannot satisfy C1.
- *Non-equilibrium phase:* φ admits phases where local mixing time is much shorter than overall recurrence time. Under the effective-mixing hypothesis and for sufficiently fast-mixing equilibria ($\tau_{\text{mix}} \ll \tau_S$), substrata in perpetual equilibrium cannot satisfy C2 in the coarse-grained description generated by iterating the one-step effective channel [Main §4.6].
- *Sufficient hidden capacity:* potential hidden sectors have $|\mathcal{C}_H|$ large enough to store interaction history without overflow. Substrata with insufficient hidden-sector capacity cannot satisfy C3.

These are structural features of the substratum's bijection φ itself, prior to any observer being instantiated. An observer-admitting substratum has the structural capacity for partial-trace observation; an observer-lacking substratum does not.

Proposition 9.10 (Observer-admitting and observer-lacking substrata are structurally distinct). *Observer-admission is determined by structural features of φ, not by whether an observer is "present" or "absent" in any specific substratum. The classes are mutually exclusive partitions of the space of substrata.*

The proposition is a clarification rather than a deep result: it states that the observer-admission criterion operates at the level of the bijection structure, separating substrata into two structurally-distinct classes. An observer-admitting substratum has the structural features that allow observation to occur; an observer-lacking substratum does not. The classes are not "the same substratum with observation added or removed" — they are different substrata characterized by different structural features of φ.

Corollary 9.11 (Universality classes are populated by observer-admitting substrata only). *The universality classes of Definition 9.4 are equivalence classes*

within the observer-admitting subset. Observer-lacking substrata are outside the framework's universality-class structure.

This corollary clarifies the framework's scope. Universality classes characterize substratum-with-observer systems by their algebra-channel structure under partial-trace observation; observer-lacking substrata do not produce algebra-channel pairs through partial trace and are therefore not in the universality-class structure. They are mathematical objects (finite sets with bijections) with dynamics, but not with extractable observational features.

The "physics" of observer-lacking substrata. Observer-lacking substrata have φ-dynamics — they evolve, they have cycle structure, they have all the formal features of permutations on finite sets. But they do not produce *physics* in the framework's sense: no quantum mechanics, no observable record, no measurement outcomes. The framework's foundational claim — that physics is the structural features of partial-trace observation — implies that physics is a feature of observer-admitting substrata, not of all substrata. This is consistent with the main paper's "observation occurs" axiom: the axiom is taken to be a structural feature of our universe (which is observer-admitting) rather than a contingent fact about whether someone happens to be looking.

Anthropic implication. The observer-admission distinction has structural-anthropic content. Most arbitrary (S, φ) pairs do not admit observers — random permutations on finite sets typically lack the coupled-dynamics, non-equilibrium-phase, and high-capacity features required for C1–C4. Observer-admitting substrata are a structurally-characterized subset. Our universe is in this subset, structurally; we are observers in an observer-admitting substratum. This is structural anthropic selection rather than contingent anthropic selection: the criterion for "physical universe" is observer-admission, with observer-lacking substrata being abstract mathematical objects rather than candidate physical universes.

The structural-anthropic reading is consistent with but more precise than the standard anthropic position. Standard anthropics says we observe a universe with parameters compatible with observers because we couldn't observe a universe without those parameters. The observer-admission reading says: substrata with observers and substrata without observers are structurally distinct kinds of objects; observer-admitting substrata are the candidate space for "physical universe"; we are in one of them.

9.7 Worked examples

We illustrate the universality-class definition with three examples.

Example 1: A selected OI realization. Fix a finite $d = 3$ cubic wave-rule model, H-link with $m = 1$ for its physical carrier, and a specified visible algebra and preparation/readout map. Its observation map supplies an algebra/channel pair. Proposition 9.5 places the established hidden-only and independent finite

ancillary transformations in the same observational class. It does not put every alphabet change, graph comparison or population-preserving relabelling in that class. Whether other OI or non-OI realizations produce the same pair is a separate classification problem. Observer admission is assessed for the specified triple, as in §9.6.

Example 2: String-theoretic class. A string-theoretic substratum (e.g., type IIA on a specific Calabi-Yau threefold with specific fluxes) produces a low-energy effective field theory. The dilation is the full string-theoretic structure (worldsheet plus target space, or matrix-model structure plus large-N limit); the visible-sector algebra is the four-dimensional effective field theory operator algebra. The universality class is the class of substrata producing the same algebra-channel pair through string-theoretic emergence. Observer-admission is a non-trivial constraint: not all string compactifications are observer-admitting, as developed in §7.9.

Example 3: Cross-class candidate. A class candidate is the union: substrata that produce the SM gauge group + three generations + SM-like Yukawa structure through any mechanism (OI's cubic-lattice derivation, string-theoretic compactification, possibly LQG-style structures, causal-set structures, or asymptotic-safety–derived structures with appropriate matter-emergence). This class is characterized by the observable target (the SM at low energies) rather than by the substratum mechanism. All members are observer-admitting (the SM-reproducing constraint forces observer-admission, since reproducing the SM requires producing the observed cosmological structure within which we are observers). Part I (§§3-7) establishes that, at the level of partial-trace observational features, OI and string theory are members of this candidate cross-class; §13.1.4 extends the per-condition analysis to LQG, causal sets, and asymptotic safety, finding that all three fail OI's A1–A6 (so are outside OI's specific class) but may share the partial-trace observational features class with OI and string theory. The question is whether the cross-class is well-defined and what additional features characterize it.

9.8 Open: does the OI-string cross-class exist?

The candidate cross-class of Example 3 is well-defined as a set: substrata producing the SM at low energies. The question is whether it is a *universality class* in the sense defined — whether all members produce equivalent algebra-channel pairs.

Part III §14 establishes partial transport: Born rule and channel-level unitarity transport directly; the memory (non-Markovian) mechanism transports, while P-indivisibility is not established cross-framework; commutant gauge-invariance pattern transports but not the specific gauge group. Specific Yukawa couplings, CP-violating phases, and other detailed predictions have not been demonstrated to transport.

If only the structural features at §14.1-14.3 transport, the OI-string cross-class is

universal at the level of these features but not at the level of detailed predictions. The class might be subdivided further: the partial-trace-features class is broader; the SM-reproducing class is narrower; the OI-Cabibbo-and-Koide-predictions class is narrower still.

Establishing definitively whether OI's specific predictions (Cabibbo, Koide, etc.) transport to the string-theoretic class would require either (a) finding a string-theoretic compactification reproducing OI's predictions, (b) showing that any SM-reproducing compactification automatically produces these predictions, or (c) showing that no string-theoretic compactification can reproduce these predictions exactly. Each of (a)-(c) is a substantial research question.

The provisional conclusion: the OI-string cross-class is well-defined at the level of partial-trace observational features (per §14) and includes specific gauge-group-and-content sub-classes, but the full structure of the cross-class — including whether OI's specific predictions are universal to the SM-reproducing sub-class or specific to OI's representative — is open.

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10. OI's framework as a worked example

This section uses OI's framework to illustrate how universality classes interact with substratum-level structure. The aim is to show that universality classes are not abstract mathematical curiosities but capture the structural content of OI's reconstruction theorem in a more general form.

10.1 OI's reconstruction theorem in universality-class language

Theorem 23 [Substratum §3] currently states conditional local lattice/gauge outputs on a selected cubic branch, under the explicit nearest-neighbor, H-link, representation, amplitude-scale and matter assumptions. It does not establish that E1–E7/A1–A6 alone select that branch or that its microscopic realizations are all related by a complete observational gauge group.

Theorem 23 currently gives conditional local lattice/gauge outputs for the selected cubic branch and does not show uniqueness modulo all admissible gauge transformations. H-Bell independently conditions the existence of a common Bell-composite realization; its uniqueness remains open.

10.2 What OI does and does not establish at the universality-class level

OI's selected cubic-branch reconstruction supplies conditional structural outputs, not uniqueness of every realization from E1–E7/A1–A6. It also does not establish Bell-inclusive uniqueness or a universal all-partitions gauge quotient; the comparisons with other universality classes remain separately scoped.

The universality-class language locates results relative to a specified observable structure. Theorem 23 gives conditional outputs for a selected realization; it does not identify a unique class from E1–E7/A1–A6. Distinct classes can agree on a restricted set of tested Standard Model quantities while differing on fuller channel or preparation/readout data. Such agreement is not equivalence under Definition 9.3 without the required map.

10.3 The structural-realism implication

The structural-realist proposal concerns branch-relative outputs and structures preserved by specified observational maps. Theorem 23 does not establish a uniquely reconstructed local residue modulo a complete substratum gauge group. Local-branch selection, all-partitions equivalence and a common Bell-compatible completion are separate open questions.

A structural-realist interpretation may treat the chosen algebra/channel structure as the content shared by equivalent descriptions under Definition 9.3. The physical warrant of the selected representative and any additional microscopic commitments require their own arguments. This interpretation does not itself prove a complete gauge classification or a unique physical substratum.

The Cabibbo, Koide and related calculations are conditional predictions of the selected OI branch. If those quantities and their comparison protocols are encoded in a completed carrier, every equivalent realization must reproduce them. A different model can share the SM gauge target yet differ on those quantities, in which case it belongs to a different observational class on that richer carrier.

10.4 Constructing OI's algebra-channel pair

The universality-class formalism requires a specified algebra, a well-defined channel and an invariance check. At a finite cutoff OI supplies an exact trace-out channel from a chosen permutation and hidden preparation. At the physical EFT scale, the following is a conditional candidate and a list of matching obligations; it is not yet a constructed continuum algebra/channel pair.

The Hilbert space. In the selected cubic branch, the finite visible configuration space gives $\mathcal{H}_{V,\mathrm{cut}} = \ell^2(\mathcal{C}_V)$. A physical SM Fock-space interpretation additionally needs a limiting construction and the stated matter/readout hypotheses, including H-spin' and H-χ'. The finite trace-out formula alone does not supply that identification. We denote a candidate EFT space by $\mathcal{H}_V^{\mathrm{OI}}$ only after those premises have been specified.

The algebra. A candidate local EFT net must be built from well-defined bounded localized gauge-invariant observables on that space. The schematic expression

$$\mathcal{A}_V^{\mathrm{OI}} = \langle \text{bounded localized gauge-invariant observables} \rangle''$$

means the von Neumann closure after their representation and relations have been supplied. Pointlike fields and their unbounded products cannot simply be inserted as bounded generators. Relevant smeared operators, their bounded functions and their domains or affiliations must be constructed. Candidate operator sectors include:

- *Gauge-field invariants.* The Yang-Mills curvature combinations $\operatorname{tr} F^a_{\mu\nu} F^{a,\mu\nu}$ for each factor of $SU(3) \times SU(2) \times U(1)$, plus topological terms $\operatorname{tr} F\tilde{F}$.
- *Fermion bilinears.* $\bar{\psi}_i \psi_j$, $\bar{\psi}_i \gamma^\mu \psi_j$, $\bar{\psi}_i \gamma^\mu \gamma^5 \psi_j$ for the SM fermion content (six quarks, six leptons, three generations), with appropriate gauge-index contractions.
- *Higgs invariants.* $\phi^\dagger \phi$, $|D_\mu \phi|^2$, $(\phi^\dagger \phi)^2$, and the Yukawa contractions $\bar{\psi}_L \phi \psi_R$ structurally fixed by the cubic-group representation theory of [SM §7].
- *Higher-dimensional operators.* A comparison must specify their Wilson coefficients and the exact assumptions behind any proposed relations, including the Cabibbo and Koide calculations of [SM §7]. A list of scalar outputs is not itself a presentation of the complete observable algebra or a derivation of every coefficient.

Gauge invariance requires the localized observables to commute with the specified gauge action, with the appropriate physical constraints imposed. The group action, localization, representation and closure are part of the construction; naming the SM factors alone does not determine them.

The channel. At a finite cutoff the chosen permutation gives the exact Schrödinger channel

$$\Phi_{\mathrm{cut}}(\rho) = \mathrm{Tr}_H \left[U_\varphi (\rho \otimes \rho_{H,\mathrm{cut}}) U_\varphi^\dagger \right].$$

The normalized uniform hidden state is available in this finite setting. A normalized scalar multiple of the identity on a hidden Hilbert space requires finite dimension. An EFT channel Φ^{OI} requires an independently specified state and a controlled limit or effective construction. Its adjoint must preserve the chosen EFT algebra before Definition 9.3 applies. A mixed hidden reference gives a unitary realization, not a minimal pure-reference Stinespring dilation; purifying it changes the hidden space. The following physical features are matching targets, not consequences of the finite trace-out identity alone:

- *SM RG flow* from the cubic-lattice scale ($1/\alpha_0 = 23.25$ structurally, [SM §6.1]) down to the electroweak scale ($1/\alpha_1 = 59.00$, $1/\alpha_2 = 29.57$, $1/\alpha_3 = 8.47$ at M_Z, [SM §6.3]).
- *Non-perturbative structural features.* Instanton-sum closure consistent with T-invariance and detailed balance.
- *Anomaly structure.* Check the anomaly constraints for the selected matter representations and charge assignment ([SM §4.6]).

- *Higher-dimensional operator coefficients.* Supply the proposed Yukawa relations and their stated premises, then determine which are actually encoded in the constructed dynamics ([SM §7]).

The pair as universality-class invariant. Once a channel and an invariant algebra have been constructed, the isomorphism class of $(\mathcal{A}_V^{\mathrm{OI}}, \Phi^{\mathrm{OI}*}|_{\mathcal{A}_V^{\mathrm{OI}}})$ is the invariant of Definition 9.4. At a fixed finite interface Proposition 9.5 proves preservation for its two named transformations. It proves no such assertion for every proposed $g \in \mathcal{G}_{\mathrm{sub}}$, nor does it automatically transport a finite-cutoff intertwiner to an EFT limit.

What the pair captures. The pair captures exactly the observables represented in its algebra and their specified adjoint dynamics. Measured masses, couplings and Wilson coefficients are invariant under equivalence only insofar as the carrier includes them. Changing a bijection in a way that changes such data is not automatically gauge. Hidden microscopic information can remain unidentifiable, but its empirical irrelevance must be shown for the chosen carrier.

A relativistic EFT comparison may use nets of local von Neumann algebras with isotony and microcausality [BrunettiFredenhagen2024]. Type-III local factors [Yngvason2005] require an appropriate infinite-system or continuum representation; a finite matrix algebra is not type III. These properties must be established for the proposed limit rather than inherited by naming an AQFT target. The finite trace-out construction and standard Stinespring theory [Stinespring1955] remain valid within their stated settings.

10.5 Status as Step 1 of §13.1.1's recipe

Section 10.4 specifies an exact finite-cutoff channel and a conditional EFT target. Step 1 of §13.1.1 remains open at the full EFT algebra/channel level. It requires the physical/limit construction, a specified algebra and state, and invariance of that algebra under the adjoint dynamics. The comparison then requires:

- *Step 2.* Construct a specific string compactification's pair $(\mathcal{H}_V^{\mathrm{string}}, \mathcal{A}_V^{\mathrm{string}}, \Phi^{\mathrm{string}})$ at the same EFT scale. Natural candidates are MSSM-like F-theory or heterotic compactifications producing the exact observed gauge group and chiral content. Particle content and gauge factors specify targets; operator relations, representation, closure, states and actual channel dynamics must also be constructed, including any proposed thresholds, instanton effects or scale dependence.
- *Step 3.* Construct and test a normal unital *-isomorphism of the specified observable algebras. Agreement on gauge factors or finitely many scalar predictions is a necessary consistency check for a carrier containing those data, not a sufficient algebra-isomorphism test.
- *Step 4.* Check $\theta \circ \Phi^{\mathrm{OI}*}|_{\mathcal{A}_V^{\mathrm{OI}}} = \Phi^{\mathrm{string}*}|_{\mathcal{A}_V^{\mathrm{string}}} \circ \theta$, for the same isomorphism θ and every channel in the specified family. Joint intervention statistics require their own process carrier.

Completing Step 1 and carrying out Steps 2–4 require a specified physical EFT on each side. The finite trace-out formula makes one part of OI's proposal concrete, but does not reduce the remaining task to comparing already constructed continuum pairs.

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11. The OI-string universality class

Part III §14 establishes that OI and string theory share structural features at the partial-trace observation level. This section examines what that establishes about cross-class membership and what additional work is needed.

11.1 What §14's findings establish

The §14 four-feature audit shows that the Born rule and channel-level unitarity transport directly between OI and string-theoretic emergence mechanisms, and that the memory (non-Markovian) mechanism transports while P-indivisibility is not established cross-framework (§14.3). The commutant gauge-invariance *pattern* transports but the specific gauge group does not.

In the terminology of §9, the audit identifies shared features of model families, not one exact universality class for all OI or string constructions. Definition 9.3 requires a chosen invariant algebra, the corresponding adjoint dynamics and an actual intertwining isomorphism. A feature comparison alone does not supply those data.

Under their respective physical assumptions, selected OI and string constructions can reproduce the same Standard-Model local gauge factors through different mechanisms. This places them in a common gauge-target family. Their matter assignments, couplings and dynamics still need comparison, and a matching target does not establish Definition 9.3 equivalence or a common global gauge-group action.

11.2 What additional work is needed

Section 10.4 gives a finite-cutoff construction and outlines the EFT target. Completing the OI algebra/channel pair at the chosen physical scale is still Step 1; the following comparison questions also remain open:

Question 1: Do OI's specific predictions transport? OI predicts the Cabibbo angle as $1/(\pi\sqrt{2})$ from cubic-group representation theory. Does any SM-reproducing string compactification reproduce this exact value through its compactification mechanism, or are string-theoretic predictions for the Cabibbo angle compactification-specific (and presumably not exactly $1/(\pi\sqrt{2})$ in general)? If the values differ, the models cannot be equivalent on a carrier containing those values. They may still share a coarser structural target.

Question 2: Do the algebras agree? OI's emergent QFT has a specific algebra of observables determined by its cubic-lattice structure (proposed conditionally in §10.4). String-theoretic compactifications produce algebras determined by the compactification specifics. Are these algebras *-isomorphic for SM-reproducing constructions, or merely structurally similar (sharing partial-trace features but differing on specific algebraic content)?

Question 3: Do the channels agree? Beyond the algebra structure, the channel Φ encodes the dynamics on visible-sector states. Does OI's Φ^{OI} (after completing the construction required in §10.4) agree with the string-theoretic Φ^{string} on identical SM gauge group plus identical particle content, or does it differ on detailed dynamical features (e.g., the specific RG flow, specific anomaly content, specific instanton effects)?

These questions require completing both pairs, constructing an algebra map and checking the adjoint intertwining. Part III's partial-trace comparison does not complete that task. Agreement of selected scalar outputs remains useful evidence, with its carrier stated explicitly.

11.3 Provisional characterization

Pending the technical work of §11.2, the OI-string relationship at the universality-class level is provisionally:

- **Shared restriction-interface features:** §14 compares quantum-probability representation and dilation architecture, with memory requiring additional dynamical structure.
- **Common gauge target:** selected constructions reproduce the Standard-Model local factors under their respective physical assumptions.
- **Equivalent restricted algebra/channel pairs:** open; requires the map and intertwining of Definition 9.3 at a specified scale.
- **Transport of quantitative predictions:** open until the relevant data and protocols are included in that comparison; once equivalence is established on such a carrier, those predictions are shared.

The provisional characterization is shared interface architecture and a common low-energy target, with exact observational equivalence unresolved. The common-core realism discussed in [DawidFranzmann2025] supplies a philosophical comparison for invariant structure across descriptions, rather than an algebra/channel isomorphism between these particular models (§15.3).

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12. Implications

12.1 Refined structural realism

The universality-class framework makes the realist commitment relative to a specified observable carrier. The invariant is the isomorphism class of

$(\mathcal{A}_V, \Phi^*|_{\mathcal{A}_V})$. Shared partial-trace architecture, a common gauge target and membership in a microscopic A1–A6 family are distinct comparisons; none by itself identifies that class or a completed $\mathcal{G}_{\text{sub}}$ quotient.

At a broad level, quantum representations share the probability dictionary and dilation machinery assumed in their construction; memory requires the additional conditions discussed in §14.3. At a gauge-target level, models can share Standard-Model local factors. At a quantitative level, predictions such as the Cabibbo value are conditional outputs of a selected branch and invariants of any completed observational carrier that contains them. They cannot change merely by replacing one equivalent representative with another.

This is an observation-relative structural realism: refining the carrier can distinguish models that agree on a coarser target, while an established isomorphism identifies exactly the content that must remain invariant. It preserves the selected branch's predictions and makes their transport assumptions explicit.

12.2 Methodology of fundamental physics

The universality-class framework suggests a methodological reframing of fundamental physics. Rather than asking "what is the substratum?" — which assumes a unique answer that committing to a specific framework presupposes — we can ask "what universality class are we in?" — which allows for multiple substratum-level descriptions consistent with the same universality class.

This is methodologically more agnostic about substratum specifics but more committed about substratum-class membership. It treats fundamental physics as a research program of identifying the universality class of our universe through observational features, with substratum descriptions being framework-specific tools for representing the class rather than competing claims about ultimate reality.

The methodological move connects to operationalist and empiricist positions in foundations of physics, but with structural-realism preserved at the universality-class level: what's real is the universality class, not just the operational data, and not the specific substratum representative.

12.3 Connection to OI's main paper

OI's main paper starts from observation and proves exact finite-law representation results under its stated setting. The algebra/channel framework packages a chosen quantum representation as an invariant observable algebra and adjoint dynamics. It does not derive that representation, non-Markovian memory or a complete operational quantum theory from the words 'observation occurs' alone.

The roles remain separate: finite-law quantum representation, the C4 memory-bearing sector, recurrence-scale indivisibility and the coherent operational lift have distinct hypotheses and evidential status in [Main]. A1–A6 and the additional local-branch assumptions supply the conditional physical model studied

in [SM] and [GR]. Definition 9.4 then compares whatever observable pairs have actually been constructed.

This broader language permits cross-framework comparison without assuming the same microscopic construction. Agreement on representation architecture or a list of scalar outputs motivates further work; a claim of the same observational class requires the normal algebra map and restricted adjoint intertwining.

12.4 Predictions stratified by hierarchy level

The following tables organize the framework's empirical comparisons by shared target and selected-branch content. These are evidential groupings, not four nested equivalence classes of Definition 9.4. Numerical predictions retain their source assumptions; exact class-level transport applies only after the relevant observables and protocols are included in an established comparison.

Level 1: Quantum representation and restriction-interface features. The first three rows concern the quantum probability/dilation framework and observed memory phenomena; memory is not forced by every partial trace. The final two rows are separately conditioned horizon claims with their gravitational assumptions, rather than consequences of Definition 9.1 alone.

Feature	Empirical status	Reference
Born rule ($p(a) = \mathrm{Tr}(\rho\Pi_a)$)	Confirmed at 10^{-3}–10^{-4}	[Sinha2010, Söllner2012, Kanthak2025]
Channel-level unitarity (Stinespring dilation)	Standard QM, theorem-level	[Stinespring1955, Barandes2026]
Non-Markovian marginal (memory; P-indivisibility not tested by these experiments)	Observed in colloidal, micromechanical, metrology	[Mehl2012, Groeblacher2015, QEC-NonMarkovian2025]
Holographic bound ($S \leq A/4$)	GW250114 confirms the classical area theorem; the entropy bound itself is untested directly	[GW250114]
Bekenstein-Hawking 1/4 coefficient	Conditional on H-slope with the horizon and frame conditions; no direct empirical test (GW250114 constrains the area theorem)	[GR §2, §8.5, GW250114]

The representation and memory comparisons support the relevant mechanisms and are shared with models satisfying the same hypotheses. They do not estab-

lish a unique microscopic realization or one algebra/channel class, and horizon claims retain their additional gravitational premises.

Level 2: Gauge-group-target features. Shared by any framework producing SM gauge structure. OI shares this level with SM-reproducing string compactifications and any other program reaching the SM target.

Feature	Empirical status	Reference
$SU(3) \times SU(2) \times U(1)$ gauge structure	Confirmed (full SM)	[PDG]
Three chiral generations	Confirmed	[PDG]
Anomaly cancellation across generations	Confirmed	[PDG]
Renormalizable Yang-Mills + Higgs sector	Confirmed	[PDG]

Level 2 confirmations are confirmations of the gauge-group-target level but do not distinguish OI from other SM-reproducing frameworks. They establish shared structure without identifying the specific universality class.

Level 3: Selected OI branch and shared effective targets. These are conditional predictions and mechanisms studied in the OI branch, some shared with broader no-GUT or running-vacuum model families. Whether each is encoded in a completed pair from §10.4 requires the physical construction and a specified observational carrier.

Feature	Empirical status	Reference
Cubic-lattice-forced higher-dim operators	Indirect (via specific predictions below)	[SM §3.1]
Type II RVM dynamical dark energy	running-vacuum *form* favoured over plain ΛCDM; $\nu \approx 0$ magnitude consistent with Bertini (2025, itself consistent with zero), in mild tension with DESI DR2's evolving-DE preference	[Bertini2025, DESI_DR2_2025, GR §7]
MOND scale $a_0 = cH/6$	Tully-Fisher relation; ongoing tests	[GR §7.3]
Holographic $A/4$ at cosmological horizon	Structural (theory-level transport; no direct test)	[GR §5]

Feature	Empirical status	Reference
No GUT-mechanism proton decay ($\tau_p \sim 10^{45}$ yr)	Super-K bounds at $\tau_p > 2.4 \times 10^{34}$ yr already exclude minimal SU(5); favors no-GUT class	[SuperK2020_proton, SM §6.7, §8.7]
No GUT-mechanism magnetic monopoles	GUT-scale monopoles undetected cosmologically; consistent with OI without inflation invocation	[MoEDAL2024, SM §6.7]
No GUT-chain cosmic strings	NANOGrav 15-year data disfavors stable cosmic-string interpretation	[NANOGravStrings2023, SM §6.7]
Non-unification of $\alpha_3, \alpha_2, \alpha_1$ below $M_{\rm Pl}$	Pure SM running confirmed; SUSY-GUT scenarios under LHC pressure ($M_{\rm SUSY} \gtrsim 3$ TeV with split-SUSY tuning)	[LHC_SUSY_2025, SM §6.4, §6.7]

A model independently reproducing these predictions can share an effective target with OI without being observationally equivalent. A comparison containing the relevant observables must preserve their values to establish Definition 9.3 equivalence. The no-GUT examples are shared phenomenological targets; the OI explanation uses the selected condensate-stabilizer and induced-coupling branch with the premises in [SM §4.6, §6.7].

Level 4: Quantitative tests of the selected OI branch. These calculations distinguish that branch from model families allowing other values under their own assumptions. If a tested quantity is encoded in an observational carrier, every equivalent representative shares its value; confirmation tests the branch prediction, rather than choosing between representatives of the same class.

Feature	Empirical status	Reference
Cabibbo angle $\sin\theta_C = 1/(\pi\sqrt{2}) \approx 0.2251$	Confirmed at 0.04σ	[SM §7.2]
Koide ratio $Q = 2/3$	Confirmed to better than 0.001%	[SM §7.3]

Feature	Empirical status	Reference
JUNO $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$	Retrodictive match at 0.07σ; exact value gives a pre-registered forward test at JUNO design precision	[Juno]
Higgs structural ratio $A = \sqrt{2/3}$	Consistent with current data	[SM §7.4]
Higgs quartic $\lambda(M_{\rm Pl}) = 0$ (composite-scalar prediction)	Confirmed at 0.6σ	[SM §7.4]
22 SM observables, classified (two parameter-free structural and five conditional structural on the empirically selected cubic substratum)	Sharpest entries sub-percent; classes per [SM §7.6]	[SM §7.5]
Majorana neutrinos, normal ordering	Awaiting LEGEND-1000 sensitivity	[SM §8.4, LEGEND2024]
$\lvert\nu\rvert \sim 10^{-32}$ in Type II RVM	Consistent with DESI DR2 $\nu \approx 0$ at 2σ	[GR §7, Bertini2025]
Universal $1/\alpha_0 = 23.25$ shared origin of three gauge couplings	Retrodictive match at few-percent (chain includes the empirical threshold δ_0; SM §6)	[SM §6.1, §6.7]

These quantitative comparisons test the selected OI branch under the assumptions and evidence classifications in [SM], [GR] and [Juno]. The proposed measurements can discriminate it from models giving other values. A successful match alone does not identify a unique substratum or an exact observational class.

Empirical strategy. A Born-rule test probes the probability framework in the specified experimental regime. Majorana searches test the conditional neutrino-sector output and other models predicting it. A precise JUNO measurement tests the selected value $1/3 - 1/(4\pi^2)$; running-vacuum and proton-decay measurements test their stated effective branches and assumptions. The relevant carrier determines which alternative models are distinguished and which predictions would be invariant under an established observational equivalence.

The most discriminating quantitative comparisons are collected at Level 4, while

the earlier tables show mechanisms or targets shared with other model families. This distinction concerns evidential reach; it does not permit an encoded observable to vary inside one class of Definition 9.4.

The tables therefore support realism about specified invariant structure and conditional empirical claims. A richer carrier can separate models that match a broad target, while an exact algebra/channel equivalence fixes the data it actually represents.

12.5 The no-GUT structural prediction

The framework predicts that no grand-unified gauge structure exists at any energy scale. This prediction has three layers, established jointly by [SM §4.6], [SM §6.4], and [SM §6.1] and brought together in [SM §6.7]:

(A) *Group-theoretic.* The selected H-link/H-cust branch reaches its low-energy gauge structure through the stabilizer of an equivariant condensate and the further physical reduction in [SM §4.6]. The no-GUT mechanism claim uses that full branch; the cubic representation alone does not exclude every larger custodial or high-energy gauge structure.

(B) *Coupling-level.* Gauge couplings do not unify. They are split at the matching scale $\mu = M_{\mathrm{Pl}}$ by C_2-dependent thresholds (induced-gauge-theory structure), with pure SM running below M_{Pl} continuing the splitting.

(C) *Mechanistic.* OI's gauge couplings emerge from the fermion determinant, with no fundamental gauge action and no moduli connecting the predicted values to other values.

This prediction is structurally distinct from string-theoretic frameworks, which traverse intermediate large gauge groups (heterotic E_8 chains, F-theory SU(5), type IIA Pati-Salam) en route to the SM. The OI-distinguishing content is layer (C) plus the universal $1/\alpha_0 = 23.25$ value (ambient; native for hypercharge under H-observer-bundle and H-Y-vertex, [SM §6.5]): the three SM gauge couplings emerge from a *single* substratum quantity, not from independent compactification moduli.

The conditional no-GUT consequences are catalogued at Level 3 in §12.4: proton-lifetime estimates, absence of GUT-mechanism monopoles and absence of GUT-chain cosmic strings under their source assumptions. These targets are shared with broader no-GUT model families. The OI-specific construction is its condensate-stabilizer and induced-coupling mechanism; shared targets do not by themselves establish one restricted algebra/channel class.

Empirical-lean assessment. The current empirical state moderately favors the no-GUT model family over standard GUT scenarios across multiple discriminators. Super-Kamiokande proton-decay bounds [SuperK2020_proton, SuperK2024_eta] at $\tau_p(p \to e^+\pi^0) > 2.4 \times 10^{34}$ years and $\tau_p(p \to \nu K^+) > 5.9 \times 10^{33}$ years already exclude minimal $SU(5)$

GUT and constrain the broader supersymmetric SU(5) parameter space [SUSY_GUT_Bourilkov2015]. LHC searches have pushed natural-SUSY parameter space to $M_{\mathrm{SUSY}} \gtrsim 3$ TeV [LHC_SUSY_2025], with the surviving SUSY-GUT scenarios requiring split-SUSY-style tuning that sacrifices the original hierarchy-problem motivation. NANOGrav 15-year pulsar timing data [NANOGravStrings2023] disfavors stable cosmic-string interpretations of the observed stochastic gravitational-wave background, with the signal preferentially explained by supermassive black-hole binary populations. Cosmological searches have found no GUT-scale magnetic monopoles, consistent with OI's prediction without requiring inflation as a dilution mechanism (inflation may still be motivated by horizon and flatness considerations, but the monopole problem is structurally absent in OI).

No single observation sharply distinguishes OI from all GUT-class scenarios — split-SUSY and topologically-suppressed F-theory models retain partial viability against current data. But the cumulative pattern across discriminators is a real empirical lean toward the no-GUT model family containing the selected OI branch, not merely consistency. The decisive future tests are Hyper-Kamiokande's projected $\tau_p \sim 10^{35}$ year sensitivity for $p \to e^+\pi^0$ [HyperK_TDR2018] and DUNE's $\sim 6.5 \times 10^{34}$ year sensitivity for $p \to K^+\nu$, plus continued LHC SUSY exclusion. Both will further sharpen the empirical position over the next decade.

Connection to the unified naturalness pattern. The no-GUT prediction is an instance of a broader pattern in the framework: substratum-level structural symmetries dissolve naturalness problems by forcing emergent parameter values that would otherwise be free. The instances catalogued in [SM §8.8] are: (i) *strong CP* — an open instance: substratum T-invariance narrows θ to $\{0, \pi\}$ and gives reciprocal transition counts but does not fix $\bar{\theta}$ ([SM §5], H-top and H-det); (ii) *hierarchy problem* — singlet-taste chiral symmetry provides composite-Higgs-style protection of the Higgs mass against cutoff-scale corrections ([SM §8.8], partial resolution); (iii) *cosmological constant* — trace-out's information compression ratio dissolves the 10^{122} discrepancy, conditional on the open coupling step ([GR §6.2]). The no-GUT structural prediction extends this pattern: rather than requiring a GUT mechanism to explain gauge-coupling unification (and inheriting all the GUT scenarios' associated empirical commitments), OI's selected condensate-stabilizer gauge branch sidesteps the unification puzzle structurally. Three of the four — hierarchy, cosmological constant, gauge structure — are dissolved by the same kind of substratum-level structural mechanism, and strong CP is narrowed but not dissolved by it, giving the framework a unified treatment of what standard physics treats as four independent open problems requiring four independent solutions.

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13. Open questions and limits

13.1 Technical questions

The main technical open questions:

Q1. Are the algebra-channel structures of OI and string theory *-isomorphic at the SM-reproducing target? (§11.2 Question 2.)

Q2. Which OI predictions are encoded in a completed comparison carrier, and which other SM-target models reproduce them? If equivalence on that carrier is established, the encoded predictions must transport; the open work is the construction and comparison, rather than whether an invariant can vary between equivalent representatives (§11.2 Question 1).

Q3. What is the precise definition of the "partial-trace observational features" class? Are the four features identified in §14 necessary and sufficient to characterize this class, or are there additional features?

Q4. Are there universality classes other than OI's and string theory's that produce the SM at low energies? (LQG-derived classes, causal-set-derived classes, asymptotic-safety classes, etc.) If so, what features characterize them?

13.1.1 Q1 developed: what *-isomorphism would entail

To establish equivalence at a specified EFT scale requires: (1) completing OI's observable algebra, channel and physical/limit bridge; (2) constructing the compactification's corresponding pair on the same observational carrier; (3) constructing a normal unital *-isomorphism θ of the algebras; and (4) checking $\theta \circ \Phi^{(1)*}|_{\mathcal{A}_1} = \Phi^{(2)*}|_{\mathcal{A}_2} \circ \theta$ for the specified channel or family. **Status:** Section 10.4 supplies an exact finite-cutoff channel and a conditional EFT specification. The complete EFT version of Step 1 and Steps 2–4 remain open.

Three possible outcomes are: (a) an algebra isomorphism intertwines the specified restricted adjoint dynamics, establishing equivalence on that carrier; (b) the algebras are isomorphic but the required intertwining fails; or (c) the observable algebras are not isomorphic. In case (a), the observables and protocols actually encoded and transported by the comparison share their predictions. Agreement at an SM target or on finitely many Wilson coefficients does not settle these alternatives, nor does a comparison on a proper algebra automatically identify the full ambient channels.

13.1.1.1 Converged comparison protocol A specific protocol for executing Steps 2-4 against a chosen compactification has been worked out as preparation for offline computational follow-up. The protocol is:

Compactification choice: heterotic line-bundle Calabi-Yau in the Anderson-Gray-Lukas-Palti class. Specifically, the Schoen-quotient model $X = \tilde{X}/(\mathbb{Z}_3 \times \mathbb{Z}_3)$ with five-line-bundle SM model [Anderson-Gray-Lukas-Palti 2011-2014]. This compactification was selected after considering three

candidates (heterotic line-bundle on Schoen quotient; type IIA Pati-Salam on intersecting D6-branes; F-theory $SU(5)$ with hypercharge flux) on multiple criteria: 4D EFT characterization completeness (heterotic AGLP has multi-loop Yukawa structure characterized in the literature); three-generation match (all three candidates produce three generations, but through structurally distinct mechanisms — see below); pure SM gauge group at IR (all three reach SM after appropriate breaking); bundle-cohomology computational tractability; phenomenological literature density (heterotic AGLP has the most extensive literature). The Schoen-$\mathbb{Z}_3 \times \mathbb{Z}_3$ model dominates on most criteria and was selected over its alternatives.

Comparison scale: below the Stueckelberg $U(1)$ mass (~10^{16} GeV). Heterotic line-bundle compactifications produce an extra Stueckelberg-massive $U(1)$ via the Green-Schwarz mechanism (anomaly cancellation in heterotic with line bundles); typical Stueckelberg masses in AGLP models are $\sim 10^{16}$ GeV, set by the compactification scale. Below this mass, the heterotic effective theory has only the SM gauge group, matching OI's structure at the IR. The comparison must therefore be conducted at scales below the Stueckelberg mass to ensure both sides have matching gauge structure. This restricts the universality-class claim to the IR EFT (scales $M_Z \lesssim \mu \lesssim 10^{16}$ GeV) but does not prevent the comparison from proceeding.

Comparison test: Yukawa textures. Both frameworks reach three generations by structurally distinct mechanisms (heterotic uses bundle-cohomology index of the relevant cohomology groups on the CY, an entirely topological invariant; OI uses the cubic group's T_1 irrep with multiplicity 3 from the spatial nearest-neighbor structure). Three generations is therefore not a discriminator; both frameworks pass this test by different routes. The natural quantitative test is whether the *Yukawa textures* arising from the cubic-group reduction in OI ([SM §7]) match the textures arising from bundle-cohomology computations on the Schoen-$\mathbb{Z}_3 \times \mathbb{Z}_3$ heterotic model. OI's Yukawa textures are characterized in [SM §§7.1-7.3] (CKM matrix, charged-lepton masses, PMNS angles); the heterotic Yukawa textures are characterized in [Anderson-Gray-Lukas-Palti 2013-2014]. The comparison reduces to: do these textures agree at the scale below the Stueckelberg mass?

Verdict criteria. Yukawa-texture agreement is a consistency check for a carrier containing those observables. A disagreement under matched conventions and scale rules out equivalence on that carrier; agreement does not establish a full algebra/channel isomorphism. Renormalizable agreement with differing higher-dimensional coefficients can support only a deliberately coarser comparison omitting the latter. A same-class verdict requires the actual algebra map and adjoint-channel intertwining.

A subtle point about scale-choice. The comparison-scale specification matters for the verdict. Below the Stueckelberg mass, both frameworks have just SM, and the comparison tests Yukawa textures. Between the Stueckelberg mass and M_{Pl}, the heterotic theory has an extra $U(1)$ + GUT-related structure that

OI does not have; the algebras differ at this scale. A comparison at the *higher* scale would yield a *different-classes* verdict not because of fundamental disagreement but because of the integrating-out structure. The honest comparison protocol must specify the scale, and the IR EFT scale (below Stueckelberg mass) is the natural choice for testing whether OI and the heterotic compactification can coexist in the same universality class at all.

Status of execution. The protocol identifies candidate models, scale conventions and initial quantities to compare. Executing it requires extracting matching OI and compactification predictions, completing the specified observable algebras and channels, constructing a normal algebra isomorphism and checking its restricted adjoint intertwining. Comparing full dilation unitaries is neither required by Definition 9.3 nor supplied by a single-channel Stinespring uniqueness theorem. The physical EFT construction and these comparison steps remain future work.

Connection to the no-GUT structural finding. The Stueckelberg $U(1)$ mass at $\sim 10^{16}$ GeV is the heterotic compactification's natural intermediate scale, set by the compactification structure. This is the string-theoretic analog of the seesaw scale (per Q8 in §13.3). The comparison scale's choice — at or below this Stueckelberg mass — implicitly settles the comparison's relationship to Q8: at the IR EFT level, OI and heterotic may share universality class even though OI lacks a structural intermediate scale (per Q10's negative resolution in §13.3). Universality-class matching at the IR is consistent with structurally-distinct intermediate-scale physics, with the structural difference being absorbed into the channel rather than being a barrier to algebra-isomorphism. This refines the §13.1.1 expected-outcome decomposition: outcome (b) is the working expectation (algebras agree at IR EFT; channels differ at intermediate scales).

13.1.2 Q2 developed: criteria for prediction transport

Q2 is a transport question relative to a specified carrier. Shared quantum representation or a common gauge group does not by itself fix Yukawa values, anomaly assignments or flavor-changing interactions; the matter representation, dynamics and measured protocols must also be specified. If a normal algebra isomorphism intertwines the chosen adjoint dynamics and transports the data defining a prediction, that prediction agrees. Microscopic combinatorial properties need not transport when they are omitted from the carrier; multi-time intervention statistics require the corresponding process data.

The Cabibbo, Koide and horizon calculations retain their stated OI branch assumptions. If their observable values are included in a completed carrier, they are class invariants and every equivalent realization reproduces them. A model producing a different value is inequivalent on that carrier, although it may share a gauge target or agree on a coarser comparison. Agreement on $A/4$ through different mechanisms is a useful shared output, not a proof of equality of the complete algebra/channel pairs.

13.1.3 Q3 developed: necessary and sufficient features for the partial-trace observational class

Q3 asks whether the four audit features identify one observational class. Within a specified quantum representation, the probability dictionary and dilation architecture have their usual roles, while memory requires retained correlations and symmetry restriction requires a specified action. None selects the quadratic probability dictionary from the observation axiom alone. The four features are not sufficient: systems can share them while their chosen observable algebras are non-isomorphic or their restricted adjoint dynamics fail to intertwine. Differences only in an ambient Hilbert-space realization, hidden dimension or unobserved multiplicity do not by themselves distinguish classes of Definition 9.4. OI's recurrence-scale P-indivisibility additionally uses a fixed finite reversible realization and a non-permutation rooted visible map; it is not automatic for an arbitrary channel.

Refinement candidates include information capacities, channel spectra and specified multi-time correlations. These can sharpen a comparison family but do not replace Definition 9.3's algebra map and intertwining. The four-feature audit is neither an unconditional necessity claim—memory and symmetry restriction require extra structure—nor a sufficient class-identification test.

13.1.4 Q4 developed: other universality classes that may produce the SM

Q4 asks which other model families can reach the SM target and whether particular members admit a comparison with the OI carrier. The following per-condition audit concerns their standard microscopic formulations and A1–A6 compatibility. Such compatibility is a different question from equivalence of their restricted observable pairs.

Loop Quantum Gravity. LQG's substratum [Rovelli2004; Ashtekar-Lewandowski2004] is the space of spin networks — graphs labeled by SU(2) representations and intertwiners — modulo spatial diffeomorphisms. Dynamics is defined either by Thiemann's Hamiltonian constraint on the kinematical Hilbert space or by spin-foam amplitudes summing over histories. Per-condition: A1 (finiteness) **fails** in the full theory — even at fixed graph, the Hilbert space is infinite-dimensional through unbounded spin labels, and the sum-over-graphs structure of the full theory makes $|S| = \infty$ generically; finite-graph and bounded-spin variants admit a finite reading but are not standard LQG. A2 (determinism / bijection) **fails** — the dynamics is quantum, with spin-foam amplitudes constructed as Feynman-like sums rather than deterministic bijections on configuration space. A3 (bounded coupling degree) holds at fixed graph valence (typically ≤ 4 in EPRL spin foams) but the bounded-coupling reading depends on the graph being fixed, which is in tension with sum-over-graphs. A4 (center independence) **holds** — diffeomorphism invariance is built in. A5 (linearity) **holds** at the operator level (the Hamiltonian constraint is a linear

operator on Hilbert space) but is complicated by the constraint structure: solutions to the constraint are highly nonlinear in the spin-network labels even when the constraint operator itself is linear. A6 (background independence) **holds** — this is LQG's defining feature. Net: A1 and A2 fail; A4 and A6 hold robustly; A3 and A5 are reframing-dependent. The structural pattern of failure is similar to string theory's matrix-model formulation in §6, with the key structural mismatches at A1 and A2 being deeper than the discretization-bridge issue of §6.1 — LQG's quantum dynamics and infinite Hilbert spaces are central to the program rather than artifacts of a particular formulation.

Causal sets. The causal-set program [BombelliLeeMeyerSorkin1987; Rideout-Sorkin1999; Surya2019] takes the substratum to be a locally finite partially-ordered set, with dynamics typically modeled by Classical Sequential Growth (CSG) — a stochastic generative process that adds events according to a probability measure on histories. Per-condition: A1 (finiteness) **holds at any finite stage** (finite causal sets are genuinely finite) but **fails** in the limit of unbounded growth. A2 (determinism / bijection) **fails** — CSG is explicitly stochastic, with each event-addition governed by a probability rather than a deterministic update rule. This is the most direct structural divergence from OI's substratum: where OI takes the substratum dynamics to be deterministic with stochasticity emerging through partial trace, causal sets take the dynamics to be intrinsically stochastic at the substratum level. A3 (bounded coupling) holds: each event has a finite number of immediate predecessors and successors. A4 (center independence) **holds** — the formalism is intrinsically relational. A5 (linearity) does not translate cleanly — CSG is described by transition probabilities rather than a wave equation, so OI's linearity-of-wave-equation framing has no direct analog. A6 (background independence) **holds**. Net: A2 fails outright, A1 fails in the limit, A5 lacks a clean translation. Causal sets and OI agree on relational structure (A4, A6) but diverge sharply at the determinism axis (A2). The structural-realist reading: OI's substratum and the causal-set substratum represent different commitments at the foundational level, with OI placing the stochasticity at the partial-trace observational layer while causal sets place it at the substratum layer.

Asymptotic safety. The asymptotic-safety program [Weinberg1979; Reuter1998; Eichhorn2018] takes gravity to be defined by a non-Gaussian UV fixed point of the renormalization group, with the metric as the dynamical variable in a continuum quantum field theory framework. Per-condition: A1 (finiteness) **fails** — continuum QFT. A2 (determinism / bijection) **fails** — quantum, path integral. A3 (bounded coupling) reframing-dependent in the same way as for matrix models; the natural reading at the spacetime level has unbounded coupling. A4 (center independence) **holds** — general covariance. A5 (linearity) **fails** — Einstein-Hilbert action plus higher-derivative terms is nonlinear in the metric, with the nonlinearity essential to producing gravitational dynamics. A6 (background independence) **holds**, modulo standard subtleties about expanding around fixed backgrounds. Net: similar pattern to string-theoretic matrix models — A2, A5, A1 fail; A4 and A6 hold; A3

reframing-dependent. The structural pattern resembles string theory closely.

Cross-program pattern. The audited formulations differ from OI's microscopic A1–A6 family, especially at determinism/bijection A2 and at finiteness A1 in their full settings. The remaining conditions need their program-specific interpretations as described above. These differences distinguish model families; they do not exclude a chosen model from sharing an OI observable pair. The finite-law quantum representation in [Main] supplies a concrete reason to keep microscopic determinism distinct from the representation used for observations.

Restriction-interface comparisons. The candidate programs can share quantum-representation or dilation architecture while differing in microscopic assumptions. Memory and divisibility require the separate analyses in §14, and symmetry comparisons require specified actions. These features organize a cross-program audit; exact observational membership remains a comparison of particular constructed restricted pairs.

Implications for §11 and Limit 1. The per-program audit broadens the comparison beyond string theory. It records microscopic differences and candidate shared mechanisms, rather than enumerating completed observational classes. For each candidate, the remaining task is to construct the chosen observable algebra and dynamics and test the map required by Definition 9.3.

13.2 Conceptual questions

Q5. Is the universality-class framework the right level of abstraction, or is it too coarse to capture features relevant for fundamental physics? (E.g., quantum gravity might depend on substratum-level features that universality classes wash out.)

Q6. How does a specified observational carrier sharpen OI's predictive content? Which branch predictions discriminate models sharing a broad SM target, and which become invariants after an exact comparison is established?

Q7. What is the relationship between universality classes of embedded observers and other equivalence classes in fundamental physics (e.g., RG universality classes, equivalent QFTs related by dualities, holographic correspondence between bulk and boundary descriptions)?

13.2.1 Q5 developed: is universality-class abstraction the right level?

Q5 concerns the choice of carrier. A comparison retaining only selected low-energy data may omit the Planck-scale or gravitational observables needed for a proposed quantum-gravity test. Including those observables can split a coarser equivalence class. Horizon entropy relations retain their gravitational and calibration assumptions; they are not consequences of partial trace alone merely because several mechanisms yield the same coefficient. Framework-specific microscopic predictions remain meaningful, while a class claim about their measured effects requires those effects to be encoded in the comparison.

13.2.2 Q6 developed: does universality-class framing weaken or strengthen OI's predictive content?

Q6 sharpens the conditional content of the prediction chain. The selected OI branch yields the Cabibbo comparison $\sin\theta_C = 1/(\pi\sqrt{2})$ under its source assumptions; sharing the SM target alone does not impose that value on another model. If a completed observational carrier includes the relevant flavor data and another realization is equivalent on it, the value necessarily agrees. A match therefore supports the conditional branch and can distinguish models giving different values, without proving a unique microscopic realization or identifying all models with the same observational class.

13.2.3 Q7 developed: relationship to other equivalence classes in fundamental physics

Q7 compares different equivalence relations. RG agreement concerns specified long-distance data; a QFT duality compares specified algebras, states and dynamics; Definition 9.3 compares a chosen invariant observable algebra and restricted adjoint channel or family. A duality yields observational equivalence when it maps those data and their restrictions. RG agreement or a holographic dictionary does not automatically supply that map for every observer choice. No general ordering of these equivalences is asserted here; the comparison depends on the scales, algebras and protocols retained.

13.3 Open questions on intermediate-scale physics

The structural investigation of OI's no-GUT prediction (developed in [SM §6.7] and [SM §8.7]) surfaced several substantive open questions that bear on the framework's predictive content but require further work to develop. They are catalogued here as research directions; their development is deferred to future work.

Q8. *The neutrino mass scale puzzle.* The framework's [SM §8.4] predicts Majorana neutrinos via the dimension-5 Weinberg operator $(LH)(LH)/\Lambda_{\text{seesaw}}$, with Λ_{seesaw} treated as a Tier 3 (solution-specific) parameter. The observed neutrino mass scale $m_\nu \sim 10^{-3}$ eV requires $\Lambda_{\text{seesaw}} \sim 10^{16}$ GeV — coincidentally the GUT scale. In standard GUT scenarios, this scale is structurally explained by gauge-coupling unification. In OI's no-GUT framework, no GUT scale exists structurally; the natural cutoff is the lattice scale $M_{\text{Pl}}/(2\sqrt{\sigma})$, of order $M_{\text{Pl}} \sim 10^{19}$ GeV when σ is of order one, which would give $m_\nu \sim 10^{-6}$ eV — three orders of magnitude too small. The framework currently treats Λ_{seesaw} as solution-specific input rather than deriving it. *Open question:* does the framework have an intermediate scale between M_Z and M_{Pl} that arises structurally from the substratum's staggered-taste structure or other lattice-level features, and that could naturally produce $\Lambda_{\text{seesaw}} \sim 10^{16}$ GeV? Speculative candidates include scales associated with the staggered-taste mass spectrum (where different tastes have different couplings $|\mu_\Gamma| = 1, |\mu_X| = 1/3$ per [SM §4.7] Theorem 8b),

or scales associated with the boundary-layer thickness in the visible-vs-hidden sector partition.

Q9. *The hierarchy problem in OI's composite-Higgs structure.* The framework's [SM §7.4] predicts $\lambda(M_{\rm Pl}) = 0$ as a structural consequence of the Higgs being a composite scalar (the singlet taste). This prediction is empirically supported at 0.6σ and identifies the Higgs sector as composite at the substratum level. Standard composite-Higgs scenarios in BSM phenomenology dissolve the hierarchy problem via chiral-symmetry protection (the Higgs as a pseudo-Nambu-Goldstone boson of an approximate chiral symmetry). *Partial resolution* (developed in [SM §8.8]): the framework structurally implements an analog of this protection through the singlet-taste chiral symmetry. The $\mu = +1$ and $\mu = -1$ singlet tastes are paired under chirality at the substratum level (exact discrete symmetry); the Higgs vev spontaneously breaks this pairing in the emergent EFT. The Higgs is the order parameter of the broken chirality, with mass protected against cutoff-scale corrections by Goldstone's theorem.

The framework *derives* cutoff insensitivity — no $O(M_{\rm Pl}^2)$ corrections to m_H^2, only explicit-breaking corrections of order $g^2 f^2/(16\pi^2)$ — together with the compositeness scale $f \sim M_{\rm Pl}$ being structurally forced. The framework *does not derive* the Higgs vev value $v \sim 246$ GeV, which remains Tier 3 (solution-specific). The boundary: the framework derives why m_H is not at the cutoff (chiral protection); it does not derive why v is at the electroweak scale rather than at the compositeness scale.

Quantitatively, the framework's partial resolution differs from "no resolution" by a factor of $\sim 10^{34}$ in the required fine-tuning. Standard SM without chiral protection requires fine-tuning m_H^2 to 34 decimal places; the framework's mechanism requires only that the Tier 3 input v be at the electroweak scale, with m_H following from $\lambda(M_{\rm Pl}) = 0$ + RG running.

Open question: is there a structural mechanism in OI that derives $v \ll f$, or is v genuinely solution-specific? Two candidate paths: (i) deriving $v \ll f$ from substratum structure through a mechanism currently not in the framework; (ii) accepting v as solution-specific, with the structural-vs-solution-specific partition being part of the framework's content. The framework's current commitment is (ii): v sits in the solution-specific category alongside CP phases, generation-specific Yukawa magnitudes, and other parameters whose values are not forced by the structural derivation chain.

Q10. *Possible unified resolution of Q8 and Q9.* Both Q8 (neutrino mass scale) and Q9 (Higgs hierarchy) point to a single underlying question: does OI's substratum produce a structural intermediate scale between M_Z and $M_{\rm Pl}$ that's not a GUT scale but plays the roles that GUT scales play in standard scenarios — setting the seesaw scale for neutrino masses and protecting the Higgs mass via chiral mechanisms? **Tentative resolution:** negative on the analysis conducted here. The framework's substratum-level scales are essentially the lattice scale $M_{\rm Pl}/(2\sqrt{\sigma})$, of order $M_{\rm Pl}$ when σ is of order one, and emergent

scales below M_Z; candidates for intermediate scales (staggered-taste mass gap, boundary-layer thickness, the $\delta_0 = \mathrm{Tr}[Y^2]$ coincidence at the matching convention $\mu = M_{\mathrm{Pl}}$, C_2-dependent threshold effective scales, cosmological-horizon-induced scales, emergent dimensional transmutation) all yield scales at M_{Pl} modulo $O(1)$ factors or scales below M_Z, but none naturally at $\sim 10^{16}$ GeV. The seesaw scale and the Higgs mass therefore remain as Tier 3 (solution-specific) parameters in the framework, set by the specific bijection φ rather than by structural derivation. *Implication:* OI's predictions on intermediate-scale physics are limited to what can be derived from M_{Pl} + EW-scale physics, with no claim of structural intermediate-scale derivation. This is consistent with the framework's existing treatment of $\Lambda_{\mathrm{seesaw}}$ as Tier 3 (per [SM §8.4]) and represents an honest scoping rather than a hidden deficiency. The negative resolution sharpens the structural-vs-solution-specific distinction: many parameters that GUT-style frameworks treat as derivable from unification structure are, in OI, genuinely solution-specific. This *may* be revisited if a structural mechanism overlooked in the chat-tractable analysis is identified by future work, but the chat-tractable conclusion is that no such mechanism is currently apparent in the framework's existing structure.

Q11. *Quantitative baryogenesis in a spectrum with no right-handed neutrino.* The framework's substratum-level baryon-number conservation (per [SM §8.7]) coexists with sphaleron-mediated $B+L$ violation in the emergent EFT. But the OI spectrum contains no right-handed neutrino — anomaly cancellation forces exactly five representations per generation and the singlet taste is the Higgs ([SM §4.9, §8.4]) — so standard type-I thermal leptogenesis, which requires heavy right-handed Majorana neutrinos, does not directly apply. *Open question:* the full quantitative calculation — connecting whatever ultraviolet physics generates the dimension-5 Weinberg operator (the $\Lambda_{\mathrm{seesaw}}$ scale of Q8) to the resulting baryon asymmetry $\eta = n_B/n_\gamma \approx 6\times 10^{-10}$ — is open, and the mechanism is not fixed by the framework. Resolution depends partly on Q8: without a structural derivation of the Weinberg-operator scale, the baryogenesis prediction inherits the same status (Tier 3 / solution-specific).

Q12. *Integrated treatment of non-perturbative emergence.* [SM §8.7]'s table of substratum-level structural precursors mapping to emergent-level non-perturbative SM features (T-invariance $\to \theta = 0$; topology of induced gauge action $\to$ sphalerons; bipartite checkerboard $\to$ chiral fermions and anomalies; block-diagonal $M \to$ substratum B conservation) provides a unified pattern that organizes the framework's treatment of non-perturbative SM physics. *Open question:* is this mapping exhaustive, or are there non-perturbative SM features (e.g., specific instanton-density distributions, specific non-perturbative QCD vacuum structure, specific anomaly-inflow mechanisms) that lack identified substratum-level precursors? A complete cataloging would either confirm the framework's structural completeness on this dimension or identify gaps requiring further development.

These five open questions (Q8-Q12) represent the framework's substantive re-

search frontier as of the no-GUT investigation. Each is a candidate research direction for future development; together they form a coherent set focused on whether OI's substratum has the structural richness needed to resolve standard SM open problems (neutrino masses, hierarchy, baryogenesis, non-perturbative completeness) without invoking new physics beyond what the framework already commits to.

13.4 Limits of the approach

The universality-class framework has limits worth flagging.

Limit 1: Programmatic with a partial model audit. The paper compares four other research programs with OI's microscopic conditions and audits candidate shared interface mechanisms. This is not an enumeration of their observational classes. A1–A6 incompatibility in a standard formulation does not rule out equality of a chosen observable pair, and shared quantum or memory architecture does not establish it. The physical constructions and explicit comparison maps remain the central unfinished tasks.

Limit 2: Depends on operator-algebra technology. The algebraic formalization adopted in §8.4 and used throughout §9 depends on operator-algebra and CPTP-channel theory. These are well-developed but place the framework in a specific mathematical setting. Frameworks that do not naturally admit operator-algebra description (e.g., theories with no clear Hilbert-space structure) may not fit the formalization without modification.

Limit 3: Depends on continuum / discrete bridge. The formalization assumes that substratum descriptions, regardless of discreteness, yield visible-sector reduced laws admitting CPTP-channel representation with operator-algebra structure — representability is universal for finite laws ([Main §3.4]); operational closure beyond it is a model hypothesis, not a framework property. For OI's discrete substratum and string theory's continuum substratum this is plausible (both yield quantum-representable observable-level descriptions). For substrata whose visible-sector observable structure is not of this form, the framework may not apply.

Limit 4: The foundational thesis is unfalsifiable on its own; the content lives in the model of the incompleteness. The framework's name and starting point — that what is known is reached through observation, which is necessarily incomplete (an embedded observer cannot access the total state) — is, read as a bare thesis, not a falsifiable physical claim. It is close to a restatement of the empirical stance common to all of science: every theory is a description constructed from inside, from partial access. So the bare thesis cannot be wrong, and therefore cannot by itself predict anything. All of the framework's predictive content lives one level down, in the *specific model* of the incompleteness: the C1–C4 architecture (a fast visible sector reading and writing to a slow, high-capacity hidden register), the cubic lattice, the wave-equation dynamics, and the ETH-genericity of the hidden sector. Each of these is either

an empirically-anchored selection or a structural stipulation (catalogued for the SM sector in [SM §8.3, *Derivation status: the three tiers*]; for the reconstruction in [Substratum §3, Refinement]). The framework's power and its circularity are accordingly the same structure seen from two sides: it is a maximally tight compression of known physics onto a small set of commitments — establishing that the Standard Model's discrete structure, the gravitational thermodynamics, and the Born-rule statistics are facets of one object rather than independent inputs (a genuine claim about the unity of physics) — while those commitments are themselves chosen to recover the physics already observed. Reading "observation is incomplete" as the *source* of the predictions overstates the framework; reading it as the *organizing principle* under which a large body of physics compresses to one structure is the defensible claim, and the one this paper's universality-class machinery is built to make precise. This limit is not a defect to be removed by further work — it is intrinsic to any framework whose foundation is the empirical stance itself — and stating it plainly is what distinguishes a structural-realist compression from a claimed derivation of the world from nothing.

13.5 Relationship to the methodology paper

The universality-class framework is closely related to a planned methodology paper articulating OI's overall posture for philosophy-of-physics venues. The methodology paper would cover the broader posture (constructive structural realism, single-foundational-commitment derivation, the architecture-level engagement with the no-go results of quantum foundations, "physics modulo gauge" reframing); the universality-class framework provides the technical machinery for the structural-realism component when applied to comparison with other unification programs. The two contributions are companions: the methodology paper situates OI's posture in the philosophy-of-physics literature; the universality-class framework provides one technical articulation of what that posture entails.

Part III: Cross-Framework Synthesis

14. Cross-framework audit of observational structures

This section develops the constructive content of recovery (v) — shared observational invariants — by examining whether structural features of OI's trace-out also follow from string theory's emergence mechanisms. The framing is that observation is a structurally privileged operation: OI's substratum and string theory's substrata are different, but the *act of observation* on each substratum produces observational features that may be substratum-independent.

We proceed feature by feature. For each, we characterize the structural feature in OI's framework, identify the analogous mechanism in string-theoretic emergence, and assess whether the feature transports across frameworks. Four feature classes are examined: Born-form quantum probability structure, channel-level unitary dilation, memory/non-Markovian marginal dynamics with divisibility tracked separately, and commutant gauge-invariance.

The discussion is technical but exploratory. The aim is not to prove cross-framework equivalence at the observational level — that would require substantial work beyond chat-track scope — but to establish whether the proposed shared invariants are structurally well-defined in both frameworks. If they are, recovery (v) admits constructive sharpening; if some don't transport, the framework's universality claims must be refined.

14.1 The Born rule

OI's framing. At the finite observable-law layer, OI supplies an exact fixed-basis Born-form representation rather than a parameter-free derivation of the quadratic probability functional [Main §3]. A finite reversible realization lifts to a permutation unitary, and with Q_{fb} defined by the dictionary $p = |U|^2$, tracing over the hidden register reproduces the visible transition law exactly. The important scope distinction is that the construction does **not select** the exponent: permutation-unitary entries are 0 or 1, so $|U|^p$ gives the same transition table for every $p > 0$. For a general quantum representation the visible density matrix $\rho_V = \mathrm{Tr}_H|\psi\rangle\langle\psi|$ gives $p(a) = \mathrm{Tr}(\Pi_a\rho_V)$ in the usual way; identifying that full coherent observable/instrument structure simultaneously across all interventions is part of the operational-lifting frontier. OI's proved contribution at this layer is therefore the exact Born-form representation of classical deterministic substrata, not an independent selection theorem for why nature chooses the quadratic dictionary.

String-theoretic analog. String theory uses standard quantum mechanics throughout — in matrix-model formulations (BFSS, IKKT), in worldsheet conformal field theory, and in the boundary CFT of AdS/CFT correspondence. The Born rule is built in as a feature of QM. When an observer in a string-theoretic context measures an observable, the probability is given by the standard quantum-mechanical trace formula on the boundary CFT or the worldsheet.

Transport assessment. The **Born-form representation structure** transports straightforwardly: once a framework is represented quantum mechanically and reduced by partial trace, subsystem probabilities use the standard quadratic trace rule. The distinction is that OI constructs an exact fixed-basis quantum representation starting from a classical reversible substratum, whereas string theory assumes the quantum formalism from the outset. This comparison should not be read as an OI derivation of the exponent 2 itself; Main §3.1/§3.4 makes explicit that the permutation construction is exponent-blind and that the

dictionary's physical origin is a separate frontier item.

The shared-invariants reading is therefore representation-theoretic rather than a new Born-rule uniqueness theorem. The quadratic Born rule itself is empirically tested at the 10^{-3}–10^{-4} level via multi-path interferometry [Sinha2010, Söllner2012], with ongoing precision improvements through ultracold-atom platforms [Kanthak2025] (further discussion in §15.5).

14.2 Channel-level unitarity

OI's framing. At the finite cutoff, a permutation-unitary lift of the bijection and the specified hidden preparation give a unitary realization of the visible CPTP channel. Stinespring's theorem separately supplies a prepared isometry for a full channel, with unitary realizations in an appropriate ancillary setting. The visible channel can be non-unitary even though its joint realization is unitary. This is a dilation/realization statement, not recovery of a unique full hidden unitary from the visible channel; the finite and separable uniqueness results in §9 concern minimal prepared isometries.

String-theoretic analog. AdS/CFT bulk reconstruction has a closely analogous structure. Boundary CFT operators are encoded in bulk operators with redundancy, and the encoding is a quantum error correcting code [HarlowBulkReconstruction2018, AlmheiriDongHarlow2014]. Bulk reconstruction from boundary subregions ("entanglement wedge reconstruction") shows that bulk operators in the entanglement wedge of a boundary subregion can be reconstructed from operators on that subregion. The reconstruction is unitary at the level of the full bulk-boundary system, with effective non-unitarity arising when restricting to a subregion.

The structural parallel is direct: in AdS/CFT, the "trace-out" is restriction to a boundary subregion, and the "visible-sector channel" is the effective dynamics on that subregion after tracing over the complement. The bulk-boundary system as a whole is unitary; the subregion-level channel is generally non-unitary; the channel admits a unitary dilation given by the full bulk reconstruction.

Transport assessment. Channel-level unitarity transports across frameworks. Both OI and AdS/CFT have the same structural feature: a CPTP channel at the observable level lifting to unitary dynamics on a dilation. The dilation in OI is the Stinespring lift to $\mathcal{H}_V \otimes \mathcal{H}_H$; in AdS/CFT, the dilation is the full bulk-boundary system before restricting to a subregion.

The mechanisms are different: OI has an explicit permutation-unitary realization at a finite cutoff with its chosen hidden state, while a holographic construction uses its own encoding and subsystem restrictions. Their common comparison is a channel/realization architecture. Minimal Stinespring uniqueness for a full channel neither identifies those physical constructions nor proves equality of their restricted observable pairs.

14.3 Memory, non-Markovianity, and recurrence-scale indivisibility

OI's framing. OI's C4 readback condition is first a statement of *conditional memory*: in general $P(a_3 \mid a_2, a_1) \neq P(a_3 \mid a_2)$ because hidden-sector correlations retain information about visible history [Main §2.3, §3.4]. That property is not identical to P-indivisibility. In Main's divisibility sense, P-indivisibility means that for some pair of rooted visible maps there is no stochastic propagator that divides the later map through the earlier one. What OI proves for one fixed finite reversible representative is more specific: a non-permutation rooted map, which C4 supplies and every non-deterministic visible step also supplies, together with finite recurrence returning the visible map to the identity, gives an indivisibility witness somewhere in the recurrence cycle, since a non-permutation stochastic map has no stochastic inverse. Bell is a separate constraint: a response-complete deterministic completion that is both measurement-independent and ontically parameter-independent obeys $|S_{\mathrm{CHSH}}| \leq 2$, so Bell-violating OI composites retain measurement independence and relinquish ontic parameter independence while preserving operational no-signaling [Main §3.3]. Thus C4 memory is the temporal mechanism, recurrence supplies an indivisibility witness that also holds without accessible memory, and neither memory nor indivisibility is a Bell selector.

String-theoretic analog. Holographic effective field theories explicitly classify modes by their Markovianity. The sound scalar in compressible relativistic fluids dual to AdS planar black holes is a non-Markovian field with Markovianity index $3 - d$ for a d-dimensional fluid. The holographic Schwinger-Keldysh formalism [HolographicSK2024, HolographicNonHydro2024] derives effective field theories that include non-Gaussian noise and non-Markovian dynamics as features of the boundary effective description, arising from integrating out bulk degrees of freedom. This supplies a close analog of the *memory/non-Markovian* part of OI's mechanism.

The structural parallel is therefore at the memory level: in both frameworks, restricting attention to a subset of degrees of freedom can produce history-dependent effective dynamics when the integrated-out degrees of freedom retain correlations across time. Partial trace or coarse-graining by itself does not force this outcome; a memoryless hidden sector can produce a Markovian reduced law.

Transport assessment. The memory/non-Markovian mechanism transports across frameworks. The stronger P-indivisibility property does **not** follow merely from the cited holographic non-Markovianity statements, and is not claimed here for string theory without a separate divisibility analysis. OI's recurrence-scale witness uses additional structure that the cross-framework analogy does not automatically supply: one fixed finite reversible representative plus a non-permutation rooted map, which C4 readback supplies.

This distinction is important for the Bell comparison as well. Holographic non-Markovian effective theories provide an analog of hidden-memory dynamics,

not by themselves a proof of the standard quantum Bell correlation set. In OI, a response-complete deterministic completion that combines the pointwise graph cone with a measurement-independent pre-setting ensemble is Bell-local and satisfies $|S_{\mathrm{CHSH}}| \leq 2$; quantum Bell violation therefore requires the separately stated ontic-parameter-dependent branch, with operational no-signaling retained. Hidden memory and stochastic indivisibility alone do not supply that branch.

The memory mechanism has independent theorem-level corroboration. [BrandnerMeyer2025] establishes a rigorous weak-memory regime for autonomous linear evolution equations with hidden-degree-of-freedom coupling, providing a controlled framework for the non-Markovian visible dynamics OI invokes. [TomizukaTakeda2026] constructs a structurally similar hidden-model decoherence picture independently, motivated by classical–quantum dynamics in gravity. [Barandes2026]'s stochastic-quantum theorem concerns indivisible stochastic processes and unitary quantum systems; Main §3.4 retains that result for the compressed form and interpretation under the translation hypothesis, while keeping indivisibility distinct from conditional-memory non-Markovianity and keeping fixed-basis representability internal. The framework's central technical move therefore sits within an established theorem-level neighborhood without identifying these different notions by definition (§15.1 develops this in detail).

14.4 Commutant gauge-invariance

OI's framing. The selected single-copy H-link/H-cust branch uses the exact six-link cubic representation to constrain an equivariant condensate. For generic distinct block values, its stabilizer in U(6) is $G_0 = \mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$; the commuting algebra of the cubic action itself is block-scalar. The physical local gauge interpretation and further central reduction use the covariance, quotient and charged-observer-action premises in [SM §4.6]. Gauge-invariant observables lie in the commutant of the specified physical gauge action, but tracing out an arbitrary hidden factor does not by itself project every surviving operator into that commutant.

String-theoretic analog. Gauge invariance in string-theoretic emergence has multiple sources, but the closest analog to OI's commutant structure is the *open-string sector* of D-brane stacks. A stack of N D-branes has $U(N)$ gauge symmetry generated by the open-string excitations on the brane worldvolume; the commutant of the worldvolume action gives the gauge group. Restricting to closed strings (graviton sector) gives the singlets under the gauge group, analogous to OI's trace-out preserving only commutant operators.

For AdS/CFT specifically, the boundary CFT has a global symmetry that becomes a bulk gauge symmetry under the duality. Boundary operators charged under the global symmetry correspond to bulk fields charged under the gauge symmetry; gauge-invariant bulk physics corresponds to singlet boundary opera-

tors. The commutant structure is preserved across the duality.

Transport assessment. The pattern is a comparison of symmetry actions and their invariant observable algebras. Selected OI and string constructions can reach the same local SM factors through different assumptions. The OI route uses the conditional condensate stabilizer and physical reduction; compactification constructions use their own gauge data. Matching the factors does not identify the full observable pairs or their global gauge actions.

The shared structure is the use of symmetry-invariant observables once a physical symmetry action has been specified. The mechanism selecting that action and its group is model-dependent. OI's exact cubic link decomposition is one input to its conditional physical construction, while a string compactification supplies different microscopic input.

This partial transport supports comparison of invariant-observable structures under stated symmetry hypotheses. It does not prove that every partial trace preserves only gauge singlets, or that cubic symmetry alone selects the physical SM action. Exact class membership still requires the restricted algebra/channel map.

14.5 Synthesis

The four-feature audit produces a mixed transport pattern:

Feature	Transports?	Caveat
Born-form quantum probability structure	Yes at the representation/formalism level	OI exhibits an exact fixed-basis representation; string theory starts from quantum formalism; the comparison does not select the exponent 2
Channel-level unitary dilation	Yes	Different dilation mechanisms (Stinespring vs holographic)
Memory / divisibility	Memory mechanism: yes; P-indivisibility: not established cross-framework	OI has C4 memory and a recurrence-scale indivisibility theorem; holographic non-Markovianity alone does not prove stochastic non-divisibility

Feature	Transports?	Caveat
Symmetry-invariant observables	Pattern compares under stated symmetry hypotheses	OI uses its conditional condensate-stabilizer/physical-reduction branch; compactifications use their own gauge data

The first two features transport directly at the representation/dilation level. The third transports at the memory/non-Markovian mechanism level, while OI's recurrence-scale P-indivisibility uses additional finite-recurrence structure and is not inferred for string theory here. The fourth transports as a *pattern* (gauge invariance via commutant) but not as a *specific output*.

This refines recovery (v) substantively. Partial trace is an architectural mechanism, not a theorem that every reduced process is memory-bearing. Once a quantum probability dictionary is in place, reduced probabilities have the usual Born form; once a visible map is CPTP, a unitary dilation is available. By contrast, non-Markovian memory requires retained hidden correlations, and P-indivisibility requires a further divisibility argument. OI supplies those extra ingredients through C4 and finite recurrence for a fixed representative; the cross-framework audit must check them separately elsewhere.

The picture that emerges is therefore narrower than a universal four-feature derivation: **observation by restriction supplies a common interface pattern, while memory, divisibility, and substratum-symmetry outputs depend on additional structure**. OI and string theory share important representation/dilation architecture and can share hidden-memory mechanisms; they do not thereby become equivalent at the full operational level.

This conclusion has implications for the universality-class question developed in Part II below. Embedded-observer universality classes should be characterized by a vector of properties rather than by "partial trace" alone: which quantum-probability representation is used, which dilation structure exists, whether the discarded sector retains operational memory, whether the reduced dynamics is actually indivisible, and which gauge structures the substratum admits. OI and string theory can intersect on several coordinates while reaching them through different mechanisms.

14.6 Implications for the OI-string relationship

The cross-framework analysis of §14.1–14.5 establishes that recovery (v) — shared observational structure — has substantive content, but it also fixes the scope of that claim. The commonality is strongest at the representation/dilation interface and becomes conditional once memory, divisibility, or substratum symmetry is involved.

The microscopic formulations audited in §6 are structurally distinct. At the observational interface they share quantum representation and dilation architecture, and they can exhibit analogous hidden-memory mechanisms. OI's recurrence-scale P-indivisibility uses C4 together with a fixed finite reversible realization; no corresponding string-theoretic theorem is inferred from holographic non-Markovianity. These findings motivate comparison without deciding Definition 9.3 equivalence.

This intersection is non-trivial but should not be overstated. A framework that already uses quantum states and partial trace inherits the standard quantum probability rule for its subsystem and the usual dilation architecture. It does **not** automatically inherit non-Markovianity, P-indivisibility, Bell causal locality, or OI's specific commutant output. Those require additional dynamical, statistical, or symmetry structure.

The constructive content of recovery (v) is therefore: **the restriction/partial-trace interface is a locus of shared architecture, not by itself a derivation of every observational feature audited here**. This connects to OI's "observation occurs" principle while preserving Main's distinction between universal representation, C4 memory, recurrence-scale indivisibility, and the still-conditional operational-local lift.

What does not transport — the specific gauge group, the specific particle content, the specific predictions — is what distinguishes substratum descriptions empirically. OI's predictions for the Cabibbo angle and Koide ratio are features of OI's specific substratum-symmetry derivation; string theory's compactifications would need to reproduce these as features of specific compactification choices. The two frameworks agree on the *kind* of physics observation produces; they differ on the *specific content*.

This is a more refined picture than the §7.1 framing of "OI and string theory as independent programs." They are independent at the substratum level, share structural features at the partial-trace level, and may or may not converge at the specific-content level depending on which compactifications string theory can realize as SM-reproducing. The recovery (v) analysis identifies precisely where the intersection lives: at the level of partial-trace structural features, not above and not below.

14.7 Loop quantum gravity: four-feature audit

The §14.1–14.6 audit was conducted for OI vs string theory. The same four features admit similar audit for other quantum-gravity / unification programs identified in §13.1.4. This section and the next two perform the audit for loop quantum gravity, causal sets, and asymptotic safety. The aim is the cross-program characterization of where the partial-trace observational features hold.

Born rule. Loop quantum gravity (LQG) [RovelliSmolin1990, Thiemann2007] uses standard quantum mechanics throughout: spin-network states form

a Hilbert space with inner products, observables are operators on this Hilbert space, and probabilities are computed via the standard Born rule on spin-network projection operators. The Born rule transports without modification.

Channel-level unitarity. LQG dynamics is implemented through spin-foam path integrals [EPRL2008, FreidelKrasnov2008] that produce transition amplitudes between spin-network states. The spin-foam amplitude is unitary at the level of the full path integral; effective dynamics on visible-sector observables (e.g., on smeared geometric observables averaging over Planck-scale spin-network structure) is CPTP, with the unitarity recovered at the spin-foam level via Stinespring-style dilation. Channel-level unitarity transports, with the dilation living in the spin-foam path integral rather than the OI substratum bijection.

Memory / divisibility. Spin-foam dynamics on visible-sector observables, after integrating over hidden internal-edge data, plausibly produces non-Markovian effective dynamics. The mechanism is structurally similar to OI's trace-out: the visible-sector marginal can carry memory of substratum-level dynamics through correlations established in earlier spin-foam evolution. This supports a plausible memory/non-Markovian analog, but it does not by itself establish P-indivisibility in Main's stochastic-propagator sense. Both the existence of the effective memory law and any stronger divisibility failure require technical work; this is the most underdeveloped of the four feature classes for LQG.

Commutant gauge-invariance. LQG uses its connection-variable gauge symmetry and diffeomorphism constraints to define physical invariant observables. This can be compared with OI's selected physical gauge action, whose local SM factors require the H-link/H-cust condensate-stabilizer construction and further reduction. Matching a symmetry-restriction pattern does not identify the specific gauge action or observable pair. The additional matter-sector construction required to reach the SM target remains a separate part of the LQG comparison.

LQG transport summary. The quantum-probability formalism, channel-level unitary-dilation pattern, and commutant gauge-invariance pattern are present in the comparison. The memory/divisibility feature is only plausible at the memory level; P-indivisibility remains open. LQG therefore shares several interface-level structures with OI without yet matching OI's recurrence-scale indivisibility result.

14.8 Causal sets: four-feature audit

Causal set theory [BombelliEtAl1987, Sorkin2003] takes the substratum to be a discrete, locally finite partially ordered set with sequential growth dynamics. The structural setup differs from OI in that the substratum dynamics is fundamentally stochastic (sequential growth via classical or quantum sequential growth) rather than deterministic.

Born rule. Classical sequential growth (CSG) [RideoutSorkin1999] is fundamentally classical-stochastic, not quantum. The Born rule does not directly arise. Quantum sequential growth (QSG) [DowkerSurya2020] adapts the framework to produce quantum probabilities through quantum measure theory. Whether QSG produces the standard Born rule or a generalization (with potentially observable higher-order interference terms) is an open technical question. Transport: open / uncertain. The current status is that the framework is structurally compatible with the Born rule but does not exhibit it as an exact internal representation identity, as OI's finite-horizon equivalence does ($S \iff D \iff Q_{\mathrm{fb}}$, [Main §3.4]). The comparison is about exhibiting the Born form, not about deriving it: neither framework selects the quadratic exponent, which in OI is part of the definition of Q_{fb} rather than a derived consequence ([Main §3.1]).

Channel-level unitarity. CSG sequential growth lacks a clean unitary lift at the substratum level — the dynamics is fundamentally stochastic. QSG's quantum-measure formulation can be reformulated through a CPTP channel structure with appropriate dilation, but the Stinespring dilation is structurally different from OI's (the dilation acts on a different Hilbert space, with different microcausality properties). Transport: structurally yes, but the dilation mechanism differs.

Memory / divisibility. Causal-set coarse graining can plausibly produce history-dependent effective dynamics when fine-grained causal structure retains correlations. That is structurally similar to OI's hidden-memory mechanism. The comparison supports a plausible memory/non-Markovian analog; it does not establish P-indivisibility without a separate divisibility analysis.

Commutant gauge-invariance. Causal sets are intrinsically diffeomorphism-invariant: the causal-order structure determines the manifold up to volume measure, and any continuous diffeomorphism preserves the order. Observable physics is determined by order-invariant quantities. The commutant gauge-invariance pattern transports, but the gauge group is the trivial diffeomorphism group rather than a Yang-Mills structure. Causal sets must reach SM gauge structure through additional matter content, similar to LQG.

Causal sets transport summary. The commutant/order-invariance pattern transports most directly. Quantum-probability and unitary-dilation structure depend on the QSG formulation, while memory is plausible and P-indivisibility remains unaudited. Causal sets are therefore related at several structural levels but do not presently match the full OI feature vector.

14.9 Asymptotic safety: four-feature audit

Asymptotic safety [Weinberg1979, Reuter1998, ReuterSaueressig2019] proposes that quantum gravity is non-perturbatively renormalizable through a non-trivial UV fixed point of the renormalization group. The framework operates within standard QFT machinery with non-perturbative RG analysis.

Born rule. Standard QFT machinery is used throughout; the Born rule is built in via the standard quantum-mechanical formalism. Transport: yes, without modification.

Channel-level unitarity. Standard QFT unitarity is inherited at the level of the asymptotic-safety effective action. The visible-sector effective dynamics — derived by integrating out high-momentum modes in the RG flow — is CPTP, with the dilation living at higher RG scales. Channel-level unitarity transports.

Memory / divisibility. Integrating over high-momentum modes in an RG flow can generate non-local-in-time or memory-bearing effective interactions. This gives a strong mechanism-level analog of OI's visible-hidden memory picture, with the split operating in momentum space rather than the spatial substratum. The existence of such non-Markovian effective structure does not by itself certify P-indivisibility in Main's divisibility sense; that stronger property would require a separate propagator analysis.

Commutant gauge-invariance. Asymptotic safety preserves gauge-invariance of the effective action under the RG flow via Ward identities. Observable physics is determined by gauge-invariant effective operators. The commutant gauge-invariance pattern transports; the specific gauge group is whatever Yang-Mills structure is included in the matter content (typically the SM gauge group, in matter-included formulations).

Asymptotic safety transport summary. Standard quantum probability and unitarity are inherited from the QFT framework, the RG split supplies a strong memory-mechanism analog, and gauge-invariance is preserved. The comparison does not establish P-indivisibility from non-Markovianity alone. Mechanistically this remains one of the closest cross-program parallels to OI, but not an all-four-feature theorem.

14.10 Cross-program synthesis

The four-feature audit across OI, string theory, LQG, causal sets, and asymptotic safety produces the following pattern:

Program	Born-form / quantum probability	Unitary dilation	Memory / indivisibility	Commutant gauge
String theory	Yes (built in)	Yes	Non-Markovian modes; indivisibility not established here	Pattern; specific differs

Program	Born-form / quantum probability	Unitary dilation	Memory / indivisibility	Commutant gauge
OI	Yes (represented)	Yes	C4 memory; indivisible at recurrence scale	SM local factors on the stated physical branch
LQG	Yes (built in)	Yes	Memory plausible; indivisibility open	Pattern; specific differs
Causal sets	Open (QSG)	Structurally	Memory plausible; indivisibility open	Pattern; trivial gauge
Asymptotic safety	Yes (built in)	Yes	Non-Markovian EFT mechanisms; indivisibility not established here	Pattern; specific differs

Three observations:

The restriction-interface features are robust, but at different logical strengths. Quantum formulations carry their probability rule and admit unitary dilation; several programs also exhibit or plausibly generate memory-bearing effective dynamics when degrees of freedom are integrated out. The stronger P-indivisibility statement does not follow from non-Markovianity alone and is proved in OI here only at recurrence scale from C4 plus a fixed finite reversible representative. Thus §15.3's universality claim is an architectural transport claim, not a theorem that partial trace forces every feature in the table. For OI specifically the split is sharper than "integrating out generates memory": partial trace alone does not force memory — the memory-free reversible witnesses are exactly the visible-decoupled maps, whose visible dynamics carries no hidden dependence. But in the uniform full-product finite reversible setting, the T-level form of (C1) is a non-permutation one-step visible matrix, and then memory follows in general rather than only by census: [Main §3.4]'s non-permutation-witness lemma gives P-indivisibility at recurrence scale, while every first-order Markov process is P-divisible. Equivalently, the visible factor of a finite deterministic bijection has zero asymptotic entropy rate (its entire history is fixed by one of finitely many initial substratum states), whereas a stationary first-order Markov chain

with non-permutation T has $H(X_{t+1} \mid X_t) > 0$. That second route needs less than uniformity and less than $|V| = 2$: it needs a φ-invariant prior — which makes the visible process stationary — giving positive weight to some visible state whose T-row is non-degenerate. Uniform-on-the-full-product is one such prior, so the ensemble censused above is a special case, and the theorem covers the non-uniform invariant priors that [Main §3.4]'s arbitrary-kernel (response-table) extension admits (check 10, `equivalence_recovery_probes.py`, with a prior supported on a visible-decoupled block as the control that correctly does not fire). The sharp form is worth stating on its own: the visible factor of a finite reversible substratum has zero entropy rate under every invariant prior, so whatever randomness an embedded observer records is bounded by $\log|\mathcal{C}_V \times \mathcal{C}_H|$ and is exhausted at recurrence scale — the same scale at which the indivisibility witness appears, and for the same reason. The exhaustive checks are therefore regressions of that theorem, not its proof: at $|V| = 2$ and $|H| = 2, 3, 4$, respectively, all 16/16, 648/648, and 39,168/39,168 T-level-C1 bijections carry visible memory; the $|H| = 4$ complement consists of exactly $1{,}152 = 2(4!)^2$ visible-decoupled bijections (checks 7–9, `equivalence_recovery_probes.py`). The 648 denominator coincides with [Main §3.1]'s entanglement-breaking census but the diagnostic is different: that census classifies channels as measure-and-prepare, while this one classifies visible processes as memory-bearing, and the two must not be merged.

Symmetry-restriction comparisons require specified actions. Several audited models use invariant observables, but their physical gauge constructions differ. The OI route is the conditional condensate-stabilizer/central-reduction branch; other programs use their own compactification or matter-sector assumptions. Shared local factors are a target match, not evidence of one exact observable class without the associated algebra map.

Asymptotic safety has the closest mechanism-level similarity to OI. Its visible-hidden split via momentum-space RG flow is structurally closest to OI's spatial-substratum trace-out. Both produce non-Markovian visible dynamics through the same architectural pattern (unitary on the dilation, CPTP on visible). This suggests asymptotic safety and OI may be more closely related than either is to string theory or LQG, and may share more universality-class structure.

The cross-program audit refines §13.1.4's per-condition A1–A6 analysis by identifying which observational structures recur even where the structural conditions differ. The evidence supports a shared *interface architecture* — coarse graining/restriction, quantum-probability form where quantum formalism is present, dilation structure, and possible hidden-memory effects — but not a blanket derivation of P-indivisibility or of the Born exponent from A1, A3, A4, A6 plus the SM target. Those stronger properties must be established by the program-specific dynamics. Section 14 is therefore a transport audit, not an independent derivation theorem.

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15. Theoretical and empirical corroboration

The framework's claims are of mixed types. Some are directly empirical (the Born rule, channel-level unitarity, non-Markovian marginal observation) and admit precision experimental tests. Others are structural — the partial-trace mechanism, the algebra-channel formalism, the universality-class hypothesis — and are corroborated through theorem-level results in independent literatures and through analog constructions in adjacent fields. This section collects the most consequential corroborations, organized by claim type. The ordering reflects strength of support: §15.1 documents the strongest external corroboration (theorem-level results for the framework's central technical mechanism); §15.5 acknowledges what remains under-supported.

15.1 The partial-trace mechanism

The framework's central technical move — that visible-sector dynamics under trace-out of a hidden sector is generically non-Markovian and admits a unitary quantum description — has theorem-level corroboration in three independent recent results.

Stochastic-quantum correspondence at theorem level. Barandes [Barandes2026], building on [Barandes2023, Barandes2025], establishes a precise correspondence theorem: every indivisible stochastic process on a configuration space corresponds to a unitarily evolving quantum system, and conversely, every quantum system can be represented as such a process. The theorem provides a first-principles explanation for why quantum theory uses complex Hilbert spaces, linear-unitary evolution, and the Born rule. This is the machinery [Main §3.4] retains — for the compressed form and its interpretation under the translation hypothesis, the fixed-basis representation's existence internal — rather than the gate from the deterministic substratum to the emergent description, established at theorem-level rigor in the foundations literature independent of the OI framework's specific content. [Main]'s finite-horizon process-dilation theorem, with its response-table extension, is consistent with the same correspondence: for each finite horizon K, every stochastic process on a finite configuration space, P-divisible or not, is reproduced on that horizon by the visible marginal of a finite deterministic bijection, depending on K, from a hidden prior independent of the visible initial state ([Main §3.4]); no single finite bijection reproduces a nonperiodic process exactly at all times.

Weak-memory regime for hidden-degree-of-freedom dynamics. Meyer and Brandner [BrandnerMeyer2025], following [Brandner2025a], prove that for autonomous linear evolution equations describing systems coupled to inaccessible degrees of freedom, integrating out those degrees of freedom yields well-defined non-Markovian visible dynamics in a weak-memory regime, with explicit error bounds and a convergent perturbation scheme. The setting is general open quantum systems with no stake in OI being right, and the conclusion is that visible-sector dynamics with hidden-sector memory is a mathematically clean

construction with a rigorous regime of validity. The framework's C2 persistence condition — in its slow-bath realization — and the resulting non-Markovian visible dynamics fall within this regime, so the framework's central technical move sits inside an established theorem.

Hidden-model decoherence producing classical–quantum dynamics. Tomizuka and Takeda [TomizukaTakeda2026] construct an explicit hidden model in which a mediator is coupled to additional unobserved environmental degrees of freedom, and derive the resulting reduced dynamics by tracing them out. The reduced dynamics is generically non-Markovian, reflecting environmental memory effects. Their motivation is the question of whether classical–quantum dynamics in gravity can arise as effective descriptions of fully quantum systems; the construction is structurally identical to OI's mechanism (visible–hidden partition, unitary on the dilation, trace-out producing non-Markovian visible dynamics) developed independently for a different physics question. The structural identity is independent corroboration that the OI mechanism is the standard way the open-systems literature now thinks about hidden-degree-of-freedom dynamics.

Experimental implementation in quantum metrology. The hidden-Markov-model framework underlying the partial-trace mechanism has direct experimental implementation in quantum metrology [QECNonMarkovian2025]: the model is Markovian on the joint system + environment and non-Markovian on the visible-sector reduced state, with correlations stored in the environment and read back producing the framework's signature non-Markovian behavior. The framework's mechanism is therefore not only mathematically rigorous but realizable in current experimental quantum-information platforms.

15.2 The algebra-channel formalism

The §8.3 / §9.1 algebra-channel formalization adopted in this paper draws on standard machinery from algebraic quantum field theory (AQFT). The formalism's reliability and expressive power are established in the AQFT literature.

AQFT canonical references. The algebraic-QFT literature cited in [BrunettiFredenhagen2024] provides the local-algebra, state and covariance framework used as an EFT target in §10.4. Type-III local factors [Yngvason2005] concern appropriate relativistic infinite-system representations. OI's exact finite-cutoff construction has finite matrix algebras; obtaining a local AQFT net, its type and its channel invariance requires the physical limit described there. The curved-spacetime algebraic literature [LabuschagneMajewski2025] supplies relevant mathematical tools, rather than completing that construction.

Stinespring dilation as universal device. Stinespring's theorem supplies a prepared isometry for the full channel, with unitary realizations available in an appropriate ancillary setting. Section 9 distinguishes that isometry from a full unitary and separates finite-dimensional kernel uniqueness (Propositions 9.7a–9.7b) from the separable written proof (Proposition 9.7c). OI's uniform

mixed hidden reference is a unitary realization rather than a minimal pure-reference dilation. These full-channel results do not identify arbitrary restricted-algebra observers' ambient channels or their multi-time generating dynamics. The channel-learning literature [RandomStinespring2025] provides comparison tools under its own operational assumptions, not the missing cross-framework intertwiner.

15.3 Structural realism in operational and dual-theory contexts

The hierarchical structural-realism position developed in §2 and §12 is supported by independent work in philosophy of physics that arrives at compatible positions through different routes.

Operational theories as structural realism. Adlam [Adlam2022] argues that operational axiomatizations of quantum mechanics (Hardy [Hardy2001], Barrett [Barrett2007], and the broader Mackey [Mackey1963] tradition) can be interpreted as a novel form of structural realism: probing the modal structure of possible operational theories, with quantum mechanics as a specific point in this space. The view aligns with Structure's two-dimensional hierarchy (observation × gauge): the observational axis corresponds to Adlam's modal-structure-of-operational-theories framing, and the gauge axis corresponds to Structure's parallel articulation of the gauge-equivalence levels at which structural commitment lives. The framework's structural-realism commitment is therefore situated within an active research program in philosophy of physics, not a novel position in isolation. Operational reconstruction theorems characterize quantum theory under their own axiom sets; [Main §3.2] distinguishes those hypotheses from the exact finite-law representation and the additional coherent operational lift. They do not establish Theorem 23's missing inverse selector or a complete substratum gauge class.

Common-core realism in QM-string dualities. Dawid and Franzmann [DawidFranzmann2025] develop a structural-realism account specifically for dualities in quantum mechanics and string theory, accommodating ontological pluralism while preserving structural realism through what they call "common core realism" — the proposal that invariant structure across dual models is the proper target of realist commitment. This is precisely the conceptual move §12.1 makes in the universality-class framing: the realist commitment is to structure invariant across substratum descriptions, with specific substratum content being framework-relative. Their work cites parallel programs by Matsubara–Johansson, Rickles, Huggett, and De Haro–Butterfield that converge on similar positions. The universality-class proposal of §9 can be read as one technical articulation of common-core realism specialized to embedded-observer systems.

15.4 Universality classes in adjacent literatures

The universality-class framework of §9 is not unique to embedded-observer systems; analogous structures appear in dissipative quantum systems and many-

body operator dynamics, where they are working analytical tools rather than conjectural propositions.

Emergent universality classes in dissipative quantum systems. Zhou et al. [ZhouEtAl2025] develop an effective field theory description for the universal dynamics of dissipative quantum systems with dipole moment conservation, identifying a strongly interacting non-equilibrium fixed point that governs equal-time phase fluctuations. The work is independent of OI but uses the same conceptual apparatus — universality classes characterized by specific structural features (dipole conservation in their case; partial-trace observational features in OI's), with framework-specific content (subdiffusive vs diffusive transport in their case; specific gauge structure in OI's). The existence of working analyses of this kind in active research programs is corroboration that the universality-class framework is a productive analytical tool, not merely a programmatic gesture.

Random-matrix universality from fast-slow decomposition. Lunt et al. [Lunt2025] prove the emergence of a universal random-matrix description of fast-mode dynamics in many-body quantum systems, via the memory-function formalism with a careful split into fast and slow modes. The fast-mode universality is direct analog to OI's partial-trace structure: hidden-sector (fast) dynamics produce universal observational features at the visible-sector (slow) level, independent of the specific microscopic content. The result is theorem-level for the random-matrix universality class; the same architecture (fast-slow decomposition producing universality at the slow-mode level) is the architecture OI's framework uses to produce SM-reproducing universality at the visible-sector level.

Holographic Schwinger-Keldysh as analog of OI's hidden memory. Recent work on holographic Schwinger-Keldysh contours [AmmonEtAl2025], extending earlier work [HolographicSK2024], establishes that the AdS/CFT correspondence prescribes a specific Schwinger-Keldysh contour structure for real-time correlation functions, with non-trivial out-of-time-order four-point functions exploring the full four-fold geometry. §14.3 identifies this as a holographic analog of OI's hidden-memory (non-Markovian) mechanism; it does not establish P-indivisibility, which §14.3 leaves unestablished cross-framework. The recent literature provides concrete prescriptions making the memory-level analog precise, supporting the §14.3 transport claim at that level.

15.5 Empirical anchors

Direct experimental confirmation applies to the partial-trace-level features identified in §14 (Born rule, channel-level unitarity, non-Markovian marginal) and to the framework's specific predictions developed in [SM], [GR], and [Juno]. The following are the most recent and precise confirmations.

Born rule precision. The cleanest test of higher-order interference is the multi-path interferometry experiment of [Sinha2010], which bounded the magnitude of three-path interference to less than 10^{-2} of the expected two-path interference,

ruling out third- and higher-order interference in optical regimes. The bound was tightened to 10^{-3}–10^{-4} by [Söllner2012] with detector-nonlinearity calibration. Recent proposals using ultracold-atom interferometry [Kanthak2025] aim for further precision via Bose–Einstein condensate platforms that avoid systematic errors from material slits. The accumulated bound is consistent with the Born rule as a partial-trace-level invariant.

Non-Markovian dynamics in physical systems. Direct experimental observation of non-Markovian visible dynamics from hidden slow degrees of freedom: [Mehl2012] in colloidal systems and [Groeblacher2015] in macroscopic micromechanical oscillators. These confirm that the trace-out mechanism is realized in physical systems and that the resulting non-Markovian behavior is operationally accessible. Recent experimental implementations in metrology [QECNonMarkovian2025] demonstrate the same mechanism in quantum-information platforms.

Cosmological tests of the framework's GR-emergence cluster. [DESI_DR2_2025] reports rejection of the cosmological constant at 2.8σ–4.2σ depending on the supernovae dataset, with potential phantom crossing at $z \approx 0.5$ and a $2\sigma+$ detection of nonzero neutrino masses in extended-parameter cosmology. The framework's [GR §7] carries dark energy in Type II running-vacuum form — the functional form the local expansion shared by every smooth model, the content the Type II channel and the magnitude; the [Bertini2025] direct fit to DESI DR2 + DES-Y5 + Planck 2018 reports $\nu = -(2.5 \pm 1.3) \times 10^{-4}$, consistent with zero at 2σ; the framework's expected magnitude $|\nu| \sim 10^{-32}$ (conditional on the Type II reading) is itself indistinguishable from zero at this precision, so the two are consistent. DESI Year 5 results, expected within the publication horizon, will be discriminating.

Neutrinoless double-beta decay. [LEGEND2024] reports LEGEND-200 sensitivity $T_{1/2} > 2.8 \times 10^{26}$ yr at 90% CL on $0\nu\beta\beta$ in ^{76}Ge (combined with GERDA + Majorana Demonstrator). The framework's [SM §8.4] predicts Majorana neutrinos, so $0\nu\beta\beta$ should be observed; LEGEND-1000 is designed to reach the inverse-ordering range. A positive observation would confirm the framework's neutrino-sector prediction; a null result at LEGEND-1000 sensitivity would constrain the framework. The swampland literature [IbañezMartinLozanoValenzuela2017] independently constrains neutrino-sector parameters via the Weak Gravity Conjecture, providing a cross-validation pathway: the framework's normal-ordering Majorana prediction sits within the WGC-allowed region.

Black-hole entropy. GW250114 (LIGO/Virgo/KAGRA, September 2025) confirmed Hawking's area theorem at 99.999% confidence — classical horizon dynamics the framework preserves; the 1/4 coefficient of [GR §5], conditional on H-slope with the horizon and frame conditions, has no direct empirical test. The observation is a non-discriminating consistency for the cosmological-horizon application of the partial-trace mechanism.

15.6 What remains under-supported

Three classes of claims remain without independent support and would benefit from focused future work.

The cross-framework class-membership proposal. Proposition 9.5 establishes a restricted positive inclusion for two channel-preserving transformations, and Definition 9.4 supplies the precise observational equivalence. Whether particular non-OI microscopic models occupy the same class, or how that class compares with the full proposed $\mathcal{G}_{\text{sub}}$ quotient, remains open. The analogous literature in §15.4 motivates the program but supplies neither the physical OI EFT pair nor the required comparison map. Completing §10.4 and the §13.1.1 construction is the concrete route to that test.

Per-condition A1–A6 audit for LQG, causal sets, and asymptotic safety (§13.1.4). Structure's analysis is the only cross-program per-condition comparison of its kind. Independent corroboration would require another author independently performing the analysis and arriving at compatible conclusions. The framework provides specific testable claims — e.g., A2 (determinism) is the most discriminating condition across candidate programs — that can be checked against each program's foundational documents.

Observer-admission distinction (§9.6). The structural distinction between observer-admitting and observer-lacking substrata is novel; the closest concept in the literature is anthropic selection (Vilenkin, Tegmark), which is statistical/probabilistic rather than structural. The distinction is not directly testable as stated; its empirical content lies in the predictions it underwrites within the SM-reproducing observer-admitting class, which are tested through the [SM §7] prediction set.

Acknowledgments

This is the fifth core paper of the OI framework, alongside [Main, SM, GR, Substratum]. It consolidates content originally developed across three separate documents: a canonical articulation of the framework's two-dimensional hierarchical structure (now §2), a technical analysis of the OI-string substratum-level relationship (now Part I), and a programmatic framework of universality classes of embedded observers (now Part II), with cross-framework synthesis in Part III. The proposal investigated in Part I was raised in discussion with [collaborator/AI-assisted analysis, depending on author preference]. Errors and speculative content are the author's responsibility.

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